

AP EAMCET-2019-Engineering-22 Apr-Shift-1
Mathematics

1. If $f : R \rightarrow R$ is defined by $f(x) = \begin{cases} \lceil [x-5] \rceil, & \text{for } x < 5 \\ \lfloor [x-5] \rfloor, & \text{for } x \geq 5 \end{cases}$

then $(f \circ f)\left(-\frac{7}{2}\right) =$

(here $[x]$ is the greatest integer not exceeding x)

(1) $(f \circ f)\left(-\frac{11}{2}\right)$

(2) $(f \circ f)\left(-\frac{9}{2}\right)$

(3) $(f \circ f)(3)$

(d) $(f \circ f)\left(\frac{9}{2}\right)$

2. If $f : A \rightarrow B$ is an onto function such that

$f(x) = \sqrt{|x| - x} + \frac{1}{\sqrt{|x| - x}}$ then A and B are respectively

(1) $(-\infty, \infty), (0, \infty)$

(2) $(-\infty, 0), [2, \infty)$

(3) $(0, \infty), (2, \infty)$

(4) $(-\infty, 0], (0, \infty)$

3. $\frac{1}{2 \cdot 5} + \frac{1}{5 \cdot 8} + \frac{1}{8 \cdot 11} + \dots 16 \text{ terms} =$

(1) $\frac{4}{25}$

(2) $\frac{8}{25}$

(3) $\frac{16}{25}$

(4) $\frac{1}{25}$

4. The maximum value of the determinant of the matrix

$$\begin{vmatrix} 1 + \sin^2 x & \cos^2 x & 4 \sin 2x \\ \sin^2 x & 1 + \cos^2 x & 4 \sin 2x \\ \sin^2 x & \cos^2 x & 1 + 4 \sin 2x \end{vmatrix} \text{ is}$$

(1) 0

(2) 2

(3) 4

(4) 6

5. If $A = \begin{bmatrix} 1 & 0 & -2 \\ -2 & -1 & 2 \\ 3 & 4 & 1 \end{bmatrix}$, then $A^{-1} =$

(1) $A^2 - 2A - 4I$

(2) $A^2 - A - 3I$

(3) $\frac{1}{2}[A^2 + A + 2I]$

(4) $A^2 + A + 2I$

6. If the system of simultaneous linear equations $x + y + z = a$, $x - y + bz = 2$, $2x + 3y - z = 1$ has infinitely many solutions then $b - 5a =$

(1) $\frac{4}{5}$

(2) 3

(3) 7

(4) -3

7. If $z = x + iy$, $x, y \in R$ and if the point P is the Argand plane represents z , then the locus of P satisfying the condition

$$\operatorname{Arg}\left(\frac{z-1}{z-3i}\right) = \frac{\pi}{2} \text{ is}$$

$$(1) \left\{ z \in C / \left| z - \frac{1+3i}{2} \right| = \frac{\sqrt{10}}{2} \right\}$$

$$(2) \{ z \in C / (3-i)z + (3+i)\bar{z} - 6 = 0 \}$$

$$(3) \left\{ z \in C / (3-i)z + (3+i)\bar{z} - 6 > 0, \left| z - \frac{1+3i}{2} \right| = \frac{\sqrt{10}}{2} \right\}$$

$$(4) \left\{ z \in C / (3-i)z + (3+i)\bar{z} - 6 < 0, \left| z - \frac{1+3i}{2} \right| = \frac{\sqrt{10}}{2} \right\}$$

8. If P , Q and R are the points, respectively representing the complex number z , $ze^{\frac{i\pi}{3}}$ and $z\left(1 + e^{\frac{i\pi}{3}}\right)$ in Argand plane, then

the area of triangle PQR is

$$(1) \sqrt{3}|z|^2$$

$$(2) \frac{\sqrt{3}}{2}|z|^2$$

$$(3) \frac{\sqrt{3}}{4}|z|^2$$

$$(4) 2\sqrt{3}|z|^2$$

9. $A(z_1)$ and $B(z_2)$ are two points in the Argand plane. Then the

locus of the complex number z satisfying $\text{Arg}\left(\frac{z - z_1}{z - z_2}\right) = 0$ or π ,

is

(1) The circle with $\overline{AB}$ as the diameter.

(2) The ellipse with A, B as extremities as the major axis.

(3) The perpendicular bisector of $\overline{AB}$.

(4) The straight line passing through the points A and B .

10. If x is the cube root of unity other than 1 the,

$$\left(x + \frac{1}{x}\right)^2 + \left(x^2 + \frac{1}{x^2}\right)^2 + \dots + \left(x^{12} + \frac{1}{x^{12}}\right)^2 =$$

(1) 12

(2) 64

(3) 24

(4) 0

11. If $3x^2 - 7x + 2 = 0$ and $15x^2 - 11x + a = 0$ have a common root and a is a positive real number, then the sum of the roots of the equation $15x^2 - ax + 7 = 0$, is

(1) $\frac{76}{15}$

(2) $\frac{38}{15}$

(3) $\frac{2}{15}$

(4) $\frac{36}{15}$

12. If α, β are the roots of $x^2 - |a|x - |b| = 0$, such that $|\alpha| < |\beta|$. If

$|\alpha| < |\beta| - 1$, then the positive root of $\log_{|\alpha|} \left(\frac{x^2}{\beta^2} \right) - 1 = 0$ is

(1) $< |\alpha|$

(2) $< \alpha$

(3) $< \beta$

(4) $> \beta$

13. If $x \in R$ and $1 \leq \frac{3x^2 - 7x + 8}{x^2 + 1} \leq 2$ then the minimum and

maximum values of x are respectively

- (1) 1, 2
- (2) 5, 12
- (3) 6, 10
- (4) 1, 6

14. Let $\phi(x) = \frac{x}{(x^2 + 1)(x + 1)}$ is a, b and c are the roots of the

equation $x^3 - 3x + \lambda = 0$ then $\phi(a)\phi(b)\phi(c) =$

- (1) λ
- (2) $\frac{-\lambda}{(\lambda + 2)(\lambda^2 + 16)}$
- (3) $\frac{\lambda}{(\lambda + 2)}$
- (4) $\frac{\lambda}{(\lambda + 2)(\lambda^2 + 16)}$

15. In an examination hall there are mn chairs in m rows and n columns. The number of ways in which m students can be seated such that no row is vacant is

- (1) $m^n n!$
- (2) $n^m m!$
- (3) $m^m n!$
- (4) $n^n m!$

16. Consider the following statements

I: The number of non trivial even divisors of the number

$$2^{\alpha_1} 3^{\alpha_2} 4^{\alpha_3} 5^{\alpha_4} 6^{\alpha_5} \text{ is } (\alpha_1 + 1)(\alpha_2 + 1)(\alpha_3 + 1)(\alpha_4 + 1)(\alpha_5 + 1) - 2$$

II: The number of non trivial odd divisors of the number

$$2^{\alpha_1} 3^{\alpha_2} 4^{\alpha_3} 5^{\alpha_4} 6^{\alpha_5} \text{ is } \alpha_2 + \alpha_4 + \alpha_5 + \alpha_2 \alpha_4 + \alpha_4 \alpha_5$$

- (1) I is false and II is false
- (2) I is true and II is true
- (3) I is false and II is true
- (4) I is true and II is false

17. The coefficient of x^5 in the expansion of $(x^2 + 2x + 3)^5$ is

- (1) 1052

- (2) 540
- (3) 480
- (4) 1020

18. If x is so small, that x^5 and higher powers of x may be neglected, then the coefficient of x^4 in the expansion of $\sqrt{x^2 + 4} - \sqrt{x^2 + 9}$ is

- (1) $\frac{19}{1728}$
- (2) $\frac{-19}{1728}$
- (3) $\frac{43}{1728}$
- (4) $\frac{-43}{1728}$

19. If $\frac{8}{(x+3)^2(x-2)} = \frac{Ax+B}{(x+3)^2} + \frac{C}{x-2}$ then $25(B+8C-A) =$ then

$$A+B+C+D =$$

- (1) 25
- (2) 1

(3) 8

(4) -8

20. Let α, β and γ such that $0 < \alpha < \beta < \gamma < 2\pi$ for any $x \in R$ if $\cos(x + \alpha) + \cos(x + \beta) + \cos(x + \gamma) = 0$ $\tan(\gamma - \alpha) =$

(1) $-\sqrt{3}$

(2) 0

(3) 1

(4) $\sqrt{3}$

21. If ABC is not a right angled triangle and

$$\sin\left(\frac{\pi}{4} - A\right)\sin\left(\frac{\pi}{4} - B\right) = -\frac{1}{2\sqrt{2}}\operatorname{cosec}\left(\frac{\pi}{4} - C\right) \quad \text{then}$$

$$\tan A \tan B + \tan B \tan C + \tan C \tan A =$$

(1) $\cot A + \cot B + \cot C$

(2) $\tan A + \tan B + \tan C$

(3) $\frac{1}{\tan A + \tan B + \tan C}$

(4) $\frac{1}{\cot A + \cot B + \cot C}$

22. If $\tan \frac{\theta}{2} = \operatorname{cosec} \theta - \sin \theta$ then $\tan^2 \frac{\theta}{2} =$

- (1) $2 - \sqrt{5}$
- (2) $-2 + \sqrt{5}$
- (3) $2 + \sqrt{5}$
- (4) $\sqrt{2} + 5$

23. The number of real value of $x \in [0, 2\pi] - \left\{ \frac{\pi}{2}, \frac{3\pi}{2} \right\}$ satisfying the equation $|\cos x|^{2 \sin^2 x - 3 \sin x + 1} = 1$ is

- (1) 3
- (2) 4
- (3) 5
- (4) 6

24. The sum of maximum and minimum values of

$$2(\cos^{-1} x)^2 - \pi \cos^{-1} x + \frac{\pi^2}{4} \text{ is}$$

- (1) $\frac{\pi^2}{8}$

(2) $\frac{3\pi^2}{8}$

(3) $\frac{3\pi^2}{2}$

(4) $4\pi^2$

25. If $y = \log_e \tan\left(\frac{\pi}{4} + \frac{x}{2}\right)$ then $\tanh\left(\frac{y}{2}\right) =$

(1) $\cot\left(\frac{x}{2}\right)$

(2) $\tan x$

(3) $\coth x$

(4) $\tan\left(\frac{x}{2}\right)$

26. In triangle ABC, if a, b, c are in arithmetic progression, then

$$\cos A + 2\cos B + \cos C =$$

(1) 1

(2) 2

(3) $\frac{3}{2}$

(4) $\sqrt{3} + 1$

27. If the area of the triangle ABC is $b^2 - (c - a)^2$ then $\tan B =$

(1) 1

(2) $\frac{13}{15}$

(3) $\frac{1}{4}$

(4) $\frac{8}{15}$

28. In ΔABC $\frac{1}{r_1^2} + \frac{1}{r_2^2} + \frac{1}{r_3^2} + \frac{1}{r^2} =$

(1) $\frac{a^2 + b^2 + c^2}{\Delta^2}$

(2) $\frac{a + b + c}{\Delta^2}$

(3) $\frac{s^2}{\Delta^2}$

(4) $\frac{4s^2}{\Delta^2}$

29. For a non zero real number x , if the points with position vectors

$$(x-u)\hat{\mathbf{i}} + x\hat{\mathbf{j}} + x\hat{\mathbf{k}}, x\hat{\mathbf{i}} + (x-v)\hat{\mathbf{j}} + x\hat{\mathbf{k}}, x\hat{\mathbf{i}} + x\hat{\mathbf{j}} + (x-w)\hat{\mathbf{k}} \quad \text{and}$$

$$(x-1)\hat{\mathbf{i}} + (x-1)\hat{\mathbf{j}} + (x-1)\hat{\mathbf{k}} \text{ are coplanar, then}$$

(1) $u + v + w = 1$

(2) $uvw = 1$

(3) $\frac{1}{u} + \frac{1}{v} + \frac{1}{w} = 1$

(4) $uv + vw + uw = 1$

30. If P is a point lying on the line passing through the points

$$A(\hat{\mathbf{i}} - \hat{\mathbf{j}} + 3\hat{\mathbf{k}}) \text{ and parallel to the vector } 2\hat{\mathbf{i}} + \hat{\mathbf{j}} - 2\hat{\mathbf{k}} \text{ such that}$$

$$|\mathbf{AP}| = 18, \text{ then a position vector of P is,}$$

(1) $-13\hat{\mathbf{i}} - 5\hat{\mathbf{j}} + 9\hat{\mathbf{k}}$

(2) $11\hat{\mathbf{i}} + 7\hat{\mathbf{j}} - 15\hat{\mathbf{k}}$

(3) $13\hat{\mathbf{i}} - 5\hat{\mathbf{j}} + 9\hat{\mathbf{k}}$

(4) $13\hat{\mathbf{i}} + 5\hat{\mathbf{j}} - 9\hat{\mathbf{k}}$

31. $\mathbf{a}, \mathbf{b}, \mathbf{c}$ are three vectors such that $|\mathbf{a}| = 1, |\mathbf{b}| = 2, |\mathbf{c}| = 3$ and $\mathbf{b} \cdot \mathbf{c} = 0$

. If the projection of $\mathbf{b}$ along $\mathbf{a}$ is equal to the projection of $\mathbf{c}$ along $\mathbf{a}$, then $|2\mathbf{a} + 3\mathbf{b} - 3\mathbf{c}| =$

- (1) 3
- (2) $\sqrt{22}$
- (3) 9
- (4) 11

32. Let $\mathbf{m}$ be a vector of magnitude $\sqrt{3}$ and perpendicular to the vectors $\hat{\mathbf{i}} + \hat{\mathbf{j}}$ and $\hat{\mathbf{j}} - \hat{\mathbf{k}}$. Let $\mathbf{n}$ be the another vector of magnitude $2\sqrt{6}$ and perpendicular to the vector $2\hat{\mathbf{i}} - \hat{\mathbf{j}}$ and $\hat{\mathbf{j}} + 2\hat{\mathbf{k}}$. The area (in sq. units) of the triangle formed with $\mathbf{m}$ and $\mathbf{n}$ as side is

- (1) $\sqrt{2}$
- (2) $\sqrt{6}$
- (3) $2\sqrt{3}$
- (4) $3\sqrt{2}$

33. $\mathbf{a} = \hat{\mathbf{i}} - \hat{\mathbf{j}} + \hat{\mathbf{k}}, \mathbf{b} = \hat{\mathbf{i}} - 2\hat{\mathbf{j}} + \hat{\mathbf{k}}, \mathbf{c} = p\hat{\mathbf{i}} + 2\hat{\mathbf{j}} + q\hat{\mathbf{k}}, \mathbf{d} = p\hat{\mathbf{i}} + q\hat{\mathbf{j}} + 2\hat{\mathbf{k}}$ are given vectors. If projection of $\mathbf{c}$ on $\mathbf{a}$ is $5\sqrt{3}$ units and if $\mathbf{a}, \mathbf{b}, \mathbf{c}$ are parallelepiped of volume 5 cubic units, then $\tan^{-1}(\mathbf{b} \cdot \mathbf{d}) =$

(1) $\frac{\pi}{2}$

(2) $\frac{\pi}{3}$

(3) $\frac{\pi}{4}$

(4) $\frac{\pi}{6}$

34. Given $\mathbf{a} = 2\hat{\mathbf{i}} + \hat{\mathbf{j}} + \hat{\mathbf{k}}, \mathbf{b} = \hat{\mathbf{i}} + 2\hat{\mathbf{j}} - \hat{\mathbf{k}}$ and a unit vector $\mathbf{c}$ are coplanar. If $\mathbf{c}$ is perpendicular to $\mathbf{a}$ then $\mathbf{c} =$

(1) $\pm \frac{1}{\sqrt{3}}(\hat{\mathbf{i}} - \hat{\mathbf{j}} - \hat{\mathbf{k}})$

(2) $\frac{1}{\sqrt{5}}(\hat{\mathbf{i}} - 2\hat{\mathbf{j}})$

(3) $\frac{-1}{\sqrt{3}}(\hat{\mathbf{i}} + \hat{\mathbf{j}} + \hat{\mathbf{k}})$

$$(4) \pm \frac{1}{\sqrt{2}}(-\hat{j} + \hat{k})$$

35. The coefficient of variation of 9,3,11,5,7 is

(1) $\frac{100\sqrt{2}}{7}$

(2) $\frac{200\sqrt{2}}{3}$

(3) $\frac{200\sqrt{2}}{7}$

(4) $\frac{100\sqrt{2}}{3}$

36. The mean deviation about the mean for the following data

Marks obtained	0-10	10-20	20-30	30-40	40-50
Number of boys	6	8	10	4	2

is

(1) 9.33

(2) 5.6

(3) 8.33

(4) 9.6

37. The probability of occurrence of an event is $\frac{2}{5}$ and the probability of non occurrence of an event is $\frac{3}{10}$. If these events are independent, then the probability that only one of the two events occur is

(1) $\frac{27}{25}$

(2) $\frac{27}{50}$

(3) $\frac{7}{25}$

(4) $\frac{14}{25}$

38. Let α be the root of $x^3 + x + 1 = 0$ and suppose that a fair die is thrown 3 times. If a, b, c are the numbers shown on die then the probability that $\alpha^a + \alpha^b + \alpha^c = 0$ is

(1) $\frac{2}{36}$

(2) $\frac{1}{27}$

(3) $\frac{1}{72}$

(4) $\frac{2}{9}$

39. Suppose that a bag A contain n red and 2 black balls and another bag B contains 2 red and n black balls. One of the two bags is selected at random and two balls are drawn from it at a time. When it is known that two balls drawn are red if the probability that those two balls drawn are from bag A is $\frac{6}{7}$, then

$n =$

(1) 6

(2) 4

(3) 8

(4) 7

40. A random variable X has its range $\{-1, 0, 1\}$. If its mean is 0.2 and $P(X = 0) = 0.2$, then $P(X = 1) =$

- (1) 0.1
- (2) 0.7
- (3) 0.4
- (4) 0.5

41. There are 800 families with four children in each family. Assuming equal chance for every child to be a boy or a girl, the number of families expected to have child of both sexes is

- (1) 700
- (2) 100
- (3) 500
- (4) 300

42. A straight line meets the X and Y axes at the points A , B respectively. If $AB = 6$ units, then the locus of the point P which divides the line segment AB such that $AP : PB = 2 : 1$, is

- (1) $3x^2 + y^2 = 36$
- (2) $4x^2 + y^2 = 36$

$$(3) 3x^2 + y^2 = 16$$

$$(4) 4x^2 + y^2 = 16$$

43. If the area of the region bounded by the curves $y = x^2$ and $x = y^2$ is k , then the area of region bounded by the curves

$$\frac{x + \sqrt{3}y}{2} = \left(\frac{\sqrt{3}x - y}{2} \right)^2 \text{ and } \frac{\sqrt{3}x - y}{2} = \left(\frac{x + \sqrt{3}y}{2} \right)^2 \text{ is}$$

$$(1) \frac{\sqrt{3}}{2}k$$

$$(2) \frac{1}{2}k$$

$$(3) k$$

$$(4) \left(\frac{(\sqrt{3} + 1)}{2} \right) k$$

44. The set of values that β can assume so that the point $(0, \beta)$ should lie on or inside the triangle having sides $3x + y + 2 = 0$, $2x - 3y + 5 = 0$ and $x + 4y - 14 = 0$ is

$$(1) \left[\frac{5}{3}, \frac{7}{2} \right]$$

(2) $\left[\frac{2}{3}, \frac{5}{2}\right]$

(3) $\left[-\frac{1}{3}, \frac{2}{3}\right]$

(4) $\left[\frac{1}{2}, \frac{5}{2}\right]$

45. If $(\lambda^2, \lambda + 1)$, $\lambda \in \mathbb{Z}$ belongs to the region between the lines $x + 2y - 5 = 0$ and $3x - y + 1 = 0$ which include the origin, then the possible number of such points is,

(1) 4

(2) 3

(3) 2

(4) infinite

46. If the mid points of the sides BC, CA and AB of a triangle ABC are respectively $(2, 1)$, $(-1, -2)$ and $(3, 3)$, then equation of side BC is

(1) $x - 2y = 0$

(2) $5x - 4y = 6$

(3) $2x + 3y = 8$

(4) $3x - 2y = 6$

47. The distance between the pair of line

$x^2 + 2\sqrt{2}xy + 2y^2 + 4x + 4\sqrt{2}y + 1 = 0$ is

(1) $4\sqrt{2}$

(2) $2\sqrt{2}$

(3) 2

(4) $6\sqrt{2}$

48. A pair of lines $S = 0$ together with the lines given by equations, $8x^2 - 14xy + 3y^2 + 10x + 10y - 25 = 0$ form a parallelogram. If its diagonal intersects at the point $(3, 2)$, then the equation of $S = 0$ is

(1) $6x^2 - 9xy + y^2 - 25x + 30y + 25 = 0$

(2) $8x^2 - 14xy + 3y^2 - 25x + 30y + 50 = 0$

(3) $8x^2 - 14xy + 3y^2 - 50x + 50y + 75 = 0$

(4) $6x^2 + 14xy - 3y^2 - 30x + 40y - 75 = 0$

49. If the equation of the circle having its center in the second quadrant touches the coordinate axis and also the line $\frac{x}{5} + \frac{y}{12} = 1$

is $x^2 + y^2 + 2\lambda x - 2\lambda y + \lambda^2 = 0$, then $\lambda =$

- (1) 3
- (2) 10
- (3) 15
- (4) -2

50. The equation of the circle passing through the point $(2,8)$, touching the lines $4x - 3y - 24 = 0$ and $4x + 3y - 42 = 0$ and having the x coordinates of its center less than or equal to 8 is

- (1) $x^2 + y^2 + 2x - 8y - 8 = 0$
- (2) $x^2 + y^2 - 4x - 6y - 12 = 0$
- (3) $x^2 + y^2 + 4x - 10y + 4 = 0$
- (4) $x^2 + y^2 - 6x - 4y + 24 = 0$

51. The point of intersection of the common tangents drawn to the circles $x^2 + y^2 - 4x - 2y + 1 = 0$ and $x^2 + y^2 - 6x - 4y + 4 = 0$ is

(1) $\left(\frac{5}{2}, \frac{3}{2}\right)$

(2) $\left(\frac{6}{5}, \frac{1}{5}\right)$

(3) $(0, -1)$

(4) $\left(\frac{12}{5}, \frac{7}{5}\right)$

52. The circle $S = 0$ cuts the circle $x^2 + y^2 - 4x + 2y - 7 = 0$ orthogonally. If $(2, 3)$ is the center of circle $S = 0$, then its radius is

(1) 2

(2) 1

(3) 3

(4) 4

53. The equation of the circle which cuts the circles

$$S_1 \equiv x^2 + y^2 - 4 = 0$$

$$S_2 \equiv x^2 + y^2 - 6x - 8y + 10 = 0$$

$$S_3 \equiv x^2 + y^2 + 2x - 4y - 2 = 0$$

at the extremities of the diameter of these circle is

(1) $x^2 + y^2 - 4x - 6y - 4 = 0$

(2) $x^2 + y^2 + 4x - 4 = 0$

(3) $x^2 + y^2 = 25$

(4) $x^2 + y^2 + x + y + 1 = 0$

54. The length of the latus rectum of the parabola

$20(x^2 + y^2 - 6x - 2y + 10) = (4x - 2y - 5)^2$ is

(1) $\frac{\sqrt{5}}{2}$

(2) $2\sqrt{5}$

(3) $\sqrt{5}$

(4) $4\sqrt{5}$

55. $y = 3x - 2$ is a straight line touching the parabola

$(y - 3)^2 = 12(x - 2)$. If a line drawn perpendicular to this line at

P on it, touches the given parabola, then the point P is

(1) $(-1, -5)$

(2) $(-1, 5)$

(3) $(-2, -8)$

(4) $(2, 4)$

56. If (l, m) is the circumcentre of an equilateral triangle inscribed

in an ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ having vertices at points with eccentric

angles θ_1 , θ_2 and θ_3 , then

$$\frac{2}{3} [\cos(\theta_1 - \theta_2) + \cos(\theta_2 - \theta_3) + \cos(\theta_3 - \theta_1)] =$$

(1) $\frac{9l^2}{2a^2} + \frac{9m^2}{b^2} - 1$

(2) $\frac{l^2}{a^2} + \frac{m^2}{b^2} - 3$

(3) $\frac{3l^2}{a^2} + \frac{3m^2}{b^2} - 1$

(4) $\frac{3l^2}{a^2} + \frac{3m^2}{b^2} - \frac{3}{2}$

57. If l and b respectively are the length and breadth of a rectangle of greatest area that can be inscribed in the ellipse $x^2 + 4y^2 = 64$ then $(l, b) =$

- (1) $(16\sqrt{2}, 4\sqrt{2})$
- (2) $(8\sqrt{2}, 6\sqrt{2})$
- (3) $(8\sqrt{2}, 4\sqrt{2})$
- (4) $(6\sqrt{2}, 4\sqrt{2})$

58. If $2x - ky + 3 = 0, 3x - y + 1 = 0$ are conjugate lines with respect to $5x^2 - 6y^2 = 15$, then $k =$

- (1) 6
- (2) 4
- (3) 3
- (4) 2

59. The points $A(2, -1, 4), B(1, 0, -1), C(1, 2, 3)$ and $D(2, 1, 8)$ form a

- (1) rectangle
- (2) square
- (3) rhombus

(4) parallelogram

60. If l_1, m_1, n_1 and l_2, m_2, n_2 are direction cosines of **OA** and **OB** such that $\angle AOB = \theta$ where O is the origin, then the direction cosines of the internal angular bisector of $\angle AOB$ are

$$(1) \frac{l_1 + l_2}{2 \sin \frac{\theta}{2}}, \frac{m_1 + m_2}{2 \sin \frac{\theta}{2}}, \frac{n_1 + n_2}{2 \sin \frac{\theta}{2}}$$

$$(2) \frac{l_1 - l_2}{2 \cos \frac{\theta}{2}}, \frac{m_1 - m_2}{2 \cos \frac{\theta}{2}}, \frac{n_1 - n_2}{2 \cos \frac{\theta}{2}}$$

$$(3) \frac{l_1 - l_2}{2 \sin \frac{\theta}{2}}, \frac{m_1 - m_2}{2 \sin \frac{\theta}{2}}, \frac{n_1 - n_2}{2 \sin \frac{\theta}{2}}$$

$$(4) \frac{l_1 + l_2}{2 \cos \frac{\theta}{2}}, \frac{m_1 + m_2}{2 \cos \frac{\theta}{2}}, \frac{n_1 + n_2}{2 \cos \frac{\theta}{2}}$$

61. The distance of the plane $3x + 4y + 5z + 19 = 0$ from the point $(1, -1, 1)$ measured along a line parallel to the line with direction ratios 2, 3, 1 is

(1) $\frac{23}{5\sqrt{2}}$

(2) $\frac{\sqrt{71}}{5\sqrt{2}}$

(3) $\sqrt{14}$

(4) $\sqrt{23}$

62. For $A \neq 0, x < 0$ $\lim_{n \rightarrow \infty} \frac{\sin x - e^{nx}}{1 + Ae^{nx}} =$

(1) $\frac{1}{A}$

(2) $\sin x$

(3) $-\frac{1}{A}$

(4) $-\sin x$

63. Define $f(x) = \begin{cases} \frac{\sqrt{1+px} - \sqrt{1-px}}{x}, & \text{if } -1 \leq x < 0 \\ \frac{2x+1}{x-2}, & \text{if } 0 \leq x < 1 \end{cases}$

If $\lim_{x \rightarrow 0} f(x)$ exists, then $p =$

(1) -1

(2) $-\frac{1}{2}$

(3) $\frac{1}{2}$

(4) 1

64. Given $\sin x = \sum_{n=1}^{\infty} (-1)^{n-1} \frac{x^{2n-1}}{(2n-1)!}$. If the function $f(x)$ is given

by $f(x) = \frac{\cos(\sin x) - \cos x}{x^4} (x \neq 0)$ and $f(0) = k$ is continuous

at $x = 0$, then $k =$

(1) $\frac{1}{6}$

(2) $\frac{1}{3}$

(3) $\frac{1}{2}$

(4) 0

65. Match for each functions in List-I to its derivative given in List-II.

List-I

A) $\sin^{-1}\left(\frac{2x}{1+x^2}\right)$

B) $\tan^{-1}\left(\frac{1-x}{1+x}\right)$

C) $e^{\log(\sin x + \cos x)}$

D) $\sqrt{1 - \sin 2x} \left(0 < x < \frac{\pi}{4}\right)$

List-II

I) $\cos x - \sin x$

II) $\frac{-1}{1+x^2}$

III) $\frac{2}{1+x^2}$

IV) $\cos x + \sin x$

V) $-\sin x - \cos x$

The correct match is

	A	B	C	D
(1)	III	II	I	V

	A	B	C	D
(2)	II	III	V	IV

	A	B	C	D
(3)	II	III	V	I

	A	B	C	D
(4)	III	II	I	IV

66. If $f(t) = \frac{1 + \operatorname{cosec} t}{1 - \operatorname{cosec} t}$ for $0 < t < \frac{\pi}{2}$ and $f'(t) = f(t)g(t)$, then

$$g(t) =$$

(1) $-4\operatorname{cosec} 2t$

(2) $4\operatorname{cosec} 2t$

(3) $2\sin 2t$

(4) $4\operatorname{cosec} t$

67. If $x = 4\cos^3 \theta$ and $y = 3\sin^2 \theta$ then $\frac{d^2y}{dx^2}$ and $\theta = \frac{\pi}{4}$ is

(1) $\frac{1}{3}$

(2) $\frac{1}{6}$

(3) $-\frac{1}{6}$

(4) $-\frac{1}{3}$

68. Tangents are drawn to the curve $y = \sin x$ from the origin. The locus of the points of contact is

(1) $xy = x + y$

$$(2) \ x^2 y^2 = x^2 - y^2$$

$$(3) \ xy = x - y$$

$$(4) \ x^2 y^2 = x^2 + y^2$$

69. A man of height 2 meters walks at a uniform speed of 7 meters per minute away from a lamp post of height 9 m. The rate (in meters per minute) at which the length of his shadow increase is

$$(1) \ 2$$

$$(2) \ \frac{5}{2}$$

$$(3) \ 3$$

$$(4) \ \frac{7}{2}$$

70. Let a, b, c be real numbers such that $2a + 3b + 6c = 0$ and

$g(x) \equiv ax^2 + bx + c = 0$ has atleast one root in the interval $(1, 2)$.

If a function $f : [1, 2] \rightarrow R$ for which Rolle's means value theorem holds is such that $f(x)$ is a primitive of $g(x)$, then

$$f(x) =$$

$$(1) \ x^3 - 3x^2 + 2x$$

$$(2) 3x^3 - 6x^2 + 2x$$

$$(3) 12x^3 - 14x^2 + 3x$$

$$(4) 3x^3 - x$$

71. The maximum volume (in cubic units) of the cylinder which can be inscribed in a sphere of diameter 6 units is

$$(1) 12\sqrt{3}\pi$$

$$(2) 4\sqrt{3}\pi$$

$$(3) 3\sqrt{3}\pi$$

$$(4) 8\sqrt{3}\pi$$

72. $\int \frac{x}{\sqrt{x+1} + \sqrt{x-1}} dx = A(x)(x+1)^{\frac{3}{2}} + B(x)(x-1)^{\frac{3}{2}} + C$, then

$$A(x) + B(x) =$$

$$(1) \frac{4}{15}$$

$$(2) -\frac{4}{15}$$

$$(3) \frac{2x}{5}$$

$$(4) -\frac{2x}{5}$$

$$73. \int \frac{x \cdot \log x}{(\sqrt{x^2 - 1})^3} dx =$$

$$(1) \sec^{-1}(x) + \frac{\log x}{\sqrt{x^2 - 1}} + c$$

$$(2) \sec^{-1}(x) - \frac{\log x}{\sqrt{x^2 - 1}} + c$$

$$(3) \frac{\log x}{\sqrt{x^2 - 1}} - \sec^{-1}(x) + c$$

$$(4) \frac{-\log x}{\sqrt{x^2 - 1}} - \sec^{-1}(x) + c$$

$$74. \int \frac{\cos 2x \cdot \sin 4x}{\cos^4 x (1 + \cos^2 2x)} dx =$$

$$(1) \log \left(\frac{1 + \cos 2x}{1 + \cos^2 2x} \right) + \sec^2 x + c$$

$$(2) \log \frac{(1 + \cos 2x)^2}{(1 + \cos^2 x)} + \sec x + c$$

$$(3) \log \frac{(1 + \cos 2x)^2}{(1 + \cos^2 2x)} + \sec^2 x + c$$

$$(4) \log \frac{1 + \cos^2 2x}{(1 + \cos 2x)^2} + \sec x + c$$

75. $A_n = \int_{\frac{\pi}{2}}^{\infty} e^{-x} \cos^n x dx$ then $\frac{A_4 - A_6}{A_4} =$

(1) $\frac{3}{2}$

(2) $\frac{7}{37}$

(3) $\frac{5}{37}$

(4) $\frac{2}{7}$

76. $\lim_{x \rightarrow \infty} \left\{ \frac{1}{n+m} + \frac{1}{n+2m} + \frac{1}{n+3m} + \dots + \frac{1}{n+nm} \right\} =$

(1) $\frac{\log_e(m)}{m}$

(2) $\frac{\log_e(1+m)}{1+m}$

$$(3) \frac{\log_e(1+m)}{m}$$

$$(4) \frac{\log_e(1+m)}{1-m}$$

$$77. \tan^{-1} \left[\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{\cos x}{1+e^x} dx \right] =$$

$$(1) \frac{\pi}{4}$$

$$(2) \frac{\pi}{3}$$

$$(3) \frac{\pi}{6}$$

$$(4) \frac{\pi}{2}$$

78. The area (in sq units) bounded by $x^2 = y$, $y = x + 2$ and the X-axis is

$$(1) \frac{2}{3}$$

$$(2) \frac{3}{5}$$

(3) $\frac{5}{6}$

(4) $\frac{4}{5}$

79. **Statement I:** The elimination of arbitrary constants from α, β and γ from $y = (\alpha + \beta + \gamma)x$ results in a differential equation of order three.

Statement II: The elimination of arbitrary constants from α, β and γ from $y = \alpha x + \beta \sin x + \gamma e^x$ results in a differential equation of order three.

Make the correct choice of the following.

- (1) I is true and II is false
- (2) I is false and II is false
- (3) I is true and II is true
- (4) I is false and II is true

80. The general solution of $\sin y \frac{dy}{dx} = \cos y (1 - x \cos y)$ is

- (1) $\sec y = x - 1 - ce^x$
- (2) $\sec y = x + 1 + ce^x$

$$(3) \sec y = x + e^x + c$$

$$(4) \sec y = x - e^x + c$$



AP EAMCET-2019-Engineering-22 Apr-Shift-1 Physics

81. If 1% and 2% are the errors in the measurement of mass and density of a cube respectively, then the error in the measurement of length is
- (1) 1%
 - (2) 3%
 - (3) 2%
 - (4) 4%
82. A body travelling along a straight line path travels first half of the distance with a velocity 7 ms^{-1} . During the travel time of the second half of distance, first half time is travelled with a velocity 14 ms^{-1} and the second half time is travelled with a velocity 21 ms^{-1} . Then the average velocity of the body during the journey is
- (1) 14 ms^{-1}
 - (2) 10 ms^{-1}
 - (3) 9 ms^{-1}

(4) 12 ms^{-1}

83. **Assertion (A)** The range of a projectile is maximum when the angle of projection is 45° .

Reason (R) The range of a projectile depends only on the angle of projection.

(1) Both (A) and (R) are true and (R) is the correct explanation of (A).

(2) Both (A) and (R) are true and (R) is not the correct explanation of (A).

(3) (A) is true but (R) is false.

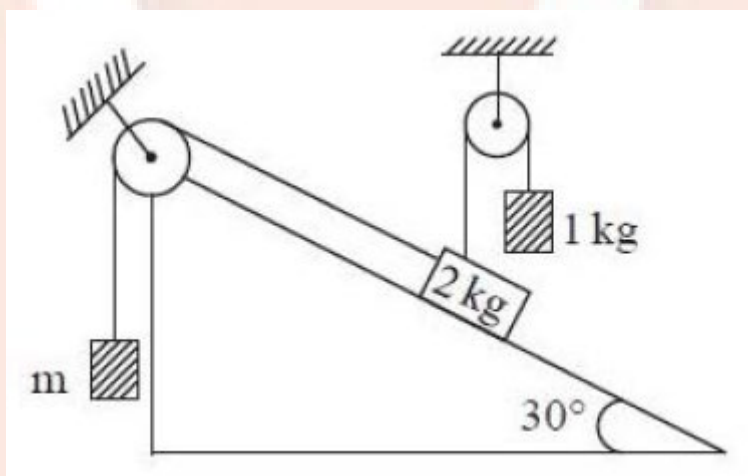
(4) (A) is false but (R) is true.

84. A car moving with a certain velocity, jumps from an inclined plane placed at one bank of a river and reaches the other bank by attaining a maximum height of 80 m. If the same car, moving with the same velocity jumps from another inclined plane having different angle of inclination and reaches the same point on the other bank by attaining maximum height of 45 m, then the width of the river is

(1) 80 m

- (2) 60 m
- (3) 125 m
- (4) 240 m

85. A system containing masses and pulleys connected on an inclined plane is shown in the figure. If the system is in equilibrium then the value of m is

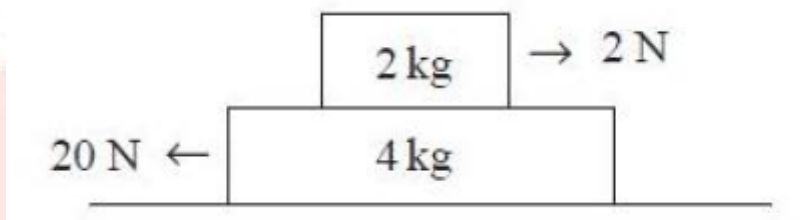


- (1) 1 kg
- (2) 0.5 kg
- (3) 0.75 kg
- (4) 0.25 kg

86. In the arrangement shown in the figure, the coefficient of friction between two blocks is 0.5. The force of friction between the two blocks is

(Assume that the 4 kg block is placed on a smooth horizontal surface.)

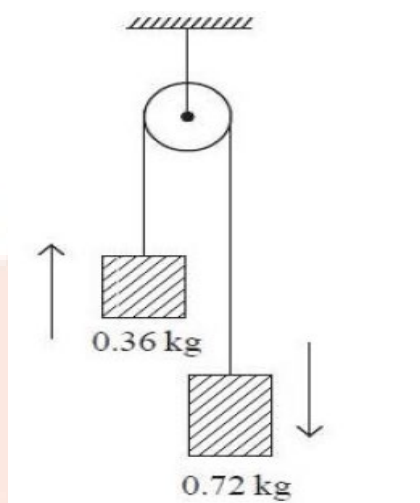
(Acceleration due to gravity = 10 ms^{-2})



- (1) 8 N
- (2) 10 N
- (3) 6 N
- (4) 4 N

87. In the arrangement shown in the figure, work done by the string on the block of mass 0.36 kg during the first second after the blocks are released from state of rest is (Ignore friction and mass of the string.)

(Acceleration due to gravity, $g = 10 \text{ ms}^{-2}$)



- (1) 8 J
- (2) 4 J
- (3) 12 J
- (4) 2 J

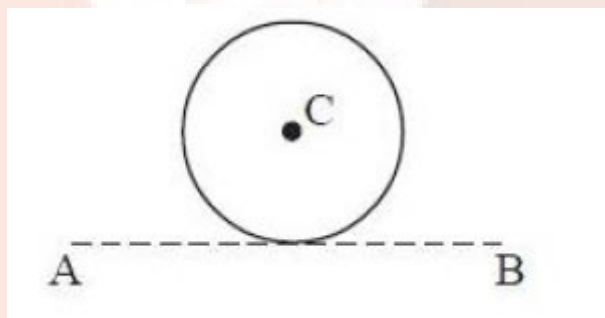
88. A man who is running has half the kinetic energy of a boy of half his mass. The man speeds up by 1 ms^{-1} and then has the same kinetic energy as the boy. The initial speed of the boy is

- (1) $(\sqrt{2} + 1) \text{ ms}^{-1}$
- (2) $2(\sqrt{2} + 1) \text{ ms}^{-1}$
- (3) $\sqrt{2} \text{ ms}^{-1}$
- (4) 2 ms^{-1}

89. A solid sphere rolls down without slipping on a smooth inclined plane of inclination $\sin^{-1}(0.42)$. If the acceleration due to gravity is 10 ms^{-2} , the acceleration of the rolling sphere is

- (1) 1 ms^{-2}
- (2) 2 ms^{-2}
- (3) 3 ms^{-2}
- (4) 4 ms^{-2}

90. A thin wire of length l having a linear density ρ is bent into a circular loop with C as its centre as shown in the figure. The moment of inertia of the loop about the line AB is



- (1) $\frac{5\rho l^3}{16\pi^2}$
- (2) $\frac{\rho l^2}{16\pi^2}$
- (3) $\frac{\rho l^2}{8\pi^2}$

$$(4) \frac{3\rho l^3}{8\pi^2}$$

91. A particle executing SHM along a straight line has zero velocity at points A and B whose distance from O on the same line OAB are a and b , respectively. If the velocity at the midpoint between A and B is v , then its time period is

$$(1) \frac{\pi(b+a)}{v}$$

$$(2) \pi \left(\frac{b-a}{v} \right)$$

$$(3) \left(\frac{b+a}{2v} \right)$$

$$(4) \left(\frac{b-a}{2v} \right)$$

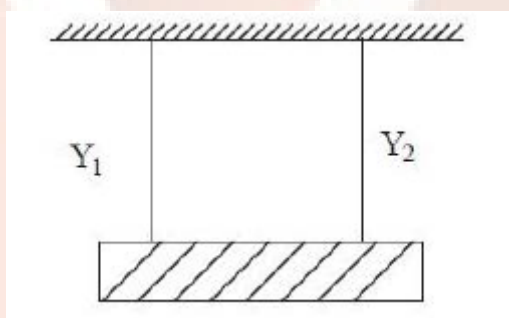
92. A rocket is launched straight up from the surface of the earth.

When its altitude is $\frac{1}{3}$ of the radius of the earth, its fuel runs out and therefore it coasts. If the rocket has to escape from the gravitational pull of the earth, the minimum velocity with which

it should coast is (Escape velocity on the surface of the earth is 11.2 kms^{-1})

- (1) 11.2 kms^{-1}
- (2) 10.7 kms^{-1}
- (3) 9.7 kms^{-1}
- (4) 8.7 kms^{-1}

93. Two wires of equal length and equal cross sectional areas are suspended as shown in the figure. Their Young's moduli are Y_1 and Y_2 , respectively. The equivalent Young's modulus is



- (1) $Y_1 + Y_2$
- (2) $\frac{Y_1 + Y_2}{2}$
- (3) $\frac{Y_1 Y_2}{Y_1 + Y_2}$
- (4) $\sqrt{Y_1 Y_2}$

94. A rain drop of radius r is falling through air, starting from rest. The work done by all the forces on the drop, when it attains terminal velocity, is proportional to

(1) r^3

(2) r^7

(3) r^5

(4) r^4

95. Which of the following statement is correct regarding P-V graph?

(a) Slope of P-V graph in an isothermal process is $-\frac{P}{V}$.

(b) Slope of P-V graph in an adiabatic process is $-\frac{P}{V}$.

(c) Slope of P-V graph in an isochoric process is $-\frac{\gamma P}{V}$.

(d) Slope of P-V graph in an isobaric process is zero.

(1) a, c, d are correct

(2) b, c are correct

(3) a, d are correct

(4) b, c, d are correct

96. Two metallic spheres P and Q are made of same material have same smoothness but the weight of P is 8 times that of Q. If the two are heated to same temperature and left to cool, then the ratio of rate of cooling of Q to that of P is

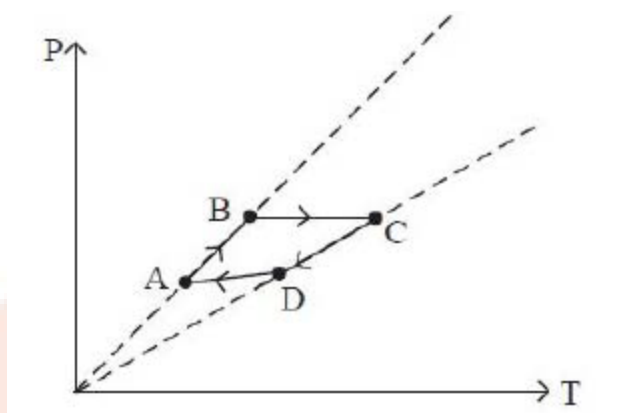
(1) 4

(2) 8

(3) 2

(4) 1

97. Three moles of an ideal monatomic gas performs a cycle ABCDA as shown in the figure. The temperatures of the gas at the states A, B, C and D are 400 K, 800 K, 2400 K and 1200 K, respectively. The work done by the gas during this cycle is
(R is universal gas constant)



- (1) $1200 R$
- (2) $3600 R$
- (3) $2400 R$
- (4) $2000 R$

98. An insulated system contains 4 moles of an ideal diatomic gas at temperature T . When a heat Q is supplied to the gas, 2 moles of the gas is dissociated into atoms and the temperature remained constant. Then the relation between Q and T is

(R = Universal gas constant)

- (1) $Q = RT$
- (2) $Q = 2RT$
- (3) $Q = 3RT$
- (4) $Q = 4RT$

99. Under standard conditions, the density of a gas is $\frac{1400}{1089} \text{ kg-m}^{-3}$

and the speed of sound propagation in it is 330 ms^{-1} , then the number of degrees of freedom of the gas molecules is

- (1) 2
- (2) 7
- (3) 5
- (4) 3

100. When the air column of a resonance tube is vibrated together with a tuning fork, 3 beats are heard per second, either the temperature of the air column is 51°C or 16°C . The frequency of the tuning fork is

- (1) 128 Hz
- (2) 98 Hz
- (3) 105 Hz
- (4) 256 Hz

101. Two sources A and B are producing notes of frequency 680 Hz. A listener moves from A to B with a constant velocity v . If

the speed of sound in air is 340 ms^{-1} , the value of ' v ' so that he hears 10 beats per second is

- (1) 2.0 ms^{-1}
- (2) 2.5 ms^{-1}
- (3) 3.0 ms^{-1}
- (4) 3.5 ms^{-1}

102. A thin converging lens of focal length 25 cm forms a sharp image of an object on a screen placed at a distance of 75 cm from the lens. Later the screen is moved closer to the lens by a distance 25 cm. The distance through which the object is to be shifted so that its image on the screen is sharp again is

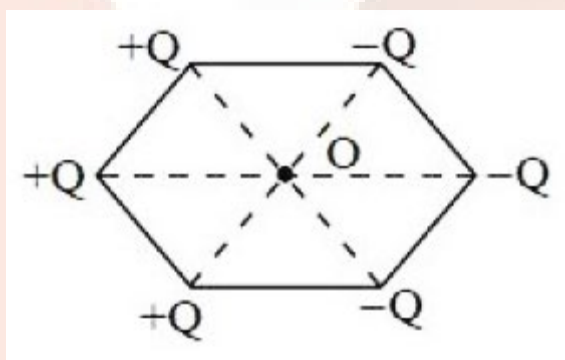
- (1) 50 cm towards the lens
- (2) 50 cm away from the lens
- (3) 12.5 cm towards the lens
- (4) 12.5 cm away from the lens

103. A ray of light is incident on the surface of a glass plate of refractive index $\sqrt{3}$ at the polarising angle. The angle of refraction of the ray is

- (1) 30°
- (2) 45°
- (3) 60°
- (4) 37°

104. Six point charges each of the magnitude Q are placed at the vertices of a regular hexagon of side 'a' as shown in the figure. Electric field intensity on the line passing through the centre O and perpendicular to the plane of the figure at a large distance $x(\gg a)$ from O is

$$\left(\frac{1}{4\pi \epsilon_0} = k \right)$$



- (1) $k \times \frac{4Qa}{x^3}$
- (2) $k \times \frac{2Qa}{x^3}$

(3) $k \times \frac{8Qa}{x^3}$

(4) 0

105. A ball of mass 1 g having a charge of $20 \mu\text{C}$ is tied to one end of a string of length 0.9 m can rotate in a vertical plane in a uniform electric field 100 NC^{-1} directed upwards. The minimum horizontal velocity that must be given to the ball at the lowest position so that it completes the vertical circle is ($g = 10 \text{ ms}^{-2}$)

(1) 9 ms^{-1}

(2) 18 ms^{-1}

(3) 36 ms^{-1}

(4) 6 ms^{-1}

106. A regular hexagon of side 5 cm has a charge $10 \mu\text{C}$ at each of its vertices. The potential at the centre of hexagon is

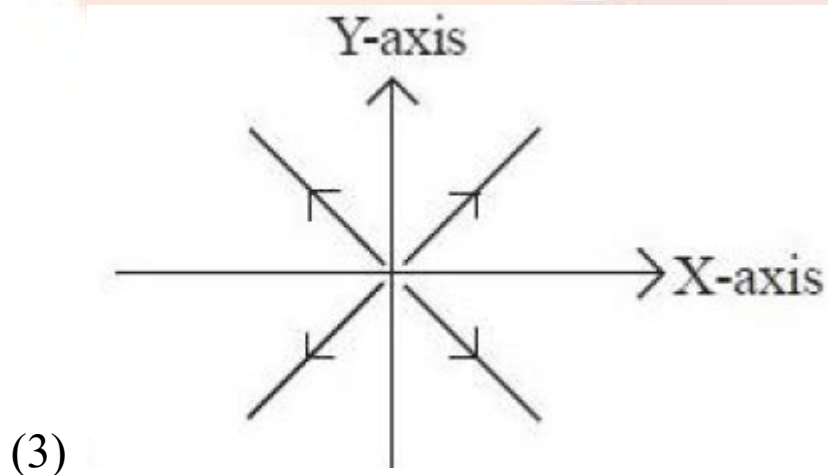
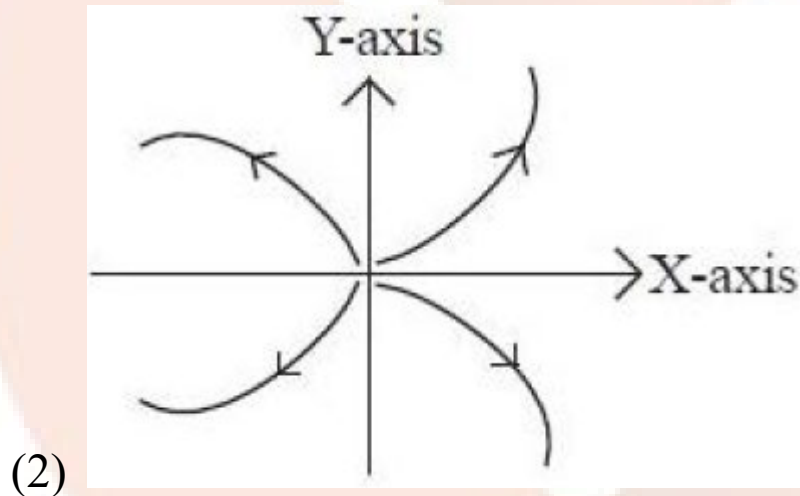
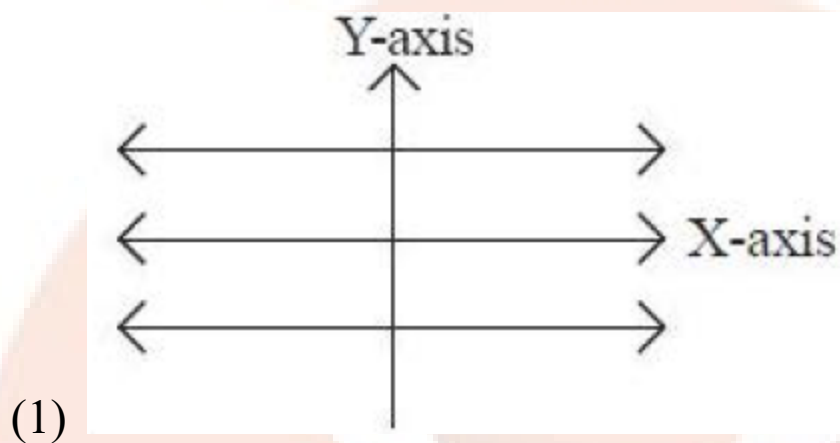
(1) 0 V

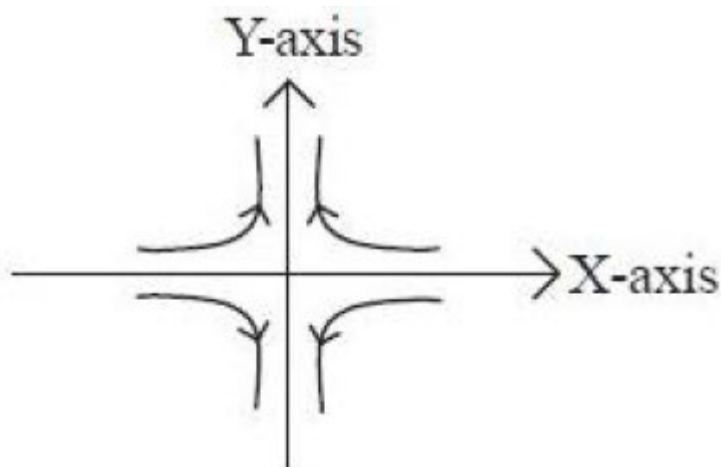
(2) $18 \times 10^5 \text{ V}$

(3) $1.08 \times 10^7 \text{ V}$

(4) $1.08 \times 10^5 \text{ V}$

107. The potential in an electric field varies as $V = (x^2 - y^2)$. The electric lines of the force in X-Y plane are





(4)

108. The length of a potentiometer wire is ' l '. A cell of emf ' E ' is balanced at a length $\left(\frac{l}{3}\right)$ from positive end of the wire. If the length of the wire is increased by $\left(\frac{l}{2}\right)$, the distance at which the same cell gives the balancing point is
(cell in the primary is ideal and no series resistance is present in the primary circuit)

(1) $\frac{2l}{3}$

(2) $\frac{l}{2}$

(3) $\frac{l}{6}$

(4) $\frac{4l}{3}$

109. n identical resistances are taken in which $\frac{n}{2}$ resistors are joined in series in the left gap and the remaining $\frac{n}{2}$ resistances are joined in parallel in the right gap of a metre bridge. Balancing length in cm is

(1) $\frac{100n^2}{n^2 + 4}$

(2) $\frac{100n^2}{n^2 + 1}$

(3) $\frac{400}{n^2 + 4}$

(4) $\frac{400}{n^2 + 1}$

110. The magnetic field normal to the plane of a coil of ' N ' turns and radius ' r ' which carries a current ' i ' is measured on the axis of the coil at a distance ' h ' from the centre of the coil. This is smaller than the field at the centre by the fraction,

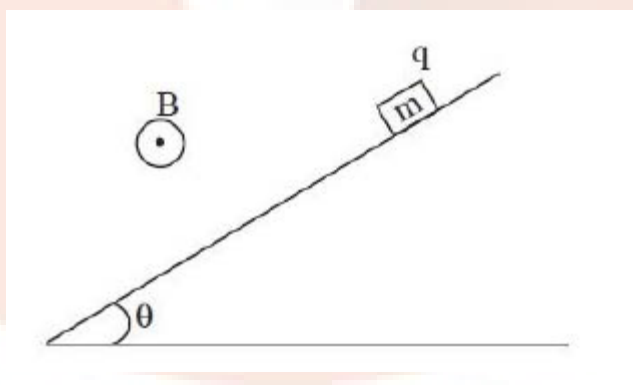
(1) $\frac{3h^2}{2r^2}$

(2) $\frac{2h^2}{3r^2}$

(3) $\frac{3r^2}{2h^2}$

(4) $\frac{2r^2}{3h^3}$

111. A small block of mass 20 g and charge 4 mC is released on a long smooth inclined plane of inclination angle of 45° . A uniform horizontal magnetic field of 1 T is acting parallel to the surface, as shown in the figure. The time from the start when the block loses contact with the surface of the plane is



(1) 2 s

(2) 3 s

(3) 5 s

(4) 6 s

112. Match the following List-I with List-II.

<u>List-I</u>	<u>List-II</u>
(A) $\oint_s EdA$	(I) 0
(B) $\oint_s BdA$	(II) $-\frac{d\phi_B}{dt}$
(C) $\oint_i Edl$	(III) $\frac{Q}{\epsilon_0}$
(D) $\oint_i Bdl$	(IV) $\mu_0(i_c + i_d)$

The correct answer is

	A	B	C	D
(1)	III	II	I	IV
	A	B	C	D
(2)	IV	I	III	II
	A	B	C	D
(3)	III	I	II	IV
	A	B	C	D
(4)	III	I	IV	II

113. Two concentric coplanar circular conducting loops have radii R and r ($R \gg r$). Their mutual inductance is proportional to

(1) $\frac{r}{R}$

(2) $\frac{R}{r}$

(3) $\frac{r^2}{R}$

(4) $\frac{R^2}{r}$

114. When an inductor L and a resistor R in series are connected across a 12 V, 50 Hz supply, a current of 0.5 A flows in the circuit. The current differs in the phase from applied voltage by $\frac{\pi}{3}$ radians. Then the value of R is

(1) 10Ω

(2) 3Ω

(3) 12Ω

(4) 15Ω

115. A point source of electromagnetic radiation has an average power output of 960 W. The peak value of the electric field at a distance 400 cm from the source is

- (1) 60 Vm^{-1}
- (2) 120 Vm^{-1}
- (3) 30 Vm^{-1}
- (4) 180 Vm^{-1}

116. All electrons ejected from a metallic surface by incident light of wavelength 400 nm travelled 1 m in the direction of uniform electric field of 2 NC^{-1} and came to rest. The work function of the surface is

- (1) 1.1 eV
- (2) 2.2 eV
- (3) 3.1 eV
- (4) 5.1 eV

117. Speed of electron in its 1st Bohr's orbit is given by $2.18 \times 10^6 \text{ ms}^{-1}$. If the time period of electron in n^{th} orbit is measured as 4.10 femto second, the value of n is

- (1) 1

(2) 2

(3) 3

(4) 4

118. A radioactive substance of half life 138.6 days is placed in a box. After n days only 20% of the substance is present, then the value of ' n ' is

(1) 693

(2) 138.6

(3) 277.2

(4) 322

119. Three amplifiers with voltage gains 10, 20 and 30 are connected in series. If the input signal has a peak value of 1 mV then the peak value of the output voltage is

(1) 6 V

(2) 60 V

(3) 0.6 V

(4) 0.06 V

120. A TV tower has a height 160 m. Its coverage range is nearly

(Earth's radius = 6400 km)

- (1) 45255 m
- (2) 55265 m
- (3) 452.55 km
- (4) 552.65 m



Chemistry

121. If the kinetic energy of an electron of mass 9.0×10^{-31} kg is 2.0×10^{-25} J, the wavelength of the electron in nm is approximately

- (1) 1004.3
- (2) 1204.3
- (3) 1104.3
- (4) 994.3

122. If a suitable photon is employed to locate an electron (mass = 9.1×10^{-31} kg) in an atom within a distance of 10.98 nm, the uncertainty involved in the measurement of its velocity in ms^{-1} is

- (1) $\frac{1.6565 \times 10^{16}}{\pi}$
- (2) $\frac{1.6565 \times 10^4}{\pi}$
- (3) $\frac{1.6565 \times 10^{-8}}{\pi}$

$$(4) \frac{1.6565 \times 10^8}{\pi}$$

123. Which one of the following is the correct order of ionic radii?

- (1) $\text{Pr}^{3+} > \text{Gd}^{3+} > \text{Tm}^{3+}$
- (2) $\text{Pr}^{3+} < \text{Gd}^{3+} < \text{Tm}^{3+}$
- (3) $\text{Pr}^{3+} > \text{Tm}^{3+} > \text{Gd}^{3+}$
- (4) $\text{Pr}^{3+} < \text{Tm}^{3+} < \text{Gd}^{3+}$

124. Observe the following molecules/ions



Identify the correct statement

- (1) $\text{H}_2, \text{N}_2, \text{O}_2, \text{F}_2$ show diamagnetic property
- (2) $\text{O}_2, \text{O}_2^+, \text{O}_2^-, \text{N}_2^+$ show paramagnetic property
- (3) $\text{N}_2, \text{F}_2, \text{O}_2^+, \text{O}_2^-$ show diamagnetic property
- (4) $\text{H}_2, \text{N}_2^+, \text{O}_2^+, \text{O}_2^-$ show paramagnetic property

125. Match the following.

List-I	List-II
--------	---------

A) BrF_5	I) AB_4E , see-saw
B) SF_4	II) AB_4E_2 , square planar
C) XeF_4	III) AB_5E , square pyramid
D) ClF_3	IV) AB_3E_2 , T-shape
	V) AB_5E_2 , square pyramid

- (1) $\frac{\underline{\text{A}}}{\underline{\text{V}}}$ $\frac{\underline{\text{B}}}{\underline{\text{I}}}$ $\frac{\underline{\text{C}}}{\underline{\text{II}}}$ $\frac{\underline{\text{D}}}{\underline{\text{IV}}}$
- (2) $\frac{\underline{\text{A}}}{\underline{\text{III}}}$ $\frac{\underline{\text{B}}}{\underline{\text{I}}}$ $\frac{\underline{\text{C}}}{\underline{\text{II}}}$ $\frac{\underline{\text{D}}}{\underline{\text{V}}}$
- (3) $\frac{\underline{\text{A}}}{\underline{\text{III}}}$ $\frac{\underline{\text{B}}}{\underline{\text{I}}}$ $\frac{\underline{\text{C}}}{\underline{\text{II}}}$ $\frac{\underline{\text{D}}}{\underline{\text{IV}}}$
- (4) $\frac{\underline{\text{A}}}{\underline{\text{V}}}$ $\frac{\underline{\text{B}}}{\underline{\text{I}}}$ $\frac{\underline{\text{C}}}{\underline{\text{III}}}$ $\frac{\underline{\text{D}}}{\underline{\text{II}}}$

126. If the most probable speed of CO_2 at 27°C is 400 m s^{-1} , the root mean square velocity of CO_2 at the same temperature in m s^{-1} is approximately

- (1) 600
(2) 490

(3) 267

(4) 245

127. 20 mL of Fe^{2+} solution of certain concentration has completely reacted with 20 mL of 0.01 M $\text{K}_2\text{Cr}_2\text{O}_7$ in an acid medium. If 20 mL of same Fe^{2+} solution has reacted completely with 20 mL of KMnO_4 solution in acid medium, the molarity of KMnO_4 solution is

(1) 0.01 M

(2) 0.12 M

(3) 0.10 M

(4) 0.012 M

128. If K_p for the conversion of oxygen to ozone at 400 K is 1.0×10^{-30} , its standard Gibbs energy change in kJ mol^{-1} is approximately

(1) 229.8

(2) 114.9

(3) -229.8

(4) -114.9

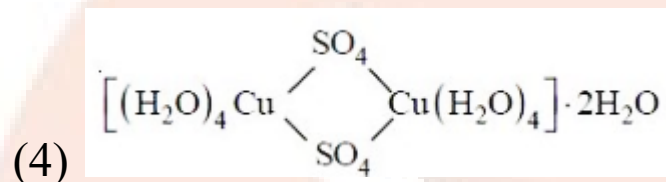
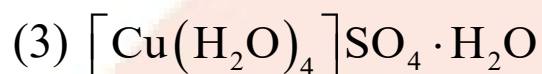
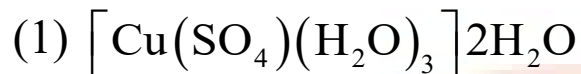
129. At 1000 K, the partial pressures of $\text{CO}_{2(g)}$ and $\text{CO}_{(g)}$ for the reaction $\text{CO}_{2(g)} + \text{C}_{(s)} \rightleftharpoons 2\text{CO}_{(g)}$ in a closed vessel at equilibrium are 0.15 and 0.60 bar respectively. The K_c for this reaction at the same temperature is approximately

- (1) 2.0×10^{-4}
- (2) 2.89×10^{-2}
- (3) 2.89×10^{-3}
- (4) 5.78×10^{-3}

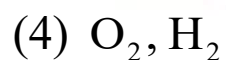
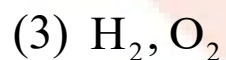
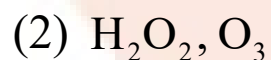
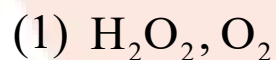
130. At T(K), if the ionization constant of ammonia in solution is 2.5×10^{-5} , the pH of 0.01 M ammonia solution and the ionization constant of its conjugate acid respectively at that temperature are ($\log 2 = 0.30$)

- (1) 10.7, 4.0×10^{-8}
- (2) 10.7, 4.0×10^{-10}
- (3) 3.3, 4.0×10^{-8}
- (4) 3.3, 4.0×10^{-8}

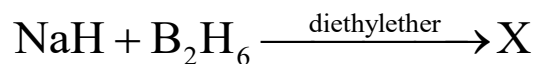
131. Which of the following correctly represents Copper sulphate pentahydrate?



132. A compound (MO_2) of group I element (M) hydrolyses to form M^+ , OH^- , X and Y. When X reacts with I_2 in basic medium, the products formed are I^- , water and Y. Then X and Y are respectively



133. What is X in the following reaction?



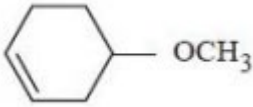
- (1) $\text{Na}[\text{BH}_4]$
- (2) NaBO_2
- (3) H_3BO_3
- (4) $(\text{C}_2\text{H}_5\text{OC}_2\text{H}_4)^+ (\text{BH}_4)^-$

134. Which one of the following is used as refrigerant for icecream and frozen foods?

- (1) Dry CO
- (2) Liquid CH_4
- (3) Dry ice
- (4) Liquid H_2

135. Acid rain is caused by the presence of X and Y in air. X, Y are respectively

- (1) SO_2 , NO_2
- (2) CFC, O_3
- (3) CO, CFC
- (4) SO_2 , O_3

136. IUPAC name of  is

- (1) 5-methoxycyclohexene
- (2) methoxycyclohex-3-ene
- (3) methoxycyclohex-4-ene
- (4) 4-methoxycyclohexene

137. Wurtz reaction of bromoethane gives n-butane. Sodium salt of X on heating with sodalime also results in n-butane. Compound X is

- (1) $\text{CH}_3\text{CH}_2\text{CH}_2\text{COOH}$
- (2) $\text{CH}_3(\text{CH}_2)_3\text{COOH}$
- (3) $\text{CH}_3(\text{CH}_2)_4\text{COOH}$
- (4) $\text{CH}_3\text{CH}_2\text{COOH}$

138. Identify ortho and para directing groups from the following

$-\text{CHO}$	$-\text{NHCOCH}_3$	$-\text{OCH}_3$	$-\text{SO}_3\text{H}$
I	II	III	IV

- (1) III, IV

(2) II, III

(3) II, IV

(4) I, IV

139. If 0.5 mol of a metal forms hexagonal close packed structure, the total number of voids, and tetrahedral voids respectively in mol are

(1) 1.5, 1.0

(2) 1.0, 0.5

(3) 1.0, 1.5

(4) 0.5, 1.0

140. At 300 K an ideal solution is formed by mixing 460 g of toluene with 390 g benzene. If the vapour pressure of pure toluene and benzene at 300 K are 32 and 40 mm respectively, the mole fraction of toluene in vapour phase is

(1) 0.196

(2) 0.588

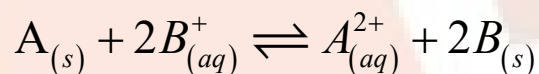
(3) 0.294

(4) 0.444

141. If the aqueous solution contains 9% and 1% (W/W) of two non-volatile, non electrolytes X (molecular weight 180) and Y (molecular weight 50) respectively, the boiling point of solution in $^{\circ}\text{C}$ approximately is ($K_b = 0.52 \text{ K kg mol}^{-1}$)

- (1) 101.4
- (2) 100.4
- (3) 102.4
- (4) 100.8

142. If the E_{cell}^0 of an equilibrium reaction



at 298 K is 0.59 V, the equilibrium constant K_c is

- (1) 1.0×10^{10}
- (2) 1.0×10^2
- (3) 1.0×10^{-20}
- (4) 1.0×10^{20}

143. Which one of the following statements is correct?

- (1) In collision theory $e^{-E_a/RT}$ corresponds to the fraction of molecules that have energy equal; or greater than E_a
- (2) The number of collisions of reacting molecules per second per unit volume of the reaction mixture is activated complex
- (3) Molecularity is the number of molecules involved in a complex reaction
- (4) A catalyst catalyses the non-spontaneous reaction

144. Which one of the following enzymes convert proteins into aminoacids?

- (1) Maltase
- (2) Pepsin
- (3) Trypsin
- (4) Zymase

145. X_2 is used in the refining of Ti metal by van-Arkele method.

Y_2 does not liberate O_2 from water and does not form HY and HOY with water. X_2 and Y_2 are respectively

- (1) I_2, Cl_2
- (2) Cl_2, I_2

(3) I_2, I_2

(4) Cl_2, Cl_2

146. Which one of the following statements is **not** correct regarding phosphine?

(1) It is a weak base

(2) It reacts with $CuSO_4$ solution to form $CuHPO_4$

(3) It is formed by the reaction of Ca_3P_2 with HCl

(4) it is used in smoke screens.

147. Aqueous sulphite reacts with dilute sulphuric acid to form $X(g)$. The liberated $X(g)$ is passed into acidified $KMnO_4$ solution. What is the oxidation state of Mn in the product formed?

(1) +6

(2) +4

(3) +2

(4) +3

148. Phosphorus reacts with SO_2Cl_2 to form PCl_5 and X. At 723 K, HCl gas reacts with O_2 in the presence of $CuCl_2$ to form water

and Y. Y reacts with X in water to form two acids A and B.

What are A and B respectively?

- (1) H_2SO_4 , HCl
- (2) H_2SO_4 , HNO_3
- (3) HCl , HNO_3
- (4) H_2SO_4 , H_3PO_4

149. Identify the correct statements from the following

i. The ionic radius of Pr^{3+} , Dy^{3+} and Sm^{3+} follow the order,

$$\text{Sm}^{3+} > \text{Pr}^{3+} > \text{Dy}^{3+}$$

ii. Eu^{2+} acts as strong reducing reagent

iii. Pu exhibits +7 oxidation state

- (1) i, ii only
- (2) i, iii only
- (3) i, ii, iii
- (4) ii, iii only

150. If the crystal field splitting energy of a tetrahedral complex

(Δ_t) of the type $[\text{ML}_4]^{n+}$ is x eV, what is the crystal field

splitting energy with respect to an octahedral complex, $[ML_6]^{n+}$?

(1) $\frac{9x}{4} \text{ eV}$

(2) $\frac{9x}{8} \text{ eV}$

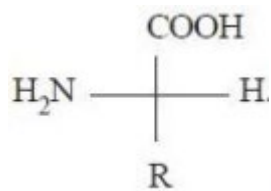
(3) $\frac{4x}{9} \text{ eV}$

(4) $\frac{4x}{5} \text{ eV}$

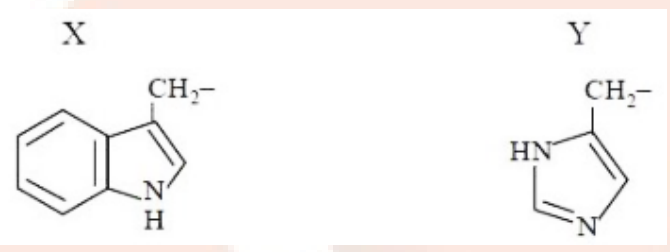
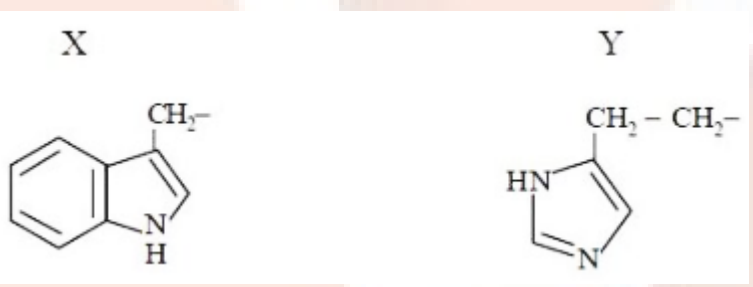
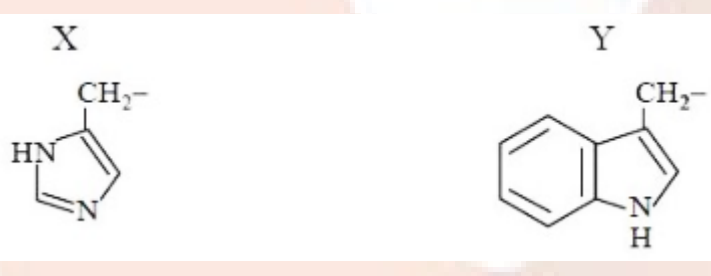
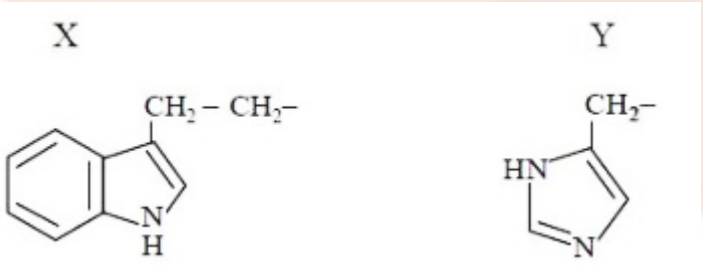
151. The monomers of nylon 6, 6 (X) and terylene (Y) are

- | | | | |
|-----|---|--|---|
| | X | | Y |
| (1) | | | |
| (2) | | | |
| (3) | | | |
| (4) | | | |

152. The general structural formula of α -Amino acid is



The group R in tryptophan (X) and histidine (Y) are respectively

- (1) 
- (2) 
- (3) 
- (4) 

153. Which of the following statements are correct?

- i. Binding of inhibitor at allosteric site changes the shape of the active site
 - ii Shape of the receptor does not changes after attachment of chemical messenger to it
 - iii A chemical messenger gives message to the cell by entering it
 - iv Erthyromycin is an example of bacteriostatic antibiotic
- (1) i, ii
 - (2) ii, iii
 - (3) i, iv
 - (4) iii, iv

154. Assertion (A) : S_N1 hydrolysis of optically active 2-bromooctane results in the formation of $(\pm)$ -octan-2-ol

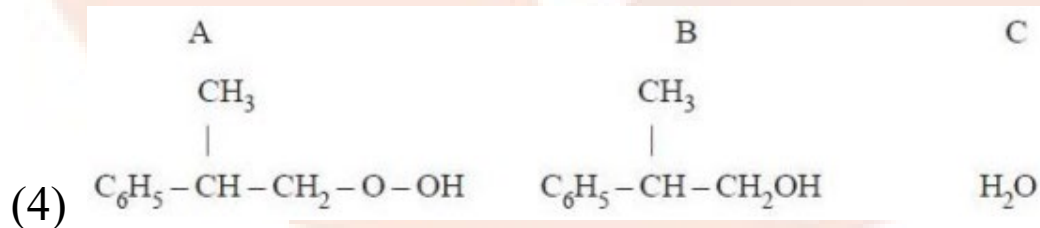
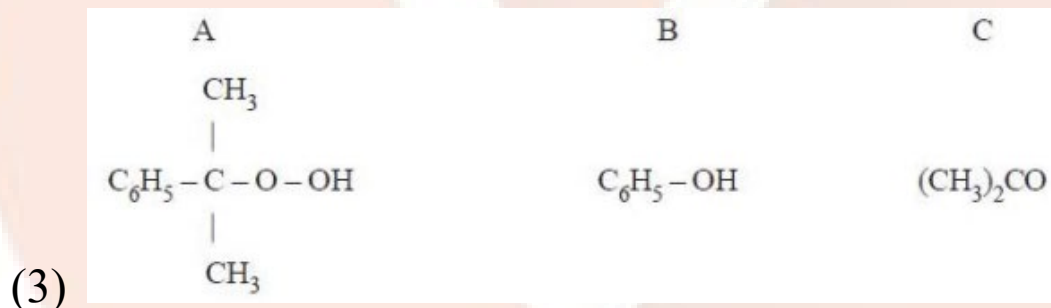
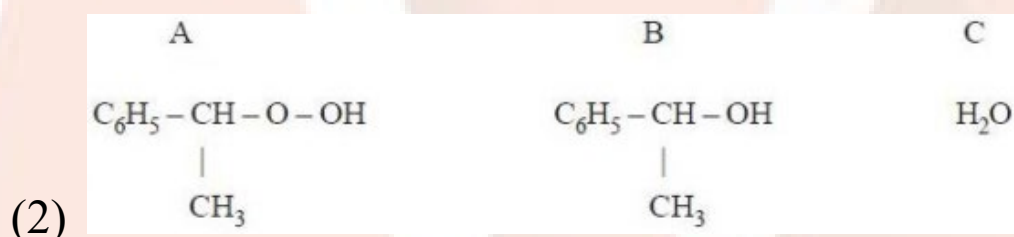
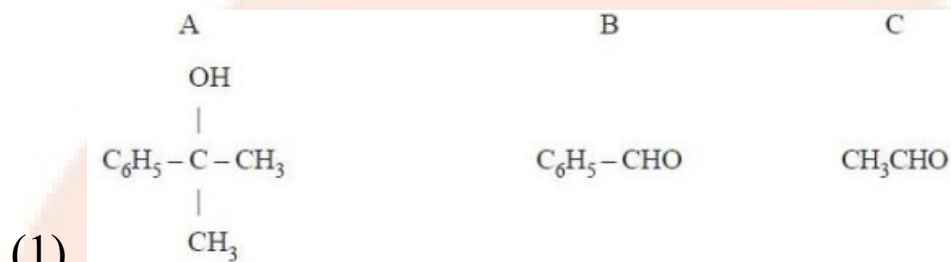
Reason (R) : The reaction proceeds through a planar carbocation which can be attacked by the nucleophile from either side

The correct answer is

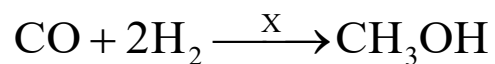
- (1) (A) and (R) are correct, (R) is the correct explanation of (A)
- (2) (A) and (R) are correct but (R) is not the correct explanation of (A)
- (3) (A) is correct but (R) is not correct

(4) (A) is not correct but (R) is correct

155. What are A, B and C in the following reactions?



156. What is X in the following reaction?



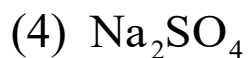
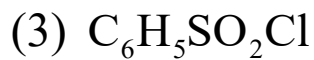
- (1) 623 K/300 atm
- (2) $\text{KMnO}_4/\text{H}^\oplus$
- (3) Zn/Δ
- (4) $\text{ZnO-Cr}_2\text{O}_3$, 200-300 atm, 573-673 K

157. Alkenes (X) and carbonyl compounds (Y) participate in which of the following addition reactions?

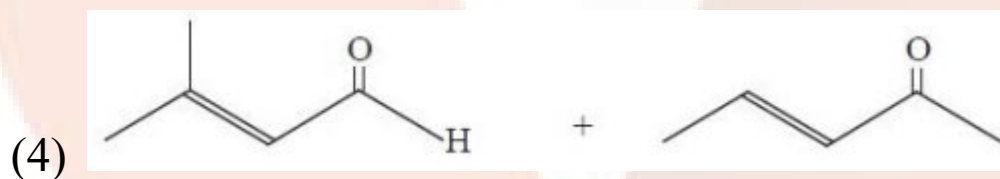
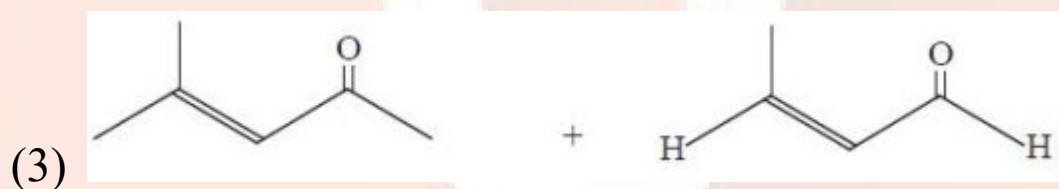
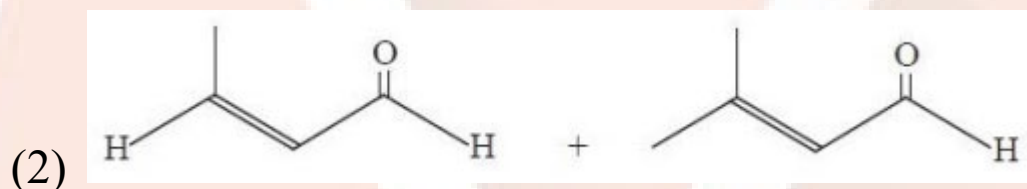
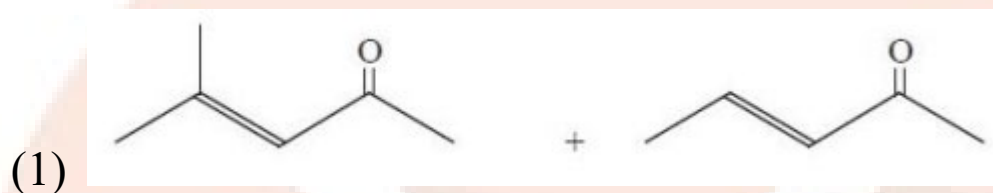
- | X | Y |
|-------------------|---------------|
| (1) Electrophilic | Electrophilic |
| X | Y |
| (2) Nucleophilic | Electrophilic |
| X | Y |
| (3) Electrophilic | Nucleophilic |
| X | Y |
| (4) Nucleophilic | Nucleophilic |

158. Which one of the following is used for purification of aldehydes?

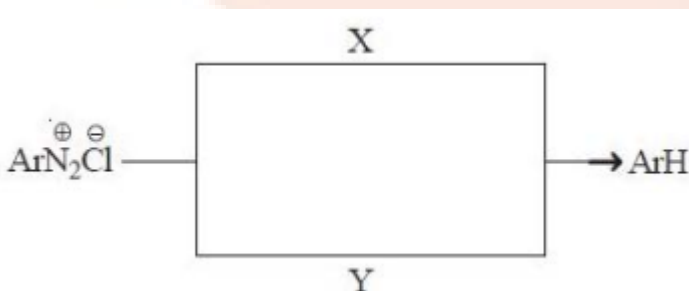
- (1) NaOCl
- (2) NaHSO_3

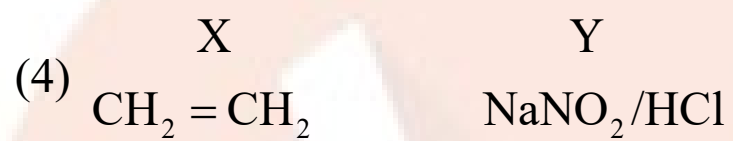


159. Mixed aldol products obtained from aldol condensation of ethanal and propanone are



160. What are X and Y in the following reactions





AP EAMCET-2019-Engineering-22 Apr-Shift-1
Answer Key

1	4	33	3	65	4	97	3	129	2
2	2	34	4	66	2	98	1	130	2
3	1	35	3	67	2	99	3	131	3
4	4	36	4	68	2	100	3	132	1
5	2	37	2	69	1	101	2	133	1
6	2	38	4	70	1	102	4	134	3
7	3	39	2	71	1	103	1	135	1
8	3	40	4	72	2	104	1	136	4
9	4	41	1	73	2	105	4	137	2
10	3	42	4	74	3	106	3	138	2
11	3	43	3	75	2	107	3	139	1
12	4	44	1	76	3	108	2	140	4
13	4	45	3	77	1	109	1	141	2
14	4	46	2	78	3	110	1	142	4
15	2	47	3	79	4	111	3	143	1
16	3	48	3	80	2	112	3	144	3
17	1	49	2	81	1	113	3	145	3

18	2	50	2	82	2	114	3	146	2
19	3	51	3	83	3	115	1	147	3
20	4	52	1	84	4	116	1	148	1
21	2	53	1	85	2	117	3	149	4
22	2	54	3	86	1	118	4	150	1
23	3	55	1	87	1	119	1	151	4
24	*	56	3	88	2	120	1	152	1
25	4	57	3	89	3	121	3	153	3
26	2	58	1	90	4	122	2	154	1
27	4	59	4	91	2	123	1	155	3
28	1	60	4	92	3	124	2	156	4
29	3	61	3	93	2	125	3	157	3
30	4	62	3	94	2	126	2	158	2
31	4	63	2	95	3	127	4	159	4
32	4	64	1	96	3	128	1	160	1

AP EAMCET-2019-Engineering-22 Apr-Shift-1
Solutions – Mathematics

1. Consider the function,

$$f \circ f\left(-\frac{7}{2}\right) = f\left[-\frac{7}{2} - 5\right]$$
$$f(9) = |19 - 15|$$
$$= 4$$

Now,

$$(A) \ f \circ f\left(-\frac{11}{2}\right) = f\left(\left[-\frac{11}{2} - 5\right]\right) = 6$$

$$(B) \ f \circ f\left(-\frac{9}{2}\right) = f\left(\left[-\frac{9}{2} - 5\right]\right) = 5$$

$$(C) \ f \circ f(3) = f\left(\left[3 - 5\right]\right) = 3$$

$$(D) \ f \circ f\left(\frac{9}{2}\right) = f\left(\left[\frac{9}{2} - 5\right]\right) = 4$$

Thus, the correct option is (4).

2. Consider the function,

$$f(x) = \sqrt{|x| - x} + \frac{1}{\sqrt{|x| - x}}$$

The above function is defined when,

$$|x| - x > 0$$

$$|x| > x$$

This implies, $x \in (-\infty, 0)$

Now,

$$\begin{aligned} f(x) &= \sqrt{|x| - x} + \frac{1}{\sqrt{|x| - x}} \\ &= \sqrt{-2x} + \frac{1}{\sqrt{-2x}} \end{aligned}$$

This implies,

$$\begin{aligned} f_{\min} &= 2\sqrt{\sqrt{-2x}, \frac{1}{\sqrt{-2x}}} \\ &= 2 \end{aligned}$$

Therefore,

$$f(x) \in [2, \infty)$$

So,

$$A = (-\infty, 0) \text{ and } B = [2, \infty)$$

3. Solve the given series.

$$\begin{aligned}
 \frac{1}{2 \cdot 5} + \frac{1}{5 \cdot 8} + \frac{1}{8 \cdot 11} + \dots 16 \text{ terms} &= \frac{1}{3} \left[\frac{3}{2 \cdot 5} + \frac{3}{5 \cdot 8} + \frac{3}{8 \cdot 11} + \dots \right] \\
 &= \frac{1}{3} \left[\frac{1}{2} - \frac{1}{5} + \frac{1}{5} - \frac{1}{8} + \frac{1}{8} - \frac{1}{11} + \dots + \frac{1}{47} - \frac{1}{50} \right] \\
 &= \frac{1}{3} \left[\frac{1}{2} - \frac{1}{50} \right] \\
 &= \frac{1}{3} \left[\frac{24}{50} \right] \\
 &= \frac{4}{25}
 \end{aligned}$$

4. Consider the matrix,

$$\begin{vmatrix} 1 + \sin^2 x & \cos^2 x & 4 \sin 2x \\ \sin^2 x & 1 + \cos^2 x & 4 \sin 2x \\ \sin^2 x & \cos^2 x & 1 + 4 \sin 2x \end{vmatrix}$$

Apply the row operations $R_1 \rightarrow R_1 - R_3$, and $R_2 \rightarrow R_2 - R_3$

$$\begin{vmatrix} 1 & 0 & -1 \\ 0 & 1 & -1 \\ \sin^2 x & \cos^2 x & 1 + 4 \sin 2x \end{vmatrix}$$

Apply the operation $C_2 \rightarrow C_2 + C_1$.

$$\begin{vmatrix} 1 & 1 & -1 \\ 0 & 1 & -1 \\ \sin^2 x & 1 & 1 + 4\sin 2x \end{vmatrix}$$

Apply the operations, $R_2 \rightarrow R_2 - R_1$ and $R_3 \rightarrow R_3 - R_1$

$$\begin{vmatrix} 1 & 1 & -1 \\ -1 & 0 & 0 \\ -1 + \sin^2 x & 0 & 2 + 4\sin 2x \end{vmatrix} = 2 + 4\sin 2x$$

The maximum value of $\sin 2x$ is 1 so,

$$2 + 4(1) = 6$$

5. The characteristic equation of the square matrix is written as,

$$|A - \lambda I| = 0$$

Then,

$$\begin{bmatrix} 1 & 0 & -2 \\ -2 & -1 & 2 \\ 3 & 4 & 1 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = 0$$

$$\begin{bmatrix} 1 & 0 & -2 \\ -2 & -1 & 2 \\ 3 & 4 & 1 \end{bmatrix} - \begin{bmatrix} \lambda & 0 & 0 \\ 0 & \lambda & 0 \\ 0 & 0 & \lambda \end{bmatrix} = 0$$

$$\begin{bmatrix} 1-\lambda & 0 & -2 \\ -2 & -1-\lambda & 2 \\ 3 & 4 & 1-\lambda \end{bmatrix} = 0$$

Simplifying the above matrix, we get,

$$(1-\lambda)[-(1+\lambda)-8]-2(-8+3(\lambda+1))=0$$

$$(1-\lambda)[-(1-\lambda^2)-8]-2(-8+3\lambda+3)=0$$

$$(1-\lambda)(\lambda^2-9)-2(-5+3\lambda)=0$$

$$\lambda^2-9-\lambda^3+9\lambda-6\lambda+10=0$$

Further simplify the above,

$$-\lambda^3+\lambda^2+3\lambda+1=0$$

$$\lambda^3-\lambda^2-3\lambda-1=0$$

From the Caley Hamilton's theorem,

$$A^3-A^2-3A-I=0$$

$$A^{-1}(A^3-A^2-3A-I)=0$$

$$A^{-1}=A^2-A-3I$$

6. Consider the system of equation,

$$x + y + z = a \quad \dots\dots (1)$$

$$x - y + bz = 2 \quad \dots\dots (2)$$

$$2x + 3y - z = 1 \quad \dots\dots (3)$$

As these equations have infinitely many solutions, therefore,

$$\begin{bmatrix} 1 & 1 & 1 \\ 1 & -1 & b \\ 2 & 3 & -1 \end{bmatrix} = 0$$

$$1(1 - 3b) - 1(-1 - 2b) + 1(3 + 2) = 0$$

$$-b = -7$$

$$b = 7$$

From equation (1) and (2),

$$2x + 8z = a + 2$$

$$x + 4z = \frac{a + 2}{2} \quad \dots\dots (4)$$

From equation (1) and (3), we get

$$x + 4z = 3a - 1 \quad \dots\dots (5)$$

From equation (4) and (5), we get

$$\frac{a+2}{2} = 3a-1$$

$$a+2 = 6a-2$$

$$5a = 4$$

So, the required value is,

$$\begin{aligned} b-5a &= 7-4 \\ &= 3 \end{aligned}$$

7. Simplifying the given expression, we get

$$\arg\left(\frac{z-1}{z-3i}\right) = \frac{\pi}{2}$$

This implies,

$$\arg\left(\frac{z-1}{z-3i}\right) = \frac{\pi}{2}$$

$$\arg(z-1) - \arg(z-3i) = \frac{\pi}{2}$$

$$\arg[(x-1) + iy] - \arg[x + (y-3)i] = \frac{\pi}{2}$$

$$\tan^{-1} \frac{y}{x-1} - \tan^{-1} \frac{y-3}{x} = \frac{\pi}{2}$$

Further simplify the above,

$$\tan^{-1} \left[\frac{\frac{y}{x-1} - \frac{y-3}{x}}{1 + \frac{y}{x-1} \frac{y-3}{x}} \right] = \frac{\pi}{2}$$

$$\frac{xy - (x-1)(y-3)}{x(x-1) + y(y-3)} = \tan \frac{\pi}{2}$$

$$\frac{xy - (x-1)(y-3)}{x(x-1) + y(y-3)} = \frac{1}{0}$$

$$x(x-1) + y(y-3) = 0$$

$$x^2 + y^2 - x - 3y = 0$$

The above equation further can be written as,

$$\left(x - \frac{1}{2}\right)^2 + \left(y - \frac{3}{2}\right)^2 = \frac{1}{4} + \frac{9}{4}$$

$$\left(x - \frac{1}{2}\right)^2 + \left(y - \frac{3}{2}\right)^2 = \left(\frac{\sqrt{10}}{2}\right)^2$$

The above represents the circle with center $\left(\frac{1}{2}, \frac{3}{2}\right)$ and radius

$$\left(\frac{\sqrt{10}}{2}\right)$$

Thus,

$$z \in C : (3-i)z + (3+i)\bar{z} - 6 > 0, \left| z - \frac{1+3i}{2} \right| = \left(x - \frac{1}{2} \right)^2 + \left(y - \frac{3}{2} \right)^2 = \frac{\sqrt{10}}{2}$$

8. According to the question,

$$\begin{aligned} PQ &= \left| ze^{\frac{i\pi}{3}} - z \right| \\ &= |z| \left| e^{\frac{i\pi}{3}} - 1 \right| \\ &= |z| \left| \cos \frac{\pi}{3} + 2i \sin \frac{\pi}{3} - 1 \right| \\ &= |z| \left| -2 \sin^2 \frac{\pi}{6} + 2i \sin \frac{\pi}{6} \cos \frac{\pi}{6} \right| \\ &= |z| \left| 2 \sin \frac{\pi}{6} \right| \left| \sin \frac{\pi}{6} - i \cos \frac{\pi}{6} \right| \\ &= |z| \cdot 2 \times \frac{1}{2} |1| \\ &= |z| \end{aligned}$$

$$\begin{aligned} \text{Again, } QR &= z \left(1 + e^{\frac{i\pi}{3}} \right) - 2e^{\frac{i\pi}{3}} \\ &= |z| \end{aligned}$$

$$\text{Similarly, } PR = |z|$$

So, the triangle PQR is an equilateral triangle.

So the area of ΔPQR is,

$$\text{Area of } \Delta PQR = \frac{\sqrt{3}}{4} |z|^2$$

9. Let the points be $z(x, y)$, $z_1(x_1, y_1)$ and $z_2(x_2, y_2)$ then,

$$\begin{aligned} z - z_1 &= x + iy - x_1 - iy_1 \\ &= x - x_1 + i(y - y_1) \end{aligned}$$

And,

$$\begin{aligned} z - z_2 &= x + iy - x_2 - iy_2 \\ &= x - x_2 + i(y - y_2) \end{aligned}$$

Now,

$$\begin{aligned} \frac{z - z_1}{z - z_2} &= \frac{[(x - x_1) + i(y - y_1)]}{[(x - x_2) + i(y - y_2)]} \times \frac{[(x - x_2) - i(y - y_2)]}{[(x - x_2) - i(y - y_2)]} \\ &= \frac{i[(x - x_2)(y - y_1) - (x - x_1)(y - y_2)]}{(x - x_2)^2 + (y - y_2)^2} \end{aligned}$$

Since, $\arg\left(\frac{z - z_1}{z - z_2}\right) = 0$ or π then,

$$\left[\frac{[(x-x_2)(y-y_1)] - [(x-x_1)(y-y_2)]}{[(x-x_1)(x-x_2)] + [(y-y_1)(y-y_2)]} \right] = 0$$

$$(x-x_2)(y-y_1) = (x-x_1)(y-y_2)$$

$$xy - xy_1 - x_2y + x_2y_1 = xy - xy_2 - x_1y + x_1y_2$$

$$x(y_2 - y_1) + y(x_1 - x_2) + (x_2y_1 - x_1y_2) = 0$$

It represents the equation of a straight line passing through A and B.

10. Since x is the cube root of unity other than 1 then,

$$\begin{aligned} & \left(x + \frac{1}{x}\right)^2 + \left(x^2 + \frac{1}{x^2}\right)^2 + \dots + \left(x^{12} + \frac{1}{x^{12}}\right)^2 \\ &= \left(\omega + \frac{1}{\omega}\right)^2 + \left(\omega^2 + \frac{1}{\omega^2}\right)^2 + \dots + \left(\omega^{12} + \frac{1}{\omega^{12}}\right)^2 \\ &= \left(\omega + \frac{1}{\omega}\right)^2 + \left(\omega^2 + \frac{1}{\omega^2}\right)^2 + \dots + \left(\omega^{11} + \frac{1}{\omega^{11}}\right)^2 \\ &= \left(\begin{aligned} & (\omega + \omega^2)^2 + (\omega + \omega^2)^2 + (1+1)^2 + (\omega + \omega^2)^2 \\ & + (\omega + \omega^2)^2 + (1+1)^2 + (\omega + \omega^2)^2 \end{aligned} \right) \\ &= 8(\omega + \omega^2)^2 + 4(1+1)^2 \\ &= 8(-1)^2 + 4(2)^2 \\ &= 8 + 16 \\ &= 24 \end{aligned}$$

11. Consider the given equations,

$$3x^2 - 7x + 2 = 0 \quad (\text{I})$$

$$15x^2 - 11x + a = 0 \quad (\text{II})$$

Here,

$$a_1 = 3, b_1 = -7, c_1 = 2$$

$$a_2 = 15, b_2 = -11, c_2 = a$$

If α be the common root, then it is given by,

$$(2(15) - 3a)^2 = (-7a + 22)(-33 + 105)$$

$$(30 - 3a^2) = (22 - 7a)(72)$$

$$[9(10 - a)^2] = (22 - 7a)8$$

$$100 + a^2 - 20a = 176 - 56a$$

$$a^2 + 36a - 76 = 0$$

$$a(a + 38) - 2(a + 38) = 0$$

$$a = 2 \quad [\because a > 0]$$

So, the sum of the roots of the equation $15x^2 - ax + 7 = 0$ can be calculated as,

$$\begin{aligned}\text{Sum of roots} &= -\frac{\text{coefficient of } x}{\text{coefficient of } x^2} \\ &= -\frac{(-a)}{15} \\ &= \frac{a}{15} \\ &= \frac{2}{15}\end{aligned}$$

12. Consider the expression,

$$\log_{|\alpha|}\left(\frac{x^2}{\beta^2}\right) = 1$$

This implies,

$$\begin{aligned}\frac{x^2}{\beta^2} &= |\alpha| \\ x^2 &= \beta^2 |\alpha|\end{aligned}$$

Since α, β are the roots of the equation,

$$x^2 - |a|x - |b| = 0$$

So,

$$\begin{aligned}\alpha + \beta &= -|a| < 0 \\ \beta &< 0\end{aligned}$$

Now,

$$x = \pm |\beta| |\alpha|^{\frac{1}{2}}$$

So, the positive root is,

$$|\beta| |\alpha|^{\frac{1}{2}} > \beta$$

13. Consider the expression,

$$1 \leq \frac{3x^2 - 7x + 8}{x^2 + 1} \leq 2$$

Now,

$$1 \leq \frac{x^2 - 7x + 8}{x^2 + 1}$$

$$0 \leq 2x^2 - 7x + 7$$

This implies,

$$f(x) = 2x^2 - 7x + 7$$

$$\Delta = 7^2 - 4(2)(7)$$

$$\Delta < 0$$

Thus,

$$f(x) = 2x^2 - 7x + 7 > 0$$

$$x \in R \text{ and } \frac{3x^2 - 7x + 8}{x^2 + 1} \leq 2$$

So,

$$\begin{aligned}
 x^2 - 7x + 6 &\leq 0 \\
 (x-1)(x-6) &\leq 0 \\
 x &\in [1, 6]
 \end{aligned}$$

Hence, the minimum and maximum values are 1 and 6.

14. The given function is,

$$\phi(x) = \frac{x}{(x^2 + 1)(x + 1)}$$

So,

$$\begin{aligned}
 \phi(a)\phi(b)\phi(c) &= \frac{abc}{(1+a)(1+b)(1+c)(1+a^2)(1+b^2)(1+c^2)} \\
 &= \frac{abc}{\left(\begin{matrix} 1+a+b+c+ab+bc \\ +ca+abc \end{matrix} \right) \left(\begin{matrix} 1+a^2+b^2+c^2+a^2b^2 \\ +b^2c^2+c^2a^2+(abc)^2 \end{matrix} \right)}
 \end{aligned}$$

Since, $a.b.c$ are the roots of given cubic equation so,

$$ab + bc + ca = -3 \quad \text{(I)}$$

$$a + b + c = 0 \quad \text{(II)}$$

$$abc = -\lambda \quad \text{(III)}$$

From equation (II),

$$(a + b + c)^2 = 0$$

$$a^2 + b^2 + c^2 + 2(ab + bc + ca) = 0$$

$$a^2 + b^2 + c^2 = 6$$

Similarly,

$$a^2b^2 + b^2c^2 + c^2a^2 = 9$$

This gives us,

$$\begin{aligned}\phi(a)\phi(b)\phi(c) &= \frac{-\lambda}{(1+0-3-\lambda)(1+6+9+\lambda^2)} \\ &= \frac{-\lambda}{(-2-\lambda)(\lambda^2+16)} \\ &= \frac{\lambda}{(2+\lambda)(\lambda^2+16)}\end{aligned}$$

15. The number of ways in which one student can seat in Ist column is n .

The number of ways in which m students can seat is n^m .

The students can be arranged in $m!$ ways.

The total number of ways is $m! \times n^m$.

16. Consider the expression,

$$2^{\alpha_1} 3^{\alpha_2} 4^{\alpha_3} 5^{\alpha_4} 6^{\alpha_5} = 2^{\alpha_1 + 2\alpha_3 + \alpha_5} 3^{\alpha_2 + \alpha_5} 5^{\alpha_4}$$

Number of non trivial even divisors is given by,

$$(\alpha_1 + 2\alpha_3 + \alpha_5)(\alpha_2 + \alpha_5 + 1)(\alpha_4 + 1) - 1$$

Number of non trivial odd divisors is given by,

$$\begin{aligned} (\alpha_2 + \alpha_5 + 1)(\alpha_4 + 1) - 1 &= \alpha_2 + \alpha_4 + \alpha_5 + \alpha_2\alpha_4 + \alpha_4\alpha_5 + 1 - 1 \\ &= \alpha_2 + \alpha_4 + \alpha_5 + \alpha_2\alpha_4 + \alpha_4\alpha_5 \end{aligned}$$

So, Statement I is false and statement II is true.

17. Consider the expression,

$$(3 + 2x + x^2)^5 = \sum \frac{n!}{(p!)(q!)(r!)} (3)^p (2x)^q (x^2)^r$$

Here, $p + q + r = n = 5$

For x^5 , $q + 2r = 5$

This gives,

$$r = 0, q = 5, p = 0$$

$$r = 1, q = 3, p = 1$$

$$r = 2, q = 1, p = 2$$

So, we get,

$$\begin{aligned}
 (3+2x+x^2)^5 &= \left(\frac{5!}{0!5!0!}(3)^0(2)^5(1)^0 + \frac{5!}{1!3!1!}(3)^1(2)^3(1)^1 \right. \\
 &\quad \left. + \frac{5!}{2!1!2!}(3)^2(2)^1(1)^2 \right) \\
 &= 32 + 20(3)(8) + 30(9)(2) \\
 &= 1052
 \end{aligned}$$

18. The given function can be written as,

$$\begin{aligned}
 \sqrt{x^2+4} - \sqrt{x^2+9} &= (x^2+4)^{\frac{1}{2}} - (x^2+9)^{\frac{1}{2}} \\
 &= 2\left(1+\frac{x^2}{4}\right)^{\frac{1}{2}} - 3\left(1+\frac{x^2}{9}\right)^{\frac{1}{2}}
 \end{aligned}$$

The coefficient of x^4 is,

$$\begin{aligned}
 &\left[2 \frac{\frac{1}{2}\left(\frac{1}{2}-1\right)}{2!} \left(\frac{x^2}{4}\right)^2 - 3 \frac{\frac{1}{2}\left(\frac{1}{2}-1\right)}{2!} \left(\frac{x^2}{9}\right)^2 \right] \\
 &= x^4 \left[-\frac{1}{4(16)} + 3\left(\frac{1}{8}\right)\left(\frac{1}{81}\right) \right] \\
 &= x^4 \left[-\frac{1}{64} + \frac{1}{216} \right]
 \end{aligned}$$

So, coefficient of x^4 is,

$$\begin{aligned}\frac{-216+64}{64(216)} &= \frac{-152}{64(216)} \\ &= \frac{-19}{1728}\end{aligned}$$

19. Consider the expression,

$$\begin{aligned}\frac{8}{(x+3)^2(x-2)} &= \frac{Ax+B}{(x+3)^2} + \frac{C}{x-2} \\ \frac{8}{(x+3)^2(x-2)} &= \frac{(Ax+B)(x-2) + C(x+3)^2}{(x+3)^2(x-2)} \\ 8 &= (Ax+B)(x-2) + C(x+3)^2\end{aligned}$$

Putting $x = 2$, we get

$$C = \frac{8}{25}$$

Putting $x = 0$, we get

$$\begin{aligned}8 &= B(-2) + C(9) \\ &= -2B + \frac{8}{25}(9)\end{aligned}$$

$$B = \frac{-64}{25}$$

Putting $x = 1$, we get

$$8 = (A + B)(-1) + C(4)^2$$

$$8 = -A - B + 16C$$

$$8 = -A + \frac{64}{25} + \frac{128}{25}$$

$$A = \frac{-8}{25}$$

Now,

$$\begin{aligned} 25(B + 8C - A) &= 25\left(\frac{-64}{25} + \frac{64}{25} + \frac{8}{25}\right) \\ &= 8 \end{aligned}$$

20. Consider the expression,

$$\cos(x + \alpha) + \cos(x + \beta) + \cos(x + \gamma) = 0$$

Take $x = -\alpha - \beta - \gamma$ so,

$$\cos(\beta + \gamma) + \cos(\alpha + \gamma) + \cos(\alpha + \beta) = 0$$

$$2\cos\left(\frac{\alpha + 2\beta + \gamma}{2}\right)\cos\left(\frac{\gamma - \alpha}{2}\right) + \cos(\alpha + \gamma) = 0 \quad \dots\dots (1)$$

Put $x = \frac{\pi}{2} - \beta$ then,

$$\sin(\beta - \alpha) + \sin(\beta - \gamma) = 0$$

$$\sin(\beta - \alpha) = \sin(\gamma - \beta)$$

$$2\beta = \alpha + \gamma \quad \dots\dots (2)$$

Solve equation (1) and (2).

$$\cos(\alpha + \gamma) \left[\frac{1}{2} \cos\left(\frac{\gamma - \alpha}{2}\right) + 1 \right] = 0$$

$$\cos\left(\frac{\gamma - \alpha}{2}\right) = -\frac{1}{2}$$

$$\frac{\gamma - \alpha}{2} = \frac{2\pi}{3}$$

$$\gamma - \alpha = \frac{\pi}{3}$$

$$\tan(\gamma - \alpha) = \sqrt{3}$$

21. Consider the expression,

$$A + B + C = \pi$$

Then,

$$\sin\left(\frac{\pi}{4} - A\right) \sin\left(\frac{\pi}{4} - B\right) \sin\left(\frac{\pi}{4} - C\right) = -\frac{1}{2\sqrt{2}}$$

$$\left(\frac{1}{\sqrt{3}}\right)^3 (\cos A - \sin A)(\cos B - \sin B)(\cos C - \sin C) = -\frac{1}{2\sqrt{2}}$$

$$\cos A \cos B \cos C (1 - \tan A)(1 - \tan B)(1 - \tan C) = -1$$

$$\begin{aligned} & \cos A \cos B \cos C \left(1 - \tan A - \tan B - \tan C + \tan A \tan B \right. \\ & \quad \left. + \tan B \tan C + \tan A \tan C - \tan A \tan B \tan C \right) \\ & = \cos A + B + C \quad [\because A + B + C = \pi] \end{aligned}$$

Further simplify above,

$$-2(\tan A + \tan B + \tan C) = -2 \left(\begin{array}{l} \tan A \tan B + \tan B \tan C \\ + \tan A \tan C \end{array} \right)$$

$$\tan A \tan B + \tan B \tan C + \tan A \tan C = \tan A + \tan B + \tan C$$

22. Simplify the given expression,

$$\tan \frac{\theta}{2} = \operatorname{cosec} \theta - \sin \theta$$

$$\frac{1}{\sin \theta} - \sin \theta$$

$$= \frac{1 - \sin^2 \theta}{\sin \theta}$$

$$= \frac{\cos^2 \theta}{\sin \theta}$$

$$= \cot \theta \cos \theta$$

$$= \frac{\left(1 - \tan^2 \frac{\theta}{2}\right)^2}{2 \tan \frac{\theta}{2} \left(1 + \tan^2 \frac{\theta}{2}\right)}$$

Further simplify the above,

$$2 \tan^2 \frac{\theta}{2} \left(1 + \tan^2 \frac{\theta}{2}\right) = 1 + \tan^4 \frac{\theta}{2} - 2 \tan^2 \frac{\theta}{2}$$

$$2 \tan^2 \frac{\theta}{2} + 2 \tan^4 \frac{\theta}{2} = 1 + \tan^4 \frac{\theta}{2} - 2 \tan^2 \frac{\theta}{2}$$

$$\tan^4 \frac{\theta}{2} + 4 \tan^2 \frac{\theta}{2} - 1 = 0$$

Solving the above quadratic equation, we get

$$\begin{aligned}\tan^2 \frac{\theta}{2} &= \frac{-4 \pm \sqrt{4^2 - (-1)4}}{2(1)} \\ &= -2 \pm \sqrt{5} \\ &= -2 + \sqrt{5} \quad \left[\because \tan^2 \frac{\theta}{2} > 0 \right]\end{aligned}$$

23. Consider the expression,

$$|\cos x|^{2\sin^2 x - 3\sin x + 1} = 1$$

So, either $|\cos x| = 1$ or $2\sin^2 x - 3\sin x + 1 = 0$

If $|\cos x| = 1$

$$\cos x = \pm 1$$

$$x = 0, \pi, 2\pi$$

$$x = 0, \pi, 2\pi$$

If $(2\sin x - 1)(\sin x - 1) = 0$

$$\sin x = \frac{1}{2} \text{ or } \sin x = 1$$

$$x = \frac{\pi}{6}, \frac{5\pi}{6} \text{ or } x = \frac{\pi}{2}$$

This implies,

$$x = 0, \pi, 2\pi, \frac{\pi}{6}, \frac{5\pi}{6}$$

So, the number of real values of x is 5.

24. Consider the expression,

$$2(\cos^{-1} x)^2 - \pi \cos^{-1} x + \frac{\pi^2}{4}$$

Let $\cos^{-1} x = y$ then,

$$\begin{aligned} 2y^2 - \pi y + \frac{\pi^2}{4} &= 2\left[y^2 - \frac{\pi}{2}y\right] + \frac{\pi^2}{4} \\ &= 2\left[\left(y - \frac{\pi}{4}\right) - \frac{\pi^2}{16}\right] + \frac{\pi^2}{4} = 2\left(y - \frac{\pi}{4}\right)^2 + \frac{\pi^2}{8} \\ &= 2\left(\cos^{-1} x - \frac{\pi}{4}\right)^2 + \frac{\pi^2}{8} \end{aligned}$$

As we know that,

$$0 \leq \cos^{-1} x \leq \pi$$

$$-\frac{\pi}{4} \leq \cos^{-1} x - \frac{\pi}{4} \leq \frac{3\pi}{4}$$

$$0 \leq \left(\cos^{-1} x - \frac{\pi}{4} \right)^2 \leq \frac{9\pi^2}{16}$$

$$0 \leq 2 \left(\cos^{-1} x - \frac{\pi}{4} \right)^2 \leq \frac{9\pi^2}{8}$$

$$\frac{\pi^2}{8} \leq \left(\cos^{-1} x - \frac{\pi}{4} \right)^2 + \frac{\pi^2}{8} \leq 5 \frac{\pi^2}{4}$$

So, the required sum is,

$$\frac{\pi^2}{8} + \frac{5\pi^2}{4} = \frac{11\pi^2}{8}$$

25. Simplify the given expression.

$$y = \log_e \tan \left(\frac{\pi}{4} + \frac{x}{2} \right)$$

Then,

$$\log_e \left(\frac{\tan\left(\frac{\pi}{4}\right) + \tan\frac{x}{2}}{1 - \tan\left(\frac{\pi}{4}\right)\tan\left(\frac{x}{2}\right)} \right) = \log_e \left(\frac{1 + \tan\frac{x}{2}}{1 - \tan\frac{x}{2}} \right)$$

$$e^y = \frac{1 + \tan\frac{x}{2}}{1 - \tan\frac{x}{2}}$$

Now,

$$\frac{e^y + 1}{e^y - 1} = \frac{2}{2 \tan\frac{x}{2}}$$

$$\frac{e^y - 1}{e^y + 1} = \tan\frac{x}{2}$$

$$\tanh\left(\frac{y}{2}\right) = \tan\frac{x}{2}$$

26. In the given triangle, a, b, c are in A.P so,

$$2b = a + c \quad (I)$$

Now,

$$\begin{aligned}
 \frac{b^2 + c^2 - a^2}{2bc} + 2 \frac{a^2 + c^2 - b^2}{2ac} + \frac{a^2 + b^2 - c^2}{2ab} &= \frac{\left(\begin{aligned} &ab^2 + ac^2 - a^3 + 2a^2b \\ &+ 2c^2b - 2b^3 \\ &+ a^2c + b^2c - c^3 \end{aligned} \right)}{2abc} \\
 &= \frac{\left(\begin{aligned} &ab^2 + ac^2 - a^3 + \\ &2b(a^2 + c^2) - 2b^3 + \\ &a^2c + b^2c - c^3 \end{aligned} \right)}{2abc} \\
 &= \frac{-2b^3 + a^2c + b^2c - c^3}{2abc}
 \end{aligned}$$

Further simplify the above,

$$\begin{aligned}
 \frac{ab^2 + 2c^2 + 2ca^2 + b^2c - 2b^3}{2abc} &= \frac{b^2(a + c) + 2ac(a + c) - 2b^3}{2abc} \\
 &= \frac{b^2(2b) + 2ac(2b) - 2b^3}{2abc} \\
 &= 2
 \end{aligned}$$

27. The area of triangle is,

$$\begin{aligned}\Delta &= b^2 - (c - a)^2 \\ &= (b + c - a)(b - c + a) \\ &= (2s - 2a)(2s - 2c) \\ &= 4(s - a)(s - c)\end{aligned}$$

This implies,

$$\begin{aligned}\sqrt{s(s - a)(s - b)(s - c)} &= 4(s - a)(s - c) \\ \sqrt{s(s - b)} &= 4\sqrt{(s - a)(s - c)} \\ \frac{1}{4} &= \sqrt{\frac{(s - a)(s - c)}{s(s - b)}} \\ &= \tan\left(\frac{B}{2}\right)\end{aligned}$$

Now,

$$\begin{aligned}\tan B &= \frac{2 \tan\left(\frac{B}{2}\right)}{1 - \tan^2\left(\frac{B}{2}\right)} \\ &= \frac{2\left(\frac{1}{4}\right)}{1 - \frac{1}{16}} \\ &= \frac{8}{15}\end{aligned}$$

28. In a triangle ABC,

$$r_1 = \frac{\Delta}{s-a}$$

$$\frac{1}{r_1} = \frac{s-a}{\Delta}$$

Similarly,

$$\frac{1}{r_2} = \frac{s-b}{\Delta}$$

$$\frac{1}{r_3} = \frac{s-c}{\Delta}$$

And,

$$\frac{1}{r} = \frac{s}{\Delta}$$

Then,

$$\begin{aligned} \frac{1}{r_1^2} + \frac{1}{r_2^2} + \frac{1}{r_3^2} + \frac{1}{r^2} &= \frac{(s-a)^2}{\Delta^2} + \frac{(s-b)^2}{\Delta^2} + \frac{(s-c)^2}{\Delta^2} + \frac{s^2}{\Delta^2} \\ &= \frac{s^2 + a^2 - 2as + s^2 + b^2 - 2sb + s^2 + c^2 - 2sc + s^2}{\Delta^2} \\ &= \frac{4s^2 + a^2 + b^2 + c^2 - 2s(a+b+c)}{\Delta^2} \\ &= \frac{a^2 + b^2 + c^2}{\Delta^2} \end{aligned}$$

29. Let us consider,

$$\mathbf{OA} = (x - u)\hat{\mathbf{i}} + x\hat{\mathbf{j}} + x\hat{\mathbf{k}}$$

$$\mathbf{OB} = x\hat{\mathbf{i}} + (x - v)\hat{\mathbf{j}} + x\hat{\mathbf{k}}$$

$$\mathbf{OC} = x\hat{\mathbf{i}} + x\hat{\mathbf{j}} + (x - w)\hat{\mathbf{k}}$$

$$\mathbf{OD} = (x - 1)\hat{\mathbf{i}} + (x - 1)\hat{\mathbf{j}} + (x - 1)\hat{\mathbf{k}}$$

Here,

$$\mathbf{DA} = (1 - u)\hat{\mathbf{i}} + \hat{\mathbf{j}} + \hat{\mathbf{k}}$$

$$\mathbf{DB} = \hat{\mathbf{i}} + (1 - v)\hat{\mathbf{j}} + \hat{\mathbf{k}}$$

$$\mathbf{DC} = \hat{\mathbf{i}} + \hat{\mathbf{j}} + (1 - w)\hat{\mathbf{k}}$$

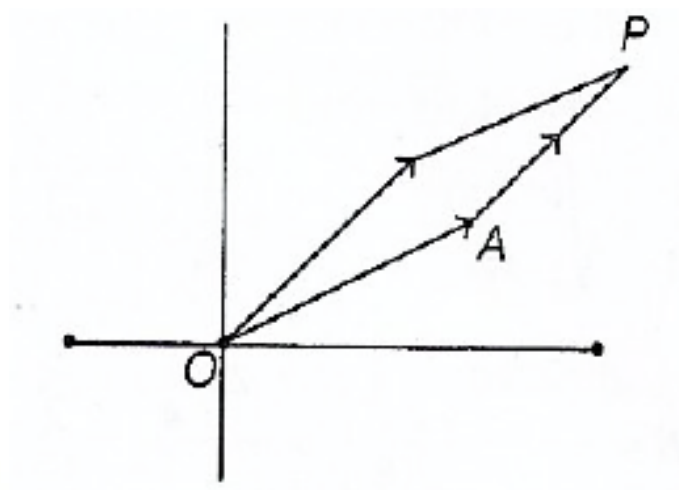
As the above points are collinear so,

$$[\mathbf{DA} \ \mathbf{BC} \ \mathbf{DC}] = 0$$

$$\begin{bmatrix} 1-u & 1 & 1 \\ 1 & 1-v & 1 \\ 1 & 1 & 1-w \end{bmatrix} = 0$$

$$\frac{1}{u} + \frac{1}{v} + \frac{1}{w} = 1$$

30. Consider the diagram,



It is given that,

$$AP = 18$$

$$\mathbf{OA} = \hat{\mathbf{i}} - \hat{\mathbf{j}} + 3\hat{\mathbf{k}}$$

Now,

$$\begin{aligned} \mathbf{AP} &= 18 \times \frac{2\hat{\mathbf{i}} + \hat{\mathbf{j}} - 2\hat{\mathbf{k}}}{\sqrt{2^2 + 1^2 + (-2)^2}} \\ &= 18 \times \frac{2\hat{\mathbf{i}} + \hat{\mathbf{j}} - 2\hat{\mathbf{k}}}{3} \\ &= 6(2\hat{\mathbf{i}} + \hat{\mathbf{j}} - 2\hat{\mathbf{k}}) \\ &= 12\hat{\mathbf{i}} + 6\hat{\mathbf{j}} - 12\hat{\mathbf{k}} \end{aligned}$$

Using the triangle law, we get

$$\mathbf{OP} = \mathbf{OA} + \mathbf{AP}$$

$$= \hat{\mathbf{i}} - \hat{\mathbf{j}} + 3\hat{\mathbf{k}} + 12\hat{\mathbf{i}} + 6\hat{\mathbf{j}} - 12\hat{\mathbf{k}}$$

$$= 13\hat{\mathbf{i}} + 5\hat{\mathbf{j}} - 9\hat{\mathbf{k}}$$

31. It is given that,

$$|\mathbf{a}| = 1, |\mathbf{b}| = 2, |\mathbf{c}| = 3, \mathbf{b} \cdot \mathbf{c} = 0$$

Then,

$$\begin{aligned} |2\mathbf{a} + 3\mathbf{b} - 3\mathbf{c}| &= \sqrt{4|\mathbf{a}|^2 + 9|\mathbf{b}|^2 + 12\mathbf{a} \cdot \mathbf{b} - 18\mathbf{b} \cdot \mathbf{c} - 12\mathbf{a} \cdot \mathbf{c}} \\ &= \sqrt{4 + 9(4) + 9(9) + 12\mathbf{a} \cdot \mathbf{b} - 0 - 12\mathbf{a} \cdot \mathbf{c}} \\ &= \sqrt{4 + 36 + 81 + 12\mathbf{a} \cdot \mathbf{b} - 12\mathbf{a} \cdot \mathbf{c}} \end{aligned}$$

The projection of $\mathbf{b}$ along $\mathbf{a}$ is equal to the projection of $\mathbf{c}$ along

$\mathbf{a}$ so, $\mathbf{a} \cdot \mathbf{b} = \mathbf{a} \cdot \mathbf{c}$. Thus,

$$\begin{aligned} |2\mathbf{a} + 3\mathbf{b} - 3\mathbf{c}| &= \sqrt{121 + 12\mathbf{a} \cdot \mathbf{b} - 12\mathbf{a} \cdot \mathbf{b}} \\ &= 11 \end{aligned}$$

32. Consider given data,

$$\mathbf{m} = \sqrt{3} \times \text{unit vectors of } \left[(\hat{\mathbf{i}} + \hat{\mathbf{j}}) \times (\hat{\mathbf{j}} - \hat{\mathbf{k}}) \right]$$

This implies,

$$\begin{aligned}(\hat{\mathbf{i}} + \hat{\mathbf{j}}) \times (\hat{\mathbf{j}} - \hat{\mathbf{k}}) &= \begin{vmatrix} \hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}} \\ 1 & 1 & 0 \\ 0 & 1 & -1 \end{vmatrix} \\&= \hat{\mathbf{i}}(-1 - 0) - \hat{\mathbf{j}}(-1 - 0) + \hat{\mathbf{k}}(1) \\&= -\hat{\mathbf{i}} + \hat{\mathbf{j}} + \hat{\mathbf{k}}\end{aligned}$$

So,

$$\begin{aligned}\mathbf{m} &= \sqrt{3} \times \frac{-\hat{\mathbf{i}} + \hat{\mathbf{j}} + \hat{\mathbf{k}}}{\sqrt{1+1+1}} \\&= \sqrt{3} \times \frac{-\hat{\mathbf{i}} + \hat{\mathbf{j}} + \hat{\mathbf{k}}}{\sqrt{3}} \\&= -\hat{\mathbf{i}} + \hat{\mathbf{j}} + \hat{\mathbf{k}}\end{aligned}$$

And,

$$\mathbf{n} = 2\sqrt{6} \times \text{unit vectors of } \left[(2\hat{\mathbf{i}} - \hat{\mathbf{j}}) \times (\hat{\mathbf{j}} + 2\hat{\mathbf{k}}) \right]$$

This implies,

$$\begin{aligned}(2\hat{\mathbf{i}} - \hat{\mathbf{j}}) \times (\hat{\mathbf{j}} + 2\hat{\mathbf{k}}) &= \begin{vmatrix} \hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}} \\ 2 & -1 & 0 \\ 0 & 1 & 2 \end{vmatrix} \\&= \hat{\mathbf{i}}(-2 - 0) - \hat{\mathbf{j}}(4 - 0) + \hat{\mathbf{k}}(2 + 0) \\&= -2\hat{\mathbf{i}} - 4\hat{\mathbf{j}} + 2\hat{\mathbf{k}}\end{aligned}$$

So,

$$\begin{aligned}\mathbf{n} &= 2\sqrt{6} \times \frac{-2\hat{\mathbf{i}} - 4\hat{\mathbf{j}} + 2\hat{\mathbf{k}}}{\sqrt{4+16+4}} \\ &= 2\sqrt{6} \times \frac{-2\hat{\mathbf{i}} - 4\hat{\mathbf{j}} + 2\hat{\mathbf{k}}}{2\sqrt{6}} \\ &= -2\hat{\mathbf{i}} - 4\hat{\mathbf{j}} + 2\hat{\mathbf{k}}\end{aligned}$$

Now, the required area is,

$$\begin{aligned}\frac{1}{2}(\mathbf{m} \times \mathbf{n}) &= \frac{1}{2}(-\hat{\mathbf{i}} + \hat{\mathbf{j}} + \hat{\mathbf{k}}) \times (-2\hat{\mathbf{i}} - 4\hat{\mathbf{j}} + 2\hat{\mathbf{k}}) \\ &= \frac{1}{2} \begin{vmatrix} \hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}} \\ -1 & 1 & 1 \\ -2 & -4 & 2 \end{vmatrix} \\ &= \frac{1}{2} \times |\hat{\mathbf{i}}(2+4) - \hat{\mathbf{j}}(-2+2) + \hat{\mathbf{k}}(4+2)| \\ &= \frac{1}{2} \times |6\hat{\mathbf{i}} + 6\hat{\mathbf{k}}| \\ &= \frac{1}{2} \sqrt{6^2 + 6^2}\end{aligned}$$

Further simplify the above,

$$\begin{aligned}\frac{1}{2}(\mathbf{m} \times \mathbf{n}) &= \frac{1}{2} \times 6 \times \sqrt{2} \\ &= 3\sqrt{2}\end{aligned}$$

33. Consider the given vectors,

$$\mathbf{a} = \hat{\mathbf{i}} - \hat{\mathbf{j}} + \hat{\mathbf{k}}$$

$$\mathbf{b} = \hat{\mathbf{i}} - 2\hat{\mathbf{j}} + \hat{\mathbf{k}}$$

$$\mathbf{c} = p\hat{\mathbf{i}} + 2\hat{\mathbf{j}} + q\hat{\mathbf{k}}$$

$$\mathbf{d} = p\hat{\mathbf{i}} + q\hat{\mathbf{j}} + 2\hat{\mathbf{k}}$$

As the projection of $\mathbf{c}$ on $\mathbf{a}$ is $5\sqrt{3}$ so,

$$\frac{\mathbf{a} \cdot \mathbf{c}}{|\mathbf{a}|} = 5\sqrt{3}$$

$$\frac{p + q - 2}{\sqrt{3}} = 5\sqrt{3}$$

$$p + q - 2 = 15$$

$$p + q = 17$$

(I)

It is also given that,

$$[\mathbf{a} \ \mathbf{b} \ \mathbf{c}] = 5$$

So,

$$\begin{bmatrix} 1 & -1 & 1 \\ 1 & -2 & 1 \\ p & 2 & q \end{bmatrix} = 5$$

$$1(-2q - 2) + 1(q - p) + 1(2 + 2p) = 5$$

$$-2q - 2 + q - p + 2 + 2p = 5$$

$$p - q = 5 \quad \text{(II)}$$

From equation (I) and (II),

$$p = 11, q = 6$$

Now,

$$\begin{aligned}\mathbf{b} \cdot \mathbf{d} &= (\hat{\mathbf{i}} - 2\hat{\mathbf{j}} + \hat{\mathbf{k}}) \cdot (p\hat{\mathbf{i}} + q\hat{\mathbf{j}} + 2\hat{\mathbf{k}}) \\ &= p - 2q + 2 \\ &= 11 - 12 + 2 \\ &= 1\end{aligned}$$

This gives,

$$\begin{aligned}\tan^{-1}(\mathbf{b} \cdot \mathbf{d}) &= \tan^{-1}(1) \\ &= \frac{\pi}{4}\end{aligned}$$

34. Consider the given data,

$$\mathbf{a} = 2\hat{\mathbf{i}} + \hat{\mathbf{j}} + \hat{\mathbf{k}}$$

$$\mathbf{b} = \hat{\mathbf{i}} + 2\hat{\mathbf{j}} - \hat{\mathbf{k}}$$

Let, $\mathbf{c} = x\hat{\mathbf{i}} + y\hat{\mathbf{j}} + z\hat{\mathbf{k}}$ so,

$$\mathbf{c} \cdot \mathbf{a} = 0$$

This implies,

$$\begin{aligned}(x\hat{\mathbf{i}} + y\hat{\mathbf{j}} + z\hat{\mathbf{k}}) \cdot (2\hat{\mathbf{i}} + \hat{\mathbf{j}} + \hat{\mathbf{k}}) &= 0 \\ 2x + y + z &= 0 \quad \quad \quad \text{(I)}\end{aligned}$$

Since, $\mathbf{a}, \mathbf{b}, \mathbf{c}$ are coplanar so,

$$[\mathbf{a} \ \mathbf{b} \ \mathbf{c}] = 0$$

Therefore,

$$\begin{bmatrix} x & y & z \\ 2 & 1 & 1 \\ 1 & 2 & -1 \end{bmatrix} = 0$$

$$x(-1-2) - y(-2-1) + z(4-1) = 0$$

$$-3x + 3y + 3z = 0$$

$$-x + y + z = 0 \quad (\text{II})$$

From equation (I) and (II),

$$x = 0, y = -z$$

So,

$$x^2 + y^2 + z^2 = 1$$

$$y = \pm \frac{1}{\sqrt{2}}$$

This gives,

$$c = \pm \frac{1}{\sqrt{2}}(-\hat{j} + \hat{k})$$

35. The mean of given data is,

$$\begin{aligned}\bar{x} &= \frac{9+3+11+5+7}{5} \\ &= \frac{35}{5} \\ &= 7\end{aligned}$$

The variance is,

$$\begin{aligned}\sigma^2 &= \frac{1}{5}(9^2 + 3^2 + 11^2 + 5^2 + 7^2) - (7)^2 \\ &= \frac{1}{5}(81 + 9 + 121 + 25 + 49) - 49 \\ &= 57 - 49 \\ &= 8\end{aligned}$$

So, the coefficient of variance is,

$$\begin{aligned}\frac{\sigma^2}{\bar{x}} \times 100 &= \frac{\sqrt{8}}{7} \times 100 \\ &= \frac{200\sqrt{2}}{7}\end{aligned}$$

36. Consider given data,

Class Interval	x_i	(f_i)	$x_i f_i$	$ x_i - 21 $	$f_i x_i - 21 $
0-10	5	6	30	16	96
10-20	15	8	120	6	48

20-30	25	8	250	4	40
30-40	35	4	140	14	56
40-50	45	2	90	24	48
	$N = \sum f_i = 30$			$\sum x_i f_i = 630$	$\sum f_i x_i - 21 = 288$

The mean is given by,

$$\begin{aligned}\bar{x} &= \frac{\sum x_i f_i}{N} \\ &= \frac{630}{30} \\ &= 21\end{aligned}$$

Thus, the mean deviation is,

$$\begin{aligned}\sigma^2 &= \frac{1}{N} \sum f_i = \frac{1}{30}(288) \\ &= 9.6\end{aligned}$$

37. As the probability of occurrence of event A is,

$$P(A) = \frac{2}{5}$$

Then, the probability of non-occurrence of event A is,

$$\begin{aligned}P(A)' &= 1 - P(A) \\&= 1 - \frac{2}{5} \\&= \frac{3}{5}\end{aligned}$$

As the probability of non-occurrence of event B is,

$$P(B)' = \frac{3}{10}$$

Then, the probability of occurrence of event B is,

$$\begin{aligned}P(B) &= 1 - P(B)' \\&= 1 - \frac{3}{10} \\&= \frac{7}{10}\end{aligned}$$

The probability that only one of the two events occur is,

$$\begin{aligned}P(A)P(B') + P(A')P(B) &= \frac{2}{5}\left(\frac{3}{10}\right) + \frac{3}{5}\left(\frac{7}{10}\right) \\&= \frac{6}{50} + \frac{21}{50} \\&= \frac{27}{50}\end{aligned}$$

38. The total number of ways for $(a.b.c)$ will be by $6 \times 6 \times 6$.

Here, $\omega^3 = 1$, $\omega^4 = \omega$, $\omega^5 = \omega^2$ and $\omega^6 = 1$

The ω, ω^2 are the roots of $x^2 + x + 1 = 0$

This implies,

$$\omega^a + \omega^b + \omega^c = 0$$

Now, suitable values for (a, b, c) are $6 \times 4 \times 2$.

So, the required probability is,

$$\begin{aligned} P &= \frac{6 \times 4 \times 2}{6 \times 6 \times 6} \\ &= \frac{8}{36} \\ &= \frac{2}{9} \end{aligned}$$

39. Let E_1 and E_2 be the events for ball drawn from bag A and from bag B respectively and E is the event that ball is red.

It is given that,

$$P\left(\frac{E_1}{E}\right) = \frac{6}{7}$$

By Baye's theorem,

$$P\left(\frac{E_1}{A}\right) = \frac{\frac{1}{2} \left(\frac{{}^n C_2}{{}^{n+2} C_2} \right)}{\frac{1}{2} \frac{{}^n C_2}{{}^{n+2} C_2} + \frac{1}{2} \frac{{}^2 C_2}{{}^{n+2} C_2}} = \frac{6}{7}$$

Simplifying the above expression, we get

$$\frac{\frac{n(n-1)}{2}}{2 + n(n-1)} = \frac{6}{7}$$

$$\frac{n(n-1)}{n^2 - n + 2} = \frac{6}{7}$$

$$7(n^2 - n) = 6n^2 - 6n + 12$$

$$7n^2 - 7n - 6n^2 + 6n = 12$$

$$n^2 - n = 12$$

$$n^2 - n - 12 = 0$$

$$(n+3)(n-4) = 0$$

$$n = 4$$

$$[\because n > 0]$$

40. Considering the given data,

$X = -1$	$X = 0$	$X = 1$
a	b	c

The mean is 0.2 so,

$$\frac{-a+c}{a+b+c} = \frac{2}{10} \quad (I)$$

And,

$$\frac{b}{a+b+c} = \frac{2}{10} \quad (II)$$

Now, from equation (I),

$$\begin{aligned} 1 + \frac{c-a}{a+b+c} &= \frac{2}{10} + 1 \\ \frac{a+b+c+c-a}{a+b+c} &= \frac{12}{10} \\ \frac{b+2c}{a+b+c} &= \frac{12}{10} \\ 2\left(\frac{c}{a+b+c}\right) + \frac{2}{10} &= \frac{12}{10} \quad [\text{from eq. (II)}] \end{aligned}$$

This implies,

$$\begin{aligned} 2P(X=1) &= \frac{12}{10} - \frac{2}{10} \\ &= 1 \end{aligned}$$

$$P(X=1) = 0.5$$

41. The event of expected to have children of both sexes is $\frac{1}{2}$.

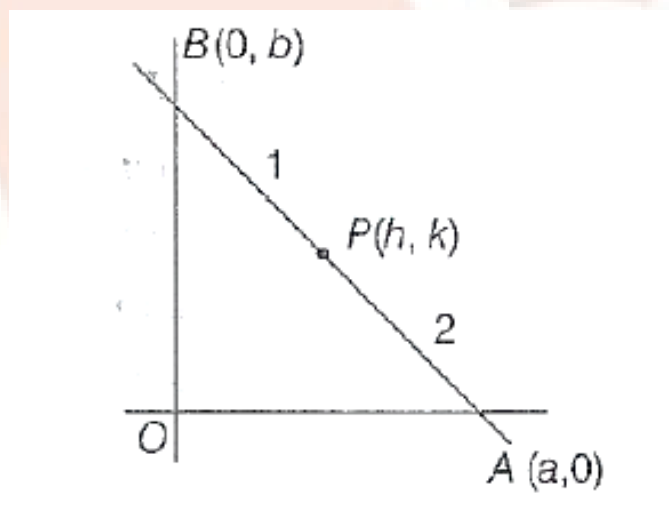
So, the probability of children of both sexes is,

$$\begin{aligned}1 - [P(\text{all boys}) + P(\text{all girls})] &= 1 - \left\{ \left(\frac{1}{2}\right)^4 + \left(\frac{1}{2}\right)^4 \right\} \\&= 1 - \frac{1}{16} - \frac{1}{16} \\&= \frac{7}{8}\end{aligned}$$

So, the number of families expected to children of both sexes is,

$$n = \frac{7}{8} \times 800 = 700$$

42. Consider the diagram,



From the above diagram,

$$P(h, k) = \left(\frac{a \times 1 + 0}{1 + 2}, \frac{0 \times 2b}{1 + 2} \right)$$

$$= \left(\frac{a}{3}, \frac{2b}{3} \right)$$

It means,

$$h = \frac{a}{3} \Rightarrow a = 3h$$

$$k = \frac{2b}{3} \Rightarrow b = \frac{3k}{2}$$

Since, $AB = 6$ So,

$$\sqrt{(a - 0)^2 + (0 - b)^2} = 6$$

$$a^2 + b^2 = 36$$

$$9h^2 + \frac{9k^2}{4} = 36$$

$$4h^2 + k^2 = 16$$

So, the required locus of point P is,

$$4x^2 + y^2 = 16$$

43. Let us consider,

$$\frac{x + \sqrt{3}y}{2} = \frac{x + \sqrt{3}y}{\sqrt{1 + (\sqrt{3})^2}} = y$$

And,

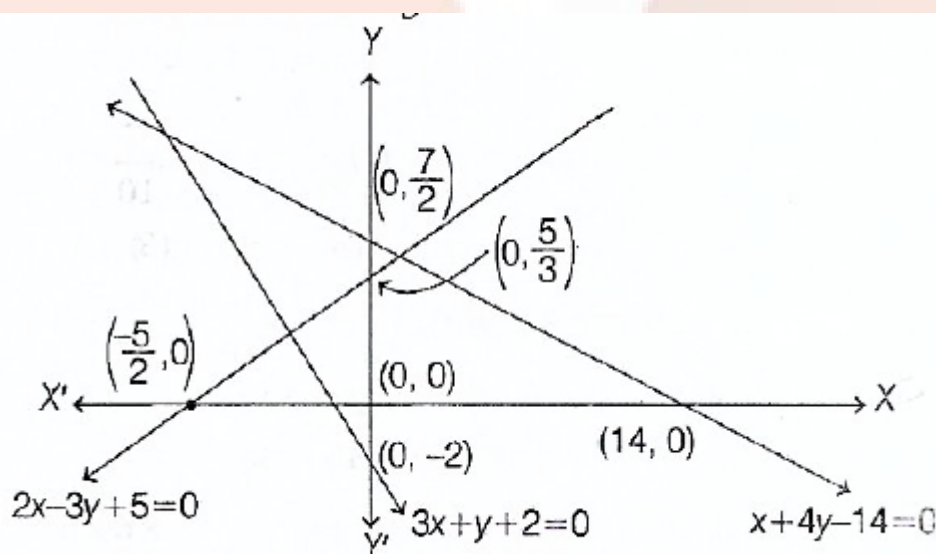
$$\frac{\sqrt{3}x - y}{2} = \frac{\sqrt{3}x - y}{\sqrt{(\sqrt{3})^2 + 1}} = x$$

So, the curve $\frac{x + \sqrt{3}y}{2} = \left(\frac{\sqrt{3}x - y}{2}\right)^2$ becomes $y = x^2$.

Similarly, the curve $\frac{\sqrt{3}x - y}{2} = \left(\frac{x + \sqrt{3}y}{2}\right)^2$ becomes $x^2 = y$.

Thus, the area of the region bounded by above curves is equal to the area bounded by the curves $y = x^2$ and $x = y^2$ which is equal to k .

44. Consider the diagram,



The line $L_1 \equiv 3x + y + 2 = 0$ can be written as,

$$\frac{x}{-\frac{2}{3}} + \frac{y}{-2} = 1$$

The line $L_2 \equiv 2x - 3y + 5 = 0$ can be written as,

$$\frac{x}{-\frac{5}{2}} + \frac{y}{\frac{5}{3}} = 1$$

And the line $L_3 \equiv x + 4y - 14 = 0$ can be written as,

$$\frac{x}{14} + \frac{y}{-\frac{7}{4}} = 1$$

From the diagram it is clear that,

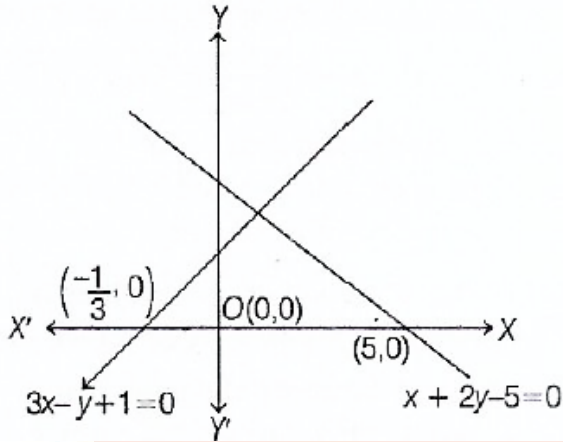
$$\beta \in \left[\frac{5}{3}, \frac{7}{2} \right]$$

45. Consider the equations,

$$x + 2y - 5 = 0 \quad (I)$$

$$3x - y + 1 = 0 \quad (II)$$

From equation (I) origin O and P are on same sides so,



This implies,

$$(-5)(\lambda^2 + 2\lambda - 3) > 0$$

$$\lambda^2 + 2\lambda - 3 < 0$$

$$\lambda(\lambda + 3) - 1(\lambda + 3) < 0$$

$$(\lambda + 3)(\lambda - 1) < 0$$

This gives,

$$\lambda \in (-3, 1)$$

From equation (II),

$$(\lambda)\left(\lambda - \frac{1}{3}\right) > 0$$

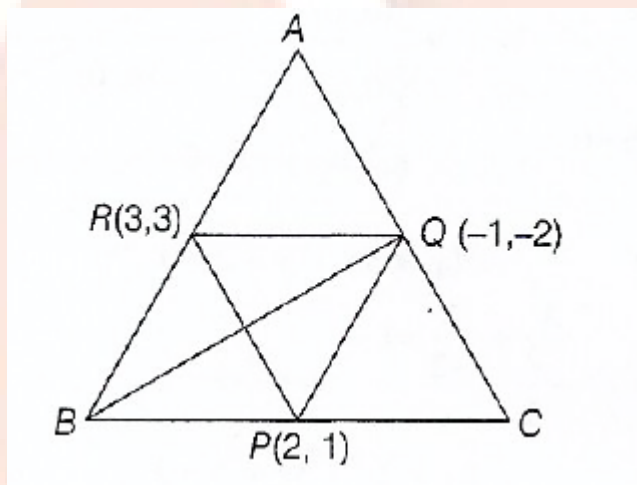
This gives,

$$\lambda \in (-\infty, 0) \cup \left(\frac{1}{3}, \infty\right)$$

$$\{-2, -1\}$$

Thus, there are two points.

46. Consider the diagram,



Let (a, b) be the coordinates of B.

From the diagram, it is clear that the midpoint of BQ is equal to the midpoint of PR so,

$$\left(\frac{a-1}{2}, \frac{b-1}{2} \right) = \left(\frac{3+2}{2}, \frac{3+1}{2} \right)$$

This implies,

$$\frac{a-1}{2} = \frac{5}{2}$$
$$a = 5 + 1$$
$$= 6$$

And,

$$\frac{b-2}{2} = \frac{4}{2}$$
$$b = 4 + 2$$
$$= 6$$

Thus, the coordinates of point B are (6,6).

Now, the equation of BC can be obtained as,

$$y - 6 = \frac{1-6}{2-6}(x-6)$$
$$= \frac{-5}{-4}(x-6)$$

$$4y - 24 = 5x - 30$$

$$5x - 4y = 6$$

47. The distance between the pair of lines

$x^2 + 2hxy + by^2 + 2gx + 4fy + c = 0$ is given by,

$$d = 2\sqrt{\frac{g^2 - ac}{a(a+b)}} \quad \dots\dots (I)$$

Consider the given equation,

$$x^2 + 2\sqrt{2}xy + 2y^2 + 4x + 4\sqrt{2}y + 1 = 0$$

Here, $a = 1, b = 2, c = 1, g = 2$

Substituting the values in equation (I), we get

$$\begin{aligned} d &= 2\sqrt{\frac{4-1}{1(1+2)}} = 2\sqrt{\frac{3}{3}} \\ &= 2 \end{aligned}$$

48. Simplify the given equation of pair of lines.

$$8x^2 - 14xy + 3y^2 + 10x + 10y - 25 = 0$$

$$(4x - y - 5)(2x - 3y + 5) = 0$$

Now, solving the two equations, $4x - y - 5 = 0$ and

$2x - 3y + 5 = 0$, we get the point of intersection as $(2, 3)$.

So, equation of $S = 0$ will be,

$$(4x - y + c_1)(2x - 3y + c_2) = 0$$

As the lines $S = 0$ passes through point $P(x_1, y_1)$ such that mid point of $P(x_1, y_1)$ and $(2, 3)$ is $(3, 2)$ therefore,

$$x_1 = 4 \text{ and } y_1 = 1$$

This gives,

$$c_1 = -15 \text{ and } c_2 = -5$$

So required equation of pair of lines is,

$$(4x - y - 15)(2x - 3y - 5) = 0$$

$$8x^2 - 14xy + 3y^2 - 50x + 50y + 75 = 0$$

49. Consider the equation of circle,

$$x^2 + y^2 + 2\lambda x - 2\lambda y + \lambda^2 = 0$$

Here, the center is $(-\lambda, \lambda)$ and radius is λ .

The circle touches $12x + 5y - 60 = 0$ so, $r = d$

This implies,

$$\lambda = \frac{|-7\lambda - 60|}{\sqrt{144 + 25}}$$

$$13\lambda = |-7\lambda - 60|$$

$$7\lambda + 60 = 13\lambda \text{ or } -7\lambda - 60 = 13\lambda$$

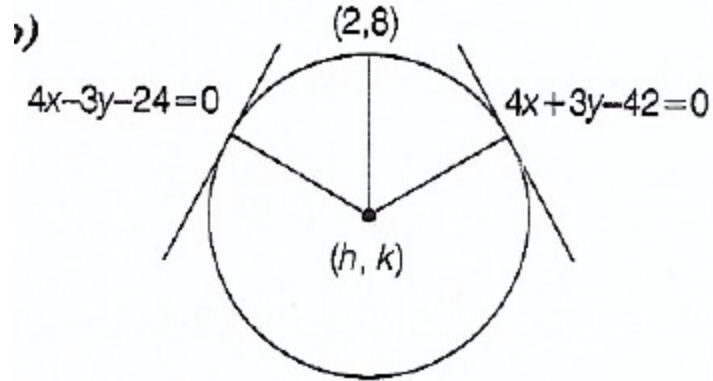
$$6\lambda = 60 \text{ or } 20\lambda = -60$$

$$\lambda = 10 \text{ or } \lambda = -3$$

As radius cannot be negative therefore,

$$\lambda = 10$$

50. Consider the diagram,



By the property of distances,

$$\left| \frac{4h - 3k - 24}{5} \right| = \left| \frac{4h + 3k - 42}{5} \right|$$

$$= \sqrt{(h - 2)^2 + (k - 8)^2}$$

This implies,

$$4h - 3k - 24 = 4h + 3k - 42$$

$$6k = 18$$

$$k = 3$$

Or,

$$4h - 3k - 24 = -4h - 3k + 42$$

$$8h = 66$$

$$h = \frac{66}{8}$$

$$= \frac{33}{4} > 8$$

Now,

$$(4h - 3k - 24)^2 = 25[(h - 2)^2 + (k - 8)^2]$$

From above,

$$h = 2$$

Thus, the centre is $(2, 3)$.

The required circle is,

$$(x - 2)^2 + (y - 3)^2 = 25$$

$$x^2 + 4 - 4x + y^2 + 9 - 6y = 25$$

$$x^2 + y^2 - 4x - 6y = 12$$

51. Consider the given circle equations,

$$x^2 + y^2 - 4x - 2y + 1 = 0 \quad \text{(I)}$$

And,

$$x^2 + y^2 - 6x - 4y + 4 = 0 \quad \text{(II)}$$

Here, $C_1 = (2, 1)$, $C_2 = (3, 2)$

And,

$$r_1 = \sqrt{4 + 1 - 1} = 2$$

$$r_2 = \sqrt{9 + 4 - 4} = 3$$

Then,

$$\begin{aligned}C_1C_2 &= \sqrt{(3-2)^2 + (2-1)^2} \\&= \sqrt{2}\end{aligned}$$

And,

$$\begin{aligned}r_1 + r_2 &= 2 + 3 \\&= 5\end{aligned}$$

This shows that,

$$C_1C_2 < r_1 + r_2$$

So, the circles intersect at two distinct points.

Let the point of intersection be $P(x, y)$ then,

$$\begin{aligned}P(x, y) &= \left(\frac{6-6}{-1}, \frac{4-3}{-1} \right) \\&= (0, -1)\end{aligned}$$

52. From the given information if two circles intersect orthogonally then,

$$2gg' + 2ff' = c + c'$$

$$\text{Here, } (g, f) = (2, 3), (g', f') = (2, -1), c' = -7$$

So,

$$\begin{aligned}2(-2)(2) + 2(3)(-1) &= c - 7 \\c &= 9\end{aligned}$$

So, the required radius is,

$$\begin{aligned}\sqrt{2^2 + 3^2 - 9} &= \sqrt{4 + 9 - 9} \\ &= 2\end{aligned}$$

53. The equations of given circles are,

$$S_1 \equiv x^2 + y^2 - 4 = 0$$

$$S_2 \equiv x^2 + y^2 - 6x - 8y + 10 = 0$$

$$S_3 \equiv x^2 + y^2 + 2x - 4y - 2 = 0$$

Let the equation of required circle be,

$$S \equiv x^2 + y^2 + 2gx + 2fy + c = 0$$

The circle $S = 0$ cuts the circles $S_1 = 0$, $S_2 = 0$, $S_3 = 0$ at extremities of the diameter so, common chord of $S = 0$ and $S_1 = 0$ passes through center of circle $S_1 = 0$ so, $c = -4$.

Similarly $(2g + 6)x + (2f + 8)y - 14 = 0$ passes through $(3, 4)$ so, $6g + 8f + 36 = 0$.

$$3g + 4f + 18 = 0 \quad \text{(I)}$$

And,

$(2g - 2)x + (2f + 4)y - 2 = 0$ passes through $(-1, 2)$ so,

$$-2g + 4f + 8 = 0 \quad \text{(II)}$$

Solving equation (I) and (II),

$$(g, f) = (-2, -3)$$

So, equation of circle is,

$$x^2 + y^2 - 4x - 6y - 4 = 0$$

54. Simplify the given equation of parabola,

$$20(x^2 + y^2 - 6x - 2y + 10) = (4x - 2y - 5)^2$$

$$(x^2 + y^2 - 6x - 2y + 10) = \left(\frac{4x - 2y - 5}{\sqrt{20}} \right)^2$$

$$(x - 3)^2 + (y - 1)^2 = \left(\frac{4x - 2y - 5}{\sqrt{20}} \right)^2$$

The focus from above is (3,1) and equation of directrix is,

$$4x - 2y - 5 = 0$$

Thus, the distance from focus to directrix is,

$$\frac{|12 - 2 - 5|}{\sqrt{20}} = \frac{\sqrt{5}}{2} = 2a$$

So, the length of latus rectum is,

$$\begin{aligned} 4a &= 2(2a) = 2 \frac{\sqrt{5}}{2} \\ &= \sqrt{5} \end{aligned}$$

55. The angle between the tangents is 90° .

The point will be on the directrix.

The equation of the directrix is,

$$x - 2 = -3$$

$$x = -1$$

From the given equation of tangent,

$$y = 3x - 2$$

$$= 3(-1) - 2$$

$$= -5$$

Thus, the required point is $P(-1, -5)$.

56. The circumcentre of an equilateral triangle is same to the centroid so,

$$(l, m) = \left(\frac{a \cos \theta_1 + a \cos \theta_2 + a \cos \theta_3}{3}, \frac{a \sin \theta_1 + a \sin \theta_2 + a \sin \theta_3}{3} \right)$$

This implies,

$$\frac{3l}{a} = \cos \theta_1 + \cos \theta_2 + \cos \theta_3 \quad (\text{I})$$

And,

$$\frac{3m}{b} = \sin \theta_1 + \sin \theta_2 + \sin \theta_3 \quad (\text{II})$$

Square and add equation (I) and (II),

$$\frac{9l^2}{a^2} + \frac{9m^2}{b^2} = \left(\sum \cos^2 \theta_1 + 2 \sum \cos \theta_1 \cos \theta_2 \right) \\ + \sum \sin^2 \theta_1 + 2 \sum \sin \theta_1 \sin \theta_2$$

$$\frac{9l^2}{a^2} + \frac{9m^2}{b^2} = 3 + 2 \sum \cos(\theta_1 - \theta_2)$$

$$\frac{2}{3} \sum \cos(\theta_1 - \theta_2) = \frac{3l^2}{a^2} + \frac{3m^2}{b^2} - 1$$

$$\frac{2}{3} [\cos(\theta_1 - \theta_2) + \cos(\theta_2 - \theta_3) + \cos(\theta_3 - \theta_1)]$$

Then,

$$\frac{9l^2}{a^2} + \frac{9m^2}{b^2} = \frac{3l^2}{a^2} + \frac{3m^2}{b^2} - 1$$

57. Let the rectangle be inscribed in the ellipse,

$$\frac{x^2}{64} + \frac{y^2}{16} = 1$$

Let the coordinates be $P(8\cos\theta, -4\sin\theta)$. Then,

$Q(-8\cos\theta, 4\sin\theta)$, $R(-8\cos\theta, 4\sin\theta)$, $S(8\cos\theta, -4\sin\theta)$

The area of rectangle is given by,

$$A = PQ \times RS$$

$$= 16\cos\theta \times 8\sin\theta$$

$$= 64\sin 2\theta$$

Differentiate w.r.t θ ,

$$\frac{dA}{d\theta} = 128 \cos 2\theta$$

$$\frac{d^2 A}{d\theta^2} = -256 \sin 2\theta$$

For the critical number of A , $\frac{dA}{d\theta} = 0$,

$$4ab \cos 2\theta = 0$$

$$\cos 2\theta = 0$$

$$2\theta = \frac{\pi}{2} \text{ or } \frac{3\pi}{2}$$

$$\theta = \frac{\pi}{4} \text{ or } \frac{3\pi}{4}$$

Then,

$$\begin{aligned} \left(\frac{d^2 A}{d\theta^2} \right)_{\theta=\frac{\pi}{4}} &= -256 \sin \frac{\pi}{2} \\ &= -256 \\ &< 0 \end{aligned}$$

So, A is maximum when $\theta = \frac{\pi}{4}$.

This implies,

$$PQ = l = 16 \cos \theta$$

$$= 16 \cos \frac{\pi}{4}$$

$$= 16 \times \frac{1}{\sqrt{2}}$$

$$= 8\sqrt{2}$$

And,

$$PS = b = 8 \sin \theta$$

$$= 8 \sin \frac{\pi}{4}$$

$$= 8 \times \frac{1}{\sqrt{2}}$$

$$= 4\sqrt{2}$$

$$\text{Therefore, } (l, b) = (8\sqrt{2}, 4\sqrt{2})$$

58. If two lines $l_1x + m_1y + n_1 = 0$ and $l_2x + m_2y + n_2 = 0$ are

conjugate lines with respect to parabola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ then,

$$a^2l_1l_2 - b^2m_1m_2 = n_1n_2$$

For the hyperbola,

$$5x^2 - 6y^2 = 15$$

$$\frac{x^3}{3} - \frac{y^2}{\frac{5}{2}} = 1$$

$$\text{Here, } a^2 = 3, b^2 = \frac{5}{2}, l_1 = 2, l_2 = 3, m_1 = -k, m_2 = -1, n_1 = 3, n_2 = 1$$

so,

$$3(2)(3) - \frac{5}{2}(-k)(-1) = 3(1)$$

$$18 - \frac{5}{2}k = 3$$

$$k = 6$$

59. From the given points,

$$\begin{aligned} AB &= \sqrt{1^2 + 1^2 + 5^2} \\ &= \sqrt{27} \end{aligned}$$

$$\begin{aligned} BC &= \sqrt{0^2 + 2^2 + 4^2} \\ &= \sqrt{20} \end{aligned}$$

$$\begin{aligned} CD &= \sqrt{1^2 + 1^2 + 5^2} \\ &= \sqrt{27} \end{aligned}$$

And

$$\begin{aligned}DA &= \sqrt{0^2 + 2^2 + 4^2} \\ &= \sqrt{20}\end{aligned}$$

These shows,

$$AB = CD \text{ and } BC = DA$$

Now,

$$\begin{aligned}AC &= \sqrt{1^2 + 3^2 + 1^2} \\ &= \sqrt{11}\end{aligned}$$

And,

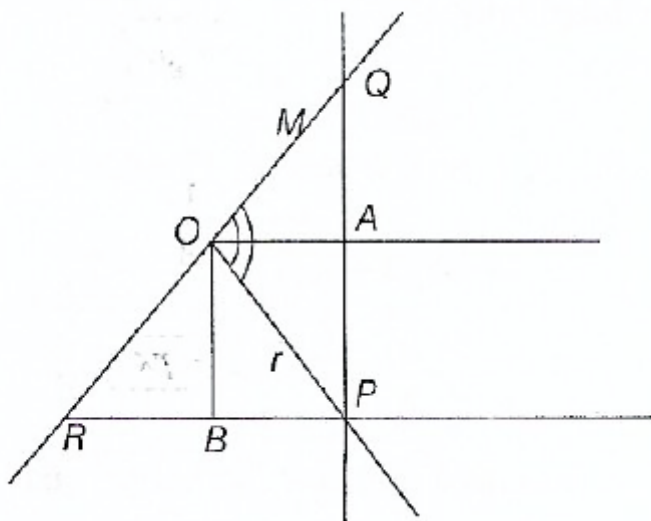
$$\begin{aligned}DB &= \sqrt{1^2 + 1^2 + 9^2} \\ &= \sqrt{83}\end{aligned}$$

It means,

$$AC \neq DB$$

Hence, A, B, C and D forms a parallelogram.

60. Consider the diagram,



The coordinates of P, Q and R are $(l_1r, m_1r, n_1r), (l_2r, m_2r, n_2r)$ and (l_3r, m_3r, n_3r) respectively.

The A and B are midpoints of PQ and PR , then OA and OB are bisectors line of direction ratios of OA are $l_1 + l_2, m_1 + m_2, n_1 + n_2$ and for OB are $l_1 - l_2, m_1 - m_2, n_1 - n_2$.

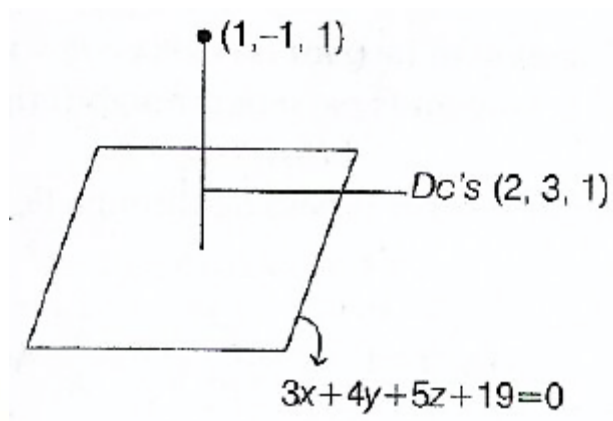
Now,

$$\begin{aligned} \sum (l_1 + l_2)^2 &= 1 + 1 + 2 \cos \theta \\ &= 2(1 + \cos \theta) \\ &= 4 \cos^2 \frac{\theta}{2} \end{aligned}$$

The DC's of internal bisector are,

$$\frac{l_1 + l_2}{2 \cos \frac{\theta}{2}}, \frac{m_1 + m_2}{2 \cos \frac{\theta}{2}}, \frac{n_1 + n_2}{2 \cos \frac{\theta}{2}}$$

61. Consider the diagram,



The equation of lines passing through $(1, -1, 1)$ and having direction cosines as $(2, 3, 1)$ are

$$\frac{x-1}{2} = \frac{y+1}{3} = \frac{z-1}{1} = r$$

From the above equations, we get

$$x = 2r + 1, y = 3r - 1, z = r + 1$$

The points, $(2r + 1, 3r - 1, r + 1)$ lie on plane.

These points satisfy the equation of the plane,

$$3(2r + 1) + 4(3r - 1) + 5(r + 1) + 19 = 0$$

$$6r + 3 + 12r - 4 + 5r + 5 + 19 = 0$$

$$23r + 23 = 0$$

$$r = -1$$

So, the point is,

$$(-2+1, -3-1, -1+1) = (-1, -4, 0)$$

Now the required distance is,

$$\begin{aligned}\sqrt{(-2)^2 + (-3)^2 + 1^2} &= \sqrt{4+9+1} \\ &= \sqrt{14}\end{aligned}$$

62. Simplify the given expression,

$$\begin{aligned}\lim_{n \rightarrow \infty} \frac{e^{nx} \left(\frac{\sin x}{e^{nx}} - 1 \right)}{e^{nx} \left(\frac{1}{e^{nx}} + A \right)} &= \lim_{n \rightarrow \infty} \frac{\left(\frac{\sin x}{e^{nx}} - 1 \right)}{\left(\frac{1}{e^{nx}} + A \right)} \\ &= \frac{0-1}{0+A} \\ &= -\frac{1}{A}\end{aligned}$$

63. Consider the function,

$$f(x) = \begin{cases} \frac{\sqrt{1+px} - \sqrt{1-px}}{x}, & \text{if } -1 \leq x < 0 \\ \frac{2x+1}{x-2}, & \text{if } 0 \leq x < 1 \end{cases}$$

At $x = 0$ both RHL and LHL are equal so,

$$\lim_{x \rightarrow 0^+} \frac{2x+1}{x-2} = \lim_{x \rightarrow 0^-} \frac{\sqrt{1+px} - \sqrt{1-px}}{x}$$

By L'Hospital rule in LHL,

$$-\frac{1}{2} = \lim_{x \rightarrow 0^-} \frac{\frac{p}{1\sqrt{1+px}} + \frac{p}{2\sqrt{1-px}}}{1}$$

$$-\frac{1}{2} = \frac{p}{2} + \frac{p}{2}$$

$$-\frac{1}{2} = p$$

64. Consider the function,

$$f(x) = \frac{\cos(\sin x) - \cos x}{x^4}$$

Simplify the above,

$$f(x) = \lim_{x \rightarrow 0} \frac{2 \sin\left(\frac{\sin x + x}{2}\right) \sin\left(\frac{x - \sin x}{2}\right)}{x^4}$$

$$= \lim_{x \rightarrow 0} \frac{2 \sin(x) \sin\left(\frac{x^3}{12}\right)}{x \times \frac{x^3}{12} \times 12}$$

Considering, $\sin x = x - \frac{x^3}{3!} + \dots$

So,

$$\lim_{x \rightarrow 0} 2 \left(\frac{\sin x}{x} \right) \times \frac{\sin \left(\frac{x^3}{12} \right)}{\frac{x^3}{12}} \times \frac{1}{12} = 2 \times 1 \times 1 \times \frac{1}{12} = \frac{1}{6}$$

65. For A, Let,

$$y = \sin^{-1} \left(\frac{2x}{1+x^2} \right)$$

$$\begin{aligned} \text{Put } x = \tan \theta &= \sin^{-1} \left(\frac{2 \tan \theta}{1 + \tan^2 \theta} \right) \\ &= \sin^{-1} (\sin 2\theta) \\ &= 2\theta \end{aligned}$$

This implies,

$$\begin{aligned} y &= 2 \tan^{-1} x \\ \frac{dy}{dx} &= \frac{2}{1+x^2} \end{aligned}$$

So, $A \rightarrow \text{III}$

For B, let,

$$\tan^{-1}\left(\frac{1-x}{1+x}\right) = y$$

$$\tan^{-1}(1) - \tan^{-1}(x) = y$$

$$y = \frac{-1}{1+x^2}$$

This implies, $B \rightarrow \Pi$

For C,

$$\begin{aligned} e^{\log(\sin x + \cos x)} &= \sin x + \cos x \\ &= y \end{aligned}$$

Differentiate above,

$$\frac{dy}{dx} = \cos x - \sin x$$

This implies, $C \rightarrow \text{I}$

66. Consider the function,

$$f(t) = \frac{1 + \operatorname{cosec} t}{1 - \operatorname{cosec} t}$$

Differentiate w.r.t t ,

$$\begin{aligned}
 f'(t) &= \frac{-(1 - \operatorname{cosec} t)(\operatorname{cosec} t \cot t)}{(1 - \operatorname{cosec} t)^2} \\
 &= -\operatorname{cosec} t \cot t - \operatorname{cosec}^2 t \cot t \\
 &= \frac{-\operatorname{cosec} t \cot t - \operatorname{cosec}^2 t \cot t}{(1 - \operatorname{cosec} t)^2} \\
 &= \frac{-\operatorname{cosec} t \cot t}{(1 - \operatorname{cosec} t)^2} \times \frac{1 + \operatorname{cosec} t}{1 + \operatorname{cosec} t}
 \end{aligned}$$

Further simplify the above,

$$\begin{aligned}
 f'(t) &= \left(\frac{1 + \operatorname{cosec} t}{1 - \operatorname{cosec} t} \right) \left(-\frac{2 \operatorname{cosec} t \cot t}{1 - \operatorname{cosec}^2 t} \right) \\
 &= f(t) \frac{2 \operatorname{cosec} t \cot t}{\cot^2 t} \\
 &= f(t) \frac{2 \operatorname{cosec} t}{\cot^2 t} \\
 &= f(t) \frac{2}{\sin t \cos t} \\
 &= f(t) (4 \operatorname{cosec} 2t) \\
 &= f(t) g(t)
 \end{aligned}$$

So,

$$g(t) = 4 \operatorname{cosec} 2t$$

67. Consider the expression,

$$x = 4\cos^3 \theta$$

Differentiate above w.r.t θ ,

$$\begin{aligned}\frac{dx}{d\theta} &= 4 \times 3\cos^2 \theta (-\sin \theta) \\ &= -12\cos^2 \theta \sin \theta\end{aligned}$$

And,

$$y = 3\sin^2 \theta$$

Differentiate w.r.t θ ,

$$\begin{aligned}\frac{dy}{d\theta} &= 3 \times 2\sin \theta (\cos \theta) \\ &= 6\sin \theta \cos \theta\end{aligned}$$

Now,

$$\begin{aligned}\frac{dy}{dx} &= \frac{\frac{dy}{d\theta}}{\frac{dx}{d\theta}} \\ &= -\frac{1}{2}\sec \theta\end{aligned}$$

Then,

$$\begin{aligned}\frac{d^2 y}{dx^2} &= -\frac{1}{2} \sec \theta \tan \theta \cdot \frac{d\theta}{dx} \\&= -\frac{1}{2} \sec \theta \tan \theta \left(\frac{-1}{12 \cos^2 \theta \sin \theta} \right) \\&= \frac{1}{24} \left(\frac{1}{\cos \theta} \right) \left(\frac{\sin \theta}{\cos \theta} \right) \left(\frac{1}{\cos^2 \theta \sin \theta} \right) \\&= \frac{1}{24 (\cos \theta)^4}\end{aligned}$$

Further simplify the above,

$$\begin{aligned}\frac{d^2 y}{dx^2} &= \frac{1}{24 \left(\cos \frac{\pi}{4} \right)^4} \\&= \frac{1}{24} \left(\frac{1}{\left(\frac{1}{\sqrt{2}} \right)^4} \right) \\&= \frac{1}{24} (4) \\&= \frac{1}{6}\end{aligned}$$

68. Consider the expression,

$$y = \sin x$$

Differentiate above w.r.t x ,

$$\frac{dy}{dx} = \cos x$$

If tangent to $y = \sin x$ meet at (h, k) then,

$$\left(\frac{dy}{dx}\right)_{(h,k)} = \cos h$$

The equation of tangent is given by,

$$\cos h(x - h) = y - k$$

As tangent is passing through origin so,

$$h \cos h = k$$

The given curve is passing through (h, k) so,

$$k = \sin h$$

$$\cos h = \sqrt{1 - k^2}$$

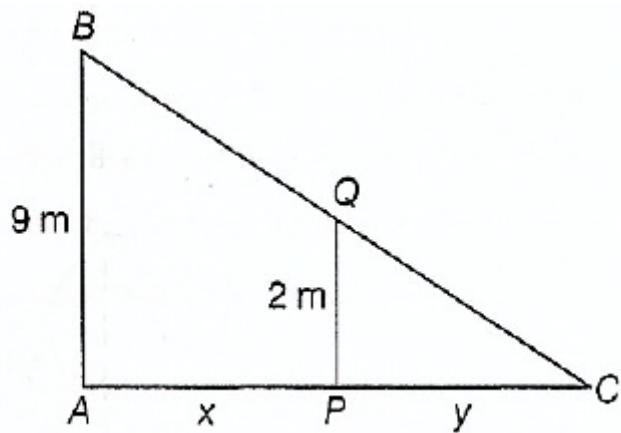
$$\cos^2 h = 1 - k^2$$

$$h^2 - k^2 = h^2 k^2$$

So, the locus of point of contact is,

$$x^2 - y^2 = x^2 y^2$$

69. Consider the diagram,



From the above diagram,

$$\frac{PC}{AC} = \frac{PQ}{AB}$$

$$\frac{y}{x+y} = \frac{2}{9}$$

$$9y = 2x + 2y$$

$$7y = 2x$$

$$x = \frac{7}{2}y$$

Differentiate above equation,

$$\frac{dx}{dt} = \frac{7}{2} \frac{dy}{dt}$$

It is given that $\frac{dx}{dt} = 7 \text{ m/min}$

So,

$$7 = \frac{7}{2} \frac{dy}{dt}$$

$$\frac{dy}{dt} = 2 \text{ m/min}$$

Hence, the length of shadow increases at 2 m/min.

70. Consider the expressions,

$$2a + 3b + 6c = 0 \quad (\text{I})$$

And,

$$g(x) = ax^2 + bx + c = 0$$

According to the given information,

$$\begin{aligned} f(x) &= \int g(x) dx \\ &= \int (ax^2 + bx + c) dx \\ &= \frac{a}{3}x^3 + \frac{b}{2}x^2 + cx \end{aligned}$$

Then,

$$\begin{aligned} f(2) &= \frac{a}{3}(2)^3 + \frac{b}{2}(2)^2 + c(2) \\ &= \frac{8}{3}a + 2b + 2c \end{aligned} \quad (\text{II})$$

And,

$$\begin{aligned}
 f(1) &= \frac{a}{3}(1)^3 + \frac{b}{2}(1)^2 + c(1) \\
 &= \frac{a}{3} + \frac{b}{2} + c
 \end{aligned}
 \tag{III}$$

Here, $f(2) = f(1)$

So,

$$\begin{aligned}
 \frac{8}{3}a + 2b + 2c &= \frac{a}{3} + \frac{b}{2} + c \\
 14a + 9b + 6c &= 0
 \end{aligned}
 \tag{IV}$$

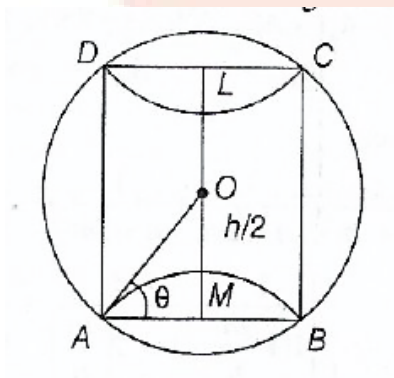
From equation (I) and (IV),

$$a = 3, b = -6 \text{ and } c = 2$$

Thus,

$$\begin{aligned}
 f(x) &= \frac{3}{3}x^3 - \frac{6}{2}x^2 + 2x \\
 &= x^3 - 3x^2 + 2x
 \end{aligned}$$

71. Consider the diagram,



In $\triangle OAM$,

$$r^2 + \left(\frac{h}{2}\right)^2 = 3^2$$

$$r^2 + \frac{h^2}{4} = 9$$

$$r^2 = 9 - \frac{h^2}{4}$$

The volume of the cylinder is given by,

$$V = \pi r^2 h$$

$$= \pi \left(9 - \frac{h^2}{4}\right) h$$

$$= \pi \left(9h - \frac{h^3}{4}\right)$$

Differentiate above with respect to h .

$$\frac{dV}{dh} = \left(9 - \frac{3h^2}{4}\right) \pi$$

For maximum or minimum volume,

$$\frac{dV}{dh} = 0$$

$$9 - \frac{3h^2}{4} = 0$$

$$\frac{3h^2}{4} = 9$$

$$h^2 = 12$$

$$h = \pm 2\sqrt{3}$$

So,

$$r^2 = 9 - 3$$

$$r^2 = 6$$

The maximum volume of cylinder is,

$$V_{\max} = \pi(6)(2\sqrt{3})$$

$$= 12\sqrt{3}\pi \text{ sq units}$$

$$\left[\because \frac{d^2V}{dh^2} = -\frac{3h}{2} < 0 \right]$$

72. Consider the expression,

$$\begin{aligned} I &= \int \frac{x}{\sqrt{x+1} + \sqrt{x-1}} dx \\ &= \int \frac{x[\sqrt{x+1} - \sqrt{x-1}]}{x+1 - x+1} dx \\ &= \frac{1}{2} \int x\sqrt{x+1} dx - \frac{1}{2} \int x\sqrt{x-1} dx \\ &= \frac{1}{2} I_1 - \frac{1}{2} I_2 \end{aligned}$$

Then,

$$I_1 = \int x\sqrt{x+1} dx$$

Put $x+1 = u \Rightarrow dx = du$

$$\begin{aligned} I_1 &= \int (u-1)\sqrt{u} du \\ &= \int \left(u^{\frac{3}{2}} - u^{\frac{1}{2}} \right) du \\ &= \frac{2}{5} u^{\frac{5}{2}} - \frac{2}{3} u^{\frac{3}{2}} + c_1 \\ &= \frac{2}{5} (x+1)^{\frac{5}{2}} - \frac{2}{3} (x+1)^{\frac{3}{2}} + c_1 \end{aligned}$$

Further simplify the above,

$$\begin{aligned} I_1 &= 2(x+1)^{\frac{3}{2}} \left[\frac{3x+3-5}{15} \right] + c_1 \\ &= \frac{2(3x-2)}{15} (x+1)^{\frac{3}{2}} + c_1 \end{aligned}$$

Now,

$$I_2 = \int x\sqrt{x-1} dx$$

Put $x-1=v \Rightarrow dx = dv$

$$\begin{aligned} I_2 &= \int (v+1)\sqrt{v} dv \\ &= \int \left(v^{\frac{3}{2}} + v^{\frac{1}{2}} \right) dv \\ &= \frac{2}{5} v^{\frac{5}{2}} + \frac{2}{3} v^{\frac{3}{2}} + c_2 \\ &= 2(x-1)^{\frac{3}{2}} \left[\frac{1}{5}(x-1) + \frac{1}{3} \right] + c_2 \\ &= 2(x-1)^{\frac{3}{2}} \left[\frac{3x-3+5}{15} \right] + c_2 \end{aligned}$$

Further simplify the above,

$$I_2 = \frac{2(3x+2)}{15} (x-1)^{\frac{3}{2}} + c_2$$

This gives,

$$\begin{aligned}
 I &= \frac{1}{2}I_1 - \frac{1}{2}I_2 \\
 &= \frac{(3x-2)}{15}(x+1)^{\frac{3}{2}} - \frac{(3x+2)}{15}(x-1)^{\frac{3}{2}} + c \quad [\because c = c_1 + c_2]
 \end{aligned}$$

This implies,

$$A(x) = \frac{1}{15}(3x-2) \text{ and } B(x) = \frac{-1}{15}(3x+2)$$

So,

$$\begin{aligned}
 A(x) + B(x) &= \frac{1}{15}(3x-2) - \frac{1}{15}(3x+2) \\
 &= \frac{-4}{15}
 \end{aligned}$$

73. Consider the integral,

$$I = \int \frac{x \log x}{\left(\sqrt{x^2-1}\right)^3} dx$$

Let,

$$\sqrt{x^2-1} = t$$

$$\frac{2x dx}{2\sqrt{x^2-1}} = dt$$

$$\frac{x dx}{\sqrt{x^2-1}} = dt$$

So,

$$\begin{aligned}
 I &= \int \frac{\log x dt}{t^2} \\
 &= \int \frac{\log(\sqrt{1+t^2})}{t^2} dt \\
 &= \int \frac{\log(\sqrt{1+t^2})}{t^2} dt \\
 &= -\frac{1}{t} \log \sqrt{1+t^2} - \int \frac{2t}{2\sqrt{1+t^2}} \left(-\frac{1}{t}\right) \frac{1}{\sqrt{1+t^2}} dt
 \end{aligned}$$

Further simplify the above,

$$\begin{aligned}
 I &= -\frac{1}{t} \log \sqrt{1+t^2} + \int \frac{dt}{1+t^2} \\
 &= -\frac{1}{t} \log \sqrt{1+t^2} + \tan^{-1}(t) + c \\
 &= \sec^{-1}(x) - \frac{\log x}{\sqrt{x^2-1}} + c
 \end{aligned}$$

74. Consider the expression,

$$I = 4 \int \frac{\cos 2x (2 \sin 2x \cos 2x)}{(1 + \cos 2x)^2 (1 + \cos^2 2x)} dx$$

Put $\cos 2x = t$ then, $-2 \sin 2x dx = dt$

Thus,

$$I = -4 \int \frac{t^2}{(1+t)^2 (1+t^2)} dt$$

Split the above into partial fractions,

$$\begin{aligned} I &= -4 \int \left[\frac{1}{2(1+t)^2} - \frac{1}{2(1+t)} + \frac{t}{2(1+t^2)} \right] dt \\ &= -2 \left[-\frac{1}{1+t} - \log(1+t) + \frac{1}{2} \log(1+t^2) \right] \\ &= \frac{2}{2 \cos^2 x} + 2 \log(1 + \cos 2x) - \log(1 + \cos^2 2x) + c \\ &= \sec^2 x + 2 \log(1 + \cos 2x) - \log(1 + \cos^2 2x) + c \\ &= \sec^2 x + \log \frac{(1 + \cos 2x)^2}{(1 + \cos^2 2x)} + c \end{aligned}$$

75. From the given expression,

$$\begin{aligned} A_6 &= \int_{\frac{\pi}{2}}^{\infty} e^{-x} \cos^6 x dx \\ &= \left[-e^{-x} \cos^6 x \right]_{\frac{\pi}{2}}^{\infty} - \int_{\frac{\pi}{2}}^{\infty} (-e^{-x}) 6 \cos^5 x (-\sin x) dx \\ &= 0 - 6 \int_{\frac{\pi}{2}}^{\infty} e^{-x} \cos^5 x \sin x dx \end{aligned}$$

This gives,

$$\frac{A_6}{-6} = \int_{\frac{\pi}{2}}^{\infty} e^{-x} \cos^5 x \sin x dx$$

$$\begin{aligned} -\frac{1}{6} A_6 &= \left(\left[-e^{-x} \cos^5 x \sin x \right]_{\frac{\pi}{2}}^{\infty} - \int_{\frac{\pi}{2}}^{\infty} (-e^{-x}) \left[\begin{array}{l} 5 \cos^4 x (-\sin x) \sin x \\ + \cos^5 x \cdot \cos x \end{array} \right] dx \right) \\ &= 0 + \int_{\frac{\pi}{2}}^{\infty} (e^{-x}) \left[-5 \cos^4 x (1 - \cos^2 x) + \cos^6 x \right] dx \\ &= -5 \int_{\frac{\pi}{2}}^{\infty} (e^{-x}) [\cos^4 x] dx + 6 \int_{\frac{\pi}{2}}^{\infty} (e^{-x}) [\cos^6 x] dx \end{aligned}$$

This implies,

$$-\frac{1}{6} A_6 = -5 A_4 + 6 A_6$$

$$5 A_4 = \frac{37}{6} A_6$$

$$A_6 = \frac{30}{37} A_4$$

Now,

$$\begin{aligned} \frac{A_4 - A_6}{A_4} &= \frac{A_4 - \frac{30}{37} A_4}{A_4} \\ &= \frac{7}{37} \end{aligned}$$

76. Consider the expression,

$$\begin{aligned}
 & \lim_{x \rightarrow \infty} \left\{ \frac{1}{n+m} + \frac{1}{n+2m} + \frac{1}{n+3m} + \dots + \frac{1}{n+nm} \right\} \\
 &= \lim_{x \rightarrow \infty} \left\{ \frac{1}{1+\frac{m}{n}} + \frac{1}{1+2\left(\frac{m}{n}\right)} + \frac{1}{1+3\left(\frac{m}{n}\right)} + \dots + \frac{1}{1+n\left(\frac{m}{n}\right)} \right\} \\
 &= \lim_{x \rightarrow \infty} \frac{1}{n} \left(\sum_{k=1}^n \frac{1}{1+m\left(\frac{k}{n}\right)} \right) \\
 &= \int_0^1 \frac{1}{1+mx} dx \\
 &= \frac{1}{m} \int_0^1 \frac{m}{1+mx} dx \\
 &= \frac{1}{m} \left[\log_e (1+mx) \right]_0^1 \\
 &= \frac{1}{m} \left[\log_e (1+m) - \log(1) \right] \\
 &= \frac{\log_e (1+m)}{m}
 \end{aligned}$$

77. Let us consider,

$$I = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{\cos x}{1 + e^x} dx \quad \dots\dots (I)$$

Further simplify the above expression,

$$\begin{aligned} I &= \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{\cos x}{1 + e^x} dx \\ &= \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{\cos x}{1 + e^x} dx \\ &= \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{\cos\left(\frac{\pi}{2} - \frac{\pi}{2} - x\right)}{1 + e^{\frac{\pi}{2} - \frac{\pi}{2} - x}} dx \quad \left[\because \int_a^b f(x) dx = \int_a^b f(a + b - x) dx \right] \\ &= \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{\cos(x)}{1 + e^{-x}} dx \\ &= \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{e^x \cos x}{(1 + e^x)} dx \\ &\quad (II) \end{aligned}$$

Add equation (I) and (II).

$$\begin{aligned} 2I &= \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{e^x \cos x}{(1+e^x)} dx + \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{\cos x}{(1+e^x)} dx \\ &= \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{(e^x + 1) \cos x}{(1+e^x)} dx \\ &= \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \cos x dx \\ &= [\sin x]_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \end{aligned}$$

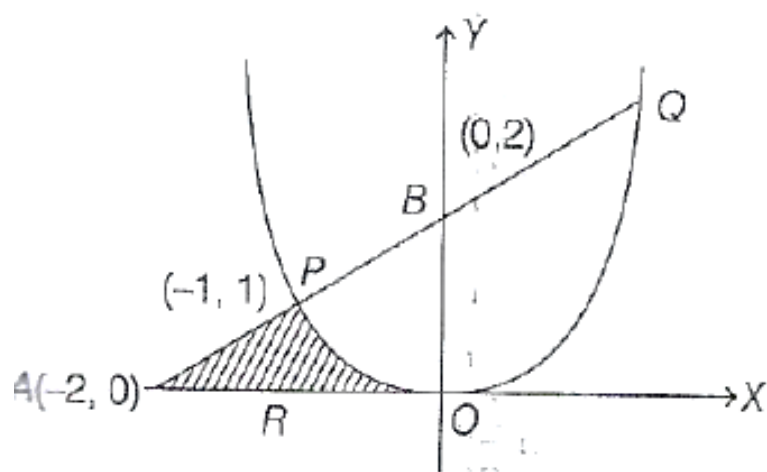
Applying the limits, we get

$$\begin{aligned} 2I &= [1 + 1] \\ &= 2 \\ I &= 1 \end{aligned}$$

Now the value of given function is,

$$\begin{aligned} \tan^{-1} \left[\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{\cos x}{1+e^x} dx \right] &= \tan^{-1}(1) \\ &= \tan^{-1} \left(\tan \frac{\pi}{4} \right) \\ &= \frac{\pi}{4} \end{aligned}$$

78. Consider the diagram,



The required area is given by,

$$\begin{aligned}
 \int_{-2}^{-1} y_{\text{line}} dx + \int_{-1}^0 y_{\text{parabola}} dx &= \int_{-2}^{-1} (x+2) dx + \int_{-1}^0 x^2 dx \\
 &= \left[\frac{x^2}{2} + 2x \right]_{-2}^{-1} + \left[\frac{1}{3} x^3 \right]_{-1}^0 \\
 &= \frac{1}{2} + \frac{1}{3} \\
 &= \frac{5}{6} \text{ sq. units}
 \end{aligned}$$

79. Consider the equation,

$$y = (\alpha + \beta + \gamma)x$$

Differentiate w.r.t x ,

$$\begin{aligned}
 \frac{dy}{dx} &= (\alpha + \beta + \gamma) \\
 &= k
 \end{aligned}$$

The order of above differential equation is 1 so. Statement I is false.

Consider the equation,

$$y = \alpha x + \beta \sin x + \gamma e^x$$

Differentiate both sides w.r.t x .

$$\frac{dy}{dx} = \alpha + \beta \cos x + \gamma e^x \quad \dots\dots (I)$$

$$\frac{d^2 y}{dx^2} = -\beta \sin x + \gamma e^x \quad \dots\dots (II)$$

And,

$$\frac{d^3 y}{dx^3} = -\beta \cos x + \gamma e^x \quad \dots\dots (III)$$

From equation (I) and (II), we get

$$\begin{aligned} \frac{d^3 y}{dx^3} - \frac{d^2 y}{dx^2} &= -\beta \cos x + \gamma e^x + \beta \sin x - \gamma e^x \\ &= \beta \sin x - \beta \cos x \\ &= \beta (\sin x - \cos x) \end{aligned}$$

The order of above differential equation is 3 so. Statement II is true.

80. Consider the given differential equation.

$$\sin y \frac{dy}{dx} = \cos y (1 - x \cos y)$$

$$\sin y \frac{dy}{dx} = \cos y - x \cos^2 y$$

$$\frac{\sin y}{\cos^2 y} \frac{dy}{dx} = \frac{1}{\cos y} - x$$

$$\sec y \tan y \frac{dy}{dx} = \sec y - x$$

Put,

$$\sec y = t$$

$$\sec y \tan y \frac{dy}{dx} = \frac{dt}{dx}$$

So,

$$\frac{dt}{dx} = t - x$$

$$\frac{dt}{dx} + (-t) = (-x)$$

The integrating factor is given by,

$$\begin{aligned} \text{IF} &= e^{\int -dx} \\ &= e^{-x} \end{aligned}$$

So, the required solution is,

$$t(\text{IF}) = \int (-x)(\text{IF})dx + c$$

$$\sec y(e^{-x}) = \int (-x)(e^{-x})dx + c$$

$$\sec y = x + 1 + ce^x$$



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Solutions – Physics

81. The expression of the density is,

$$\rho = \frac{m}{V}$$

$$\rho = \frac{m}{l^3}$$

The length from the above expression is,

$$l = \left(\frac{m}{\rho} \right)^{\frac{1}{3}}$$

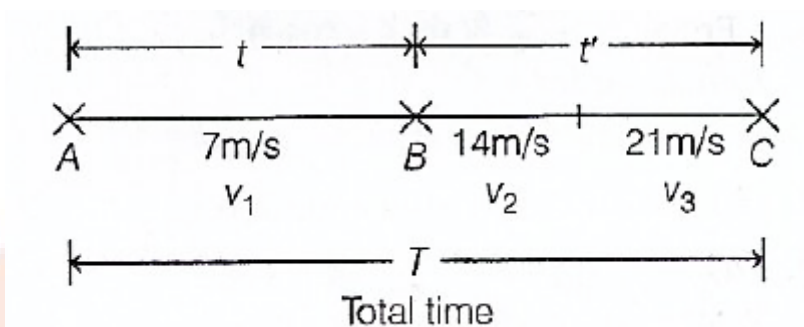
The error in measurement of length can be calculated as,

$$\frac{\Delta l}{l} \times 100 = \frac{1}{3} \left(\frac{\Delta m}{m} \times 100 + \frac{\Delta \rho}{\rho} \times 100 \right)$$

$$\frac{\Delta l}{l} \times 100 = \frac{1}{3} (1 \times 100 + 2 \times 100)$$

$$\frac{\Delta l}{l} = 1\%$$

82. Consider the diagram,



Given, $v_{AB} = 7 \text{ m/s}$, $(v_1)_{BC} = 14 \text{ m/s}$, $(v_2)_{BC} = 21 \text{ m/s}$

The distance between A and B is,

$$d_{AB} = v_{AB}t$$

$$d_{AB} = 7t$$

The distance between B and C is,

$$d_{BC} = (v_{BC})_{avg} t'$$

$$d_{BC} = \left[\frac{(v_{BC})_1 + (v_{BC})_2}{2} \right] t'$$

$$d_{BC} = \left[\frac{14 + 21}{2} \right] t'$$

$$d_{BC} = \frac{35}{2} t'$$

According to question,

$$d_{AB} = d_{BC}$$

$$7t = \frac{35}{2}t'$$

$$t' = \frac{2}{5}t$$

The total time taken from A to C is,

$$T = t + t'$$

$$T = t + \frac{2t}{5}$$

$$T = \frac{7t}{5}$$

The distance AC is,

$$d_{AC} = vt$$

$$d_{AC} = \left[\frac{v_{AB} + (v_{BC})_1 + (v_{BC})_2}{3} \right] t$$

$$d_{AC} = \left[\frac{7 + 14 + 21}{3} \right] t$$

$$d_{AC} = 14t$$

The average velocity during the whole journey is,

$$v_{avg} = \frac{d_{AC}}{T}$$

$$v_{avg} = \frac{14t}{7t} \times 5$$

$$v_{avg} = 10 \text{ m/s}$$

83. The expression of the range of the projectile is,

$$R = \frac{u^2 \sin 2\theta}{g}$$

The maximum range of the projectile occurs at 45° angle of projection.

Thus,

$$R_{\max} = \frac{u^2 \sin(2 \times 45^\circ)}{g}$$

$$R_{\max} = \frac{u^2}{g}$$

Thus, assertion is true but reason is false.

84. The expression of the maximum height of projectile is,

$$H = \frac{u^2 \sin^2 \theta}{2g} \quad \text{..... (I)}$$

The equation (I) for 80 m height is,

$$80 = \frac{u^2 \sin^2 \theta_1}{2g} \quad \dots\dots (II)$$

The equation (I) for 45 m height is,

$$45 = \frac{u^2 \sin^2 \theta_2}{2g} \quad \dots\dots (III)$$

From equation (II) and (III)

$$\frac{80}{40} = \frac{\sin^2 \theta_1}{\sin^2 \theta_2}$$

$$\frac{\sin \theta_1}{\sin \theta_2} = \frac{4}{3}$$

$$\frac{\sin \theta_1}{\sin \theta_2} = \frac{\frac{4}{5}}{\frac{3}{5}}$$

Thus,

$$\sin \theta_1 = \frac{4}{5} \Rightarrow \theta_1 = 53^\circ$$

$$\sin \theta_2 = \frac{3}{5} \Rightarrow \theta_2 = 37^\circ$$

From equation (II),

$$80 = \frac{u^2 \sin^2 53^\circ}{2g}$$

$$\frac{u^2}{g} = 250$$

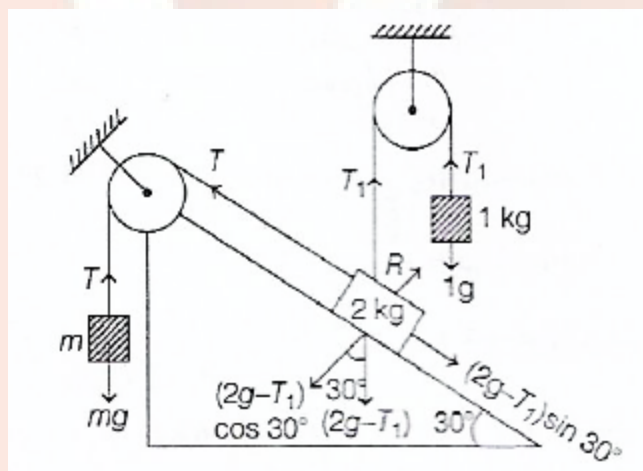
The width of the river is calculated as,

$$d = \left(\frac{u^2}{g} \right) \sin 2\theta_1$$

$$d = (250) \sin(2 \times 53^\circ)$$

$$d \approx 240 \text{ m}$$

85. Consider the free body diagram of the system,



The equilibrium equations for the system is,

$$T = mg \quad \text{..... (I)}$$

$$T = (2g - T_1) \sin 30^\circ \quad \text{..... (II)}$$

$$T_1 = 1g \quad \text{..... (III)}$$

Substitute $T_1 = 1g$ in equation (II),

$$T = (2g - 1g) \sin 30^\circ$$

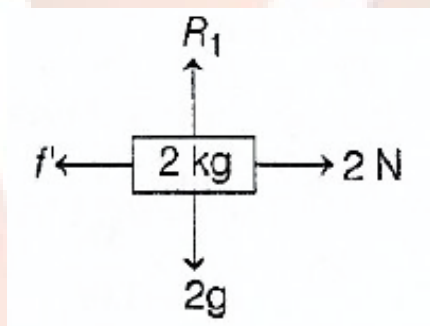
$$T = \frac{g}{2}$$

From equation (I),

$$\frac{g}{2} = mg$$

$$m = 0.5 \text{ kg}$$

86. The free body diagram of 2 kg block is shown below,



The equilibrium equation for the block is,

$$f' = \mu R_1$$

$$f' = (0.5)2g$$

$$f' = 10 \text{ N}$$

Since $f' > 2 \text{ N}$, so 2 kg block will move along the direction of 4 kg body.

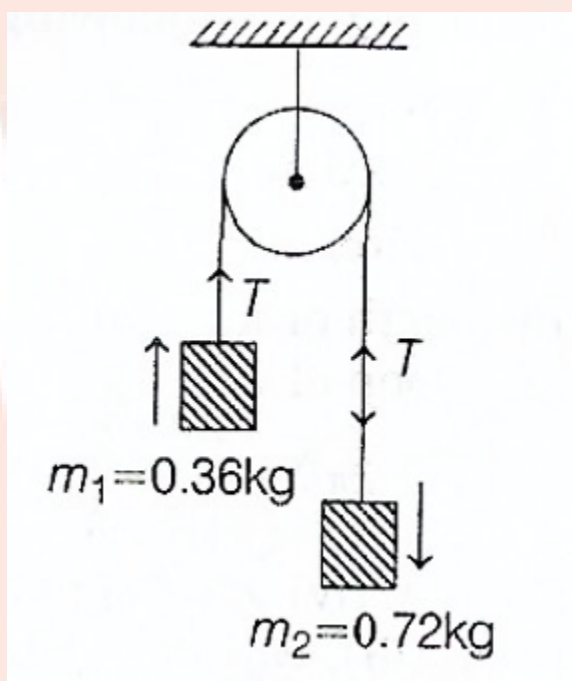
The net friction force on 2 kg block is,

$$f = f' - 2$$

$$f = 10 - 2$$

$$f = 8 \text{ N}$$

87. Consider the diagram,



The dynamic equation for mass m_1 is,

$$T - m_1 g = m_1 a \quad \text{..... (I)}$$

The dynamic equation for mass m_2 is,

$$m_2 g - T = m_2 a \quad \text{..... (II)}$$

From equation (I) and (II), we get

$$T - m_1g + m_2g - T = m_1a + m_2a$$

$$a = \frac{(m_2 - m_1)g}{(m_1 + m_2)}$$

$$a = \frac{(0.72 - 0.36)(10)}{(0.72 + 0.36)}$$

$$a = \frac{3.6}{1.08} \text{ m/s}^2$$

Substitute values in equation (I),

$$T - (0.36)(10) = (0.36)\left(\frac{3.6}{1.08}\right)$$

$$T = 4.8 \text{ N}$$

The distance travelled by mass m_1 is,

$$s = ut + \frac{1}{2}at^2$$

$$s = (0)(1) + \frac{1}{2}\left(\frac{3.6}{1.08}\right)(1)^2$$

$$s = \frac{1}{2}\left(\frac{3.6}{1.08}\right)(1)^2$$

$$s = \frac{1.8}{1.08}$$

The required work done is,

$$W = T \times s$$

$$W = (4.8) \times \left(\frac{1.8}{1.08} \right)$$

$$W = 8 \text{ J}$$

88. Let the mass of the man is m and speed is v_1 , thus the mass of the boy is $\frac{m}{2}$ and let his initial velocity is v_2 .

According to given condition,

$$(KE)_{\text{man}} = \frac{1}{2}(KE)_{\text{boy}}$$

$$\frac{1}{2}mv_1^2 = \frac{1}{2}\left(\frac{1}{2}\frac{m}{2}v_2^2\right)$$

$$\frac{v_1^2}{v_2^2} = \frac{1}{4}$$

$$v_1 = \frac{v_2}{2}$$

After the increase in velocity of man by 1 m/s, the new velocity of the man is $v'_1 = \left(\frac{v_2}{2} + 1 \right)$

According to given condition,

$$(KE)'_{\text{man}} = (KE)_{\text{boy}}$$

$$\frac{1}{2}mv_1'^2 = \frac{1}{2}\left(\frac{m}{2}\right)v_2^2$$

$$v_1'^2 = \frac{v_2^2}{2}$$

$$v_1' = \frac{v_2}{\sqrt{2}}$$

Substituting the values, we get

$$\left(\frac{v_2}{2} + 1\right) = \frac{v_2}{\sqrt{2}}$$

$$\sqrt{2}v_2 + 2\sqrt{2} = 2v_2$$

$$v_2 = \frac{2}{(\sqrt{2} - 1)}$$

$$v_2 = 2(\sqrt{2} + 1)$$

89. Given inclination of the plane is,

$$\theta = \sin^{-1}(0.42)$$

$$\sin \theta = 0.42$$

The expression of the acceleration of rolling solid sphere on the inclined plane without slipping is,

$$a = \frac{g \sin \theta}{1 + \frac{I}{MR^2}}$$

$$a = \frac{g \sin \theta}{1 + \frac{\frac{2}{5}MR^2}{MR^2}}$$

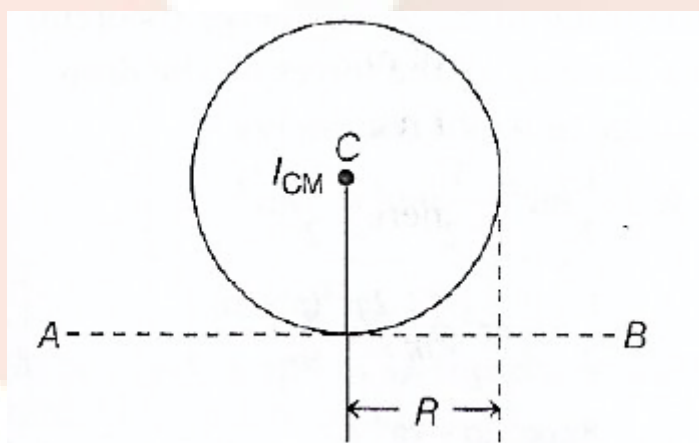
$$a = \frac{5}{7} g \sin \theta$$

Substituting the values, we get

$$a = \frac{5}{7}(10)(0.42)$$

$$a \approx 3 \text{ m/s}^2$$

90. Consider the figure below.



The total moment of inertia for circular loop of wire is given as,

$$I = I_{AB} + I_{CM}$$

$$I = MR^2 + \frac{1}{2}MR^2$$

$$I = \frac{3}{2}MR^2 \quad \text{..... (I)}$$

The length of the wire is,

$$l = 2\pi R$$

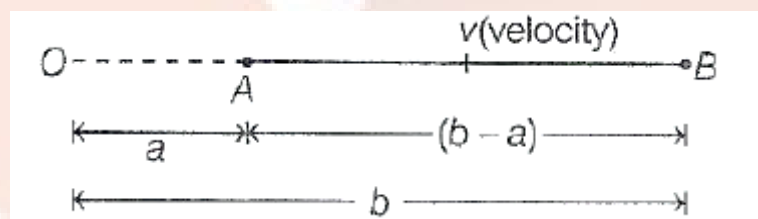
$$R = \frac{l}{2\pi}$$

From equation (I),

$$I = \frac{3}{2}(\rho l) \left(\frac{l}{2\pi} \right)^2$$

$$I = \frac{3\rho l^3}{8\pi^2}$$

91. Consider the below diagram,



The expression of the amplitude of particle along AB is,

$$A = \frac{b - a}{2}$$

The angular velocity of the particle is,

$$v = A\omega$$

$$v = \left(\frac{b-a}{2} \right) \omega$$

$$\omega = \frac{2v}{b-a}$$

The time period is calculated as,

$$T = \frac{2\pi}{\omega}$$

$$T = \frac{2\pi}{\frac{2v}{b-a}}$$

$$T = \left(\frac{b-a}{v} \right) \pi$$

92. Given the escape velocity on the surface of earth is,

$$v_e = 11.2 \text{ km/s}$$

$$\text{or, } \sqrt{2gR_e} = 11.2$$

$$\text{Also, } h = \frac{R_e}{3}$$

For the equilibrium, the potential energy at the altitude $\frac{R_e}{3}$

should be equal to the kinetic energy.

Thus,

$$\frac{GM_e m}{R_e + h} = \frac{1}{2} m v_{e1}^2$$

$$\frac{GM_e}{R_e + \frac{R_e}{3}} = \frac{1}{2} v_{e1}^2$$

$$\frac{\frac{gR_e^2}{4R_e}}{3} = \frac{1}{2} v_{e1}^2 \quad \left(\text{As, } g = \frac{GM_e}{R_e^2} \right)$$

$$\frac{3}{4} g R_e = \frac{v_{e1}^2}{2}$$

Solve further,

$$\frac{3}{4} (2gR_e) = v_{e1}^2$$

$$\frac{\sqrt{3}}{2} \sqrt{2gR_e} = v_{e1}$$

$$\frac{\sqrt{3}}{2} (11.2) = v_{e1}$$

$$v_{e1} = 9.7 \text{ km/s}$$

93. Let the equivalent Young's modulus of the combination is Y and the equivalent area of the combination will be $2A$.

For parallel combination,

$$k_{eq} = k_1 + k_2$$

$$\frac{Y(2A)}{L} = \frac{Y_1 A}{L} + \frac{Y_2 A}{L}$$

$$Y = \frac{Y_1 + Y_2}{2}$$

94. The initial velocity of rain drop will be zero, so $u = 0$.

The expression of the final velocity of rain drop is,

$$v = \frac{2gr^2(\rho - \sigma)}{9\eta}$$

By work energy theorem,

$$W = \frac{1}{2}mv^2 - \frac{1}{2}mu^2$$

$$W = \frac{1}{2}mv^2 - \frac{1}{2}m(0)^2$$

$$W = \frac{1}{2}(\rho V)v^2$$

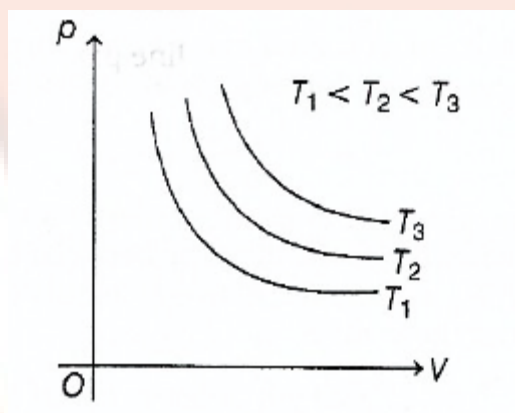
$$W = \frac{1}{2}\rho\left(\frac{4}{3}\pi r^3\right)v^2$$

Substitute the value of v in the above expression.

$$W = \frac{1}{2} \rho \left(\frac{4}{3} \pi r^3 \right) \left[\frac{2gr^2(\rho - \sigma)^2}{9\eta} \right]^2$$

$$W \propto r^7$$

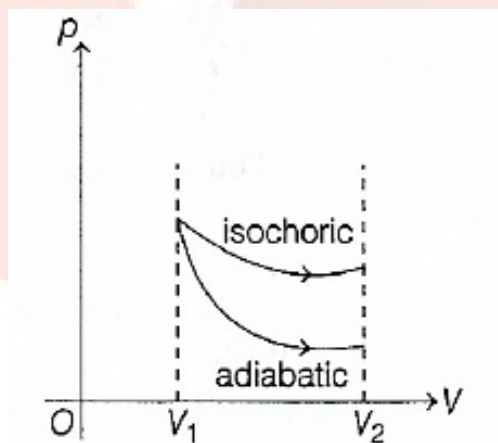
95. The isothermal process on PV graph as shown below.



For isothermal process, the slope is

$$\frac{dP}{dV} = -\frac{P}{V}$$

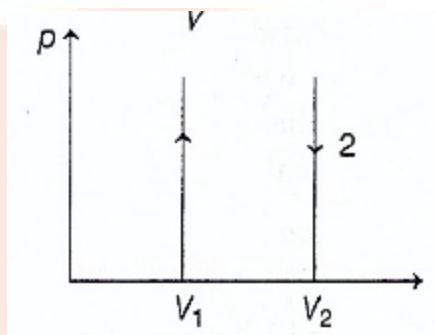
The adiabatic process on PV graph as shown below.



For adiabatic process, the slope is

$$\frac{dP}{dV} = -\frac{\gamma P}{V}$$

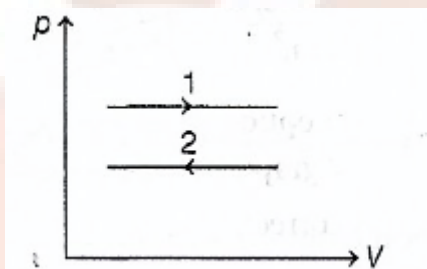
The isochoric process on PV graph as shown below.



For isochoric process, the slope is

$$\frac{dP}{dV} = \infty$$

The isobaric process on PV graph as shown below.



For isobaric process, the slope is

$$\frac{dP}{dV} = 0$$

Thus option (3) is correct.

96. By mass and density relation,

$$m = \rho V$$

$$m = \rho \left(\frac{4}{3} \pi r^3 \right)$$

$$r = \left(\frac{3m}{4\pi\rho} \right)^{\frac{1}{3}}$$

The area is calculated as,

$$A = 4\pi r^2$$

$$A = 4\pi \left(\frac{3m}{4\pi\rho} \right)^{\frac{2}{3}}$$

From Stefan's law,

$$\frac{\Delta Q}{\Delta t} = e\sigma AT^4$$

$$\frac{mc\Delta T}{\Delta t} = e\sigma AT^4$$

$$\frac{\Delta T}{\Delta t} = \frac{e\sigma AT^4}{mc}$$

$$\frac{\Delta T}{\Delta t} = \frac{4e\sigma\pi T^4}{mc} \left(\frac{3m}{4\pi\rho} \right)^{\frac{2}{3}}$$

$$\frac{\Delta T}{\Delta t} \propto \left(\frac{1}{m} \right)^{\frac{1}{3}}$$

The ratio of cooling rate of Q and P is,

$$\frac{\left(\frac{\Delta T}{\Delta t} \right)_Q}{\left(\frac{\Delta T}{\Delta t} \right)_P} = \left(\frac{m_P}{n_Q} \right)^{\frac{1}{3}}$$

$$= \left(\frac{8m_Q}{n_Q} \right)^{\frac{1}{3}}$$

$$= 2$$

97. The expression of the net work is,

$$W_{net} = nR \left[(T_B - T_A) + (T_C - T_B) + (T_D - T_C) - (T_D - T_A) \right]$$

Substitute values.

$$W_{net} = 3R[0 + (2400 - 800) + 0 - (1200 - 400)]$$

$$W_{net} = 2400R$$

98. The expression of the heat supplied is,

$$Q = u_f - u_i$$

$$Q = (u_{4 \text{ moles}} + u'_{2 \text{ moles}}) - u'_{4 \text{ moles}}$$

Substitute values.

$$Q = \left(4 \times \frac{3}{2} RT + 2 \times \frac{5}{2} RT \right) - \left(4 \times \frac{5}{2} RT \right)$$

$$Q = RT$$

99. The expression of speed of sound is,

$$v = \sqrt{\frac{\gamma P}{\rho}}$$

Thus,

$$\gamma = \frac{v^2 \rho}{P}$$

Substitute values.

$$\gamma = \frac{330 \times 300}{10^5} \times \frac{1400}{1089}$$

$$\gamma = 1.4$$

The number of degree of freedom is,

$$\gamma = 1 + \frac{2}{f}$$

$$1.4 = 1 + \frac{2}{f}$$

$$f = 5$$

100. The frequency of tuning fork is directly proportional to square root of temperature, therefore

$$n \propto \sqrt{T}$$

$$\frac{n_1}{n_2} = \sqrt{\frac{T_1}{T_2}} \quad \text{..... (I)}$$

$$\text{At } T_1 = 51^\circ\text{C} = 324 \text{ K}, n_1 = n + 3$$

$$\text{At } T_2 = 16^\circ\text{C} = 289 \text{ K}, n_2 = n - 3$$

From equation (I)

$$\frac{n_1}{n_2} = \sqrt{\frac{324}{289}}$$

$$\frac{n+3}{n-3} = \frac{18}{17}$$

$$n = 105 \text{ Hz}$$

101. The expression of the beats per second from point A to B is,

$$n = f_A \left(\frac{v_s + u}{v_s} \right) - f_s \left(\frac{v_s - u}{v_s} \right)$$

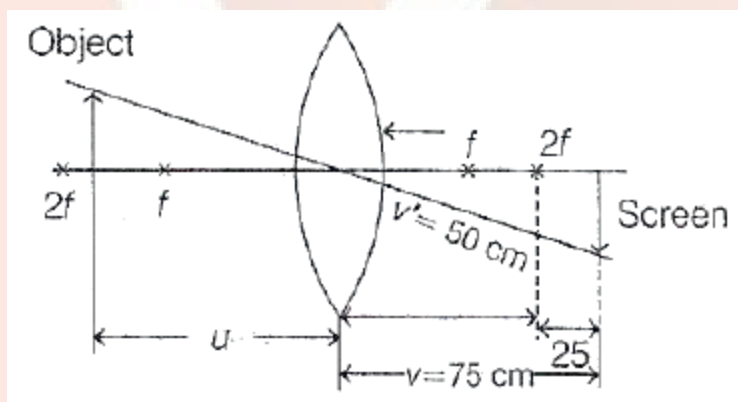
Substitute values.

$$10 = 680 \left(\frac{340 + u}{340} \right) - 680 \left(\frac{340 - u}{340} \right)$$

$$1 = 68 \left[\frac{340 + u - 340 + u}{340} \right]$$

$$u = 2.5 \text{ m/s}$$

102. Consider the figure,



Using the lens formula, we get

$$\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$$

$$\frac{1}{25} = \frac{1}{75} - \frac{1}{(-u)}$$

$$u = \frac{75}{2} \text{ cm}$$

The screen will be at $2f$ when it shift up to by 25 cm.

Thus the distance is calculated as,

$$d = v - u$$

$$d = 2f - u$$

$$d = 2(25) - \frac{75}{2}$$

$$d = 12.5 \text{ cm}$$

103. The angle of incident is calculated as,

$$\mu = \tan i$$

$$\sqrt{3} = \tan i$$

$$i = 60^\circ$$

The angle of refraction of the ray is calculated as,

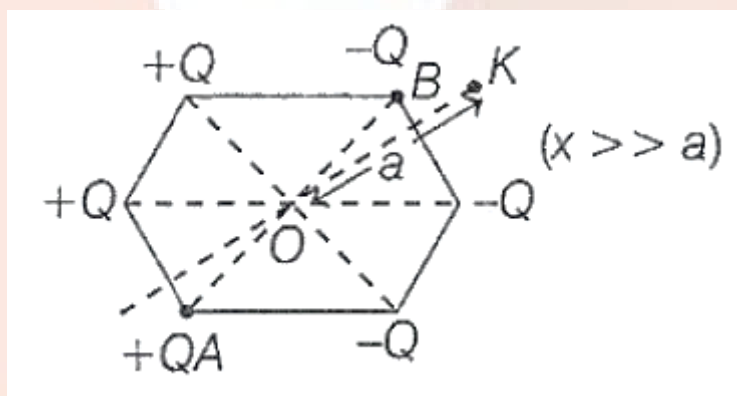
$$\mu = \frac{\sin i}{\sin r}$$

$$\sqrt{3} = \frac{\sin 60^\circ}{\sin r}$$

$$\sin r = \frac{\sin 60^\circ}{\sqrt{3}}$$

$$r = 30^\circ$$

104. Consider the figure of the system as shown below,



The force at the point K is,

$$F_K = \frac{2kQa}{(a^2 + x^2)^{\frac{3}{2}}}$$

There are three such forces as F_K .

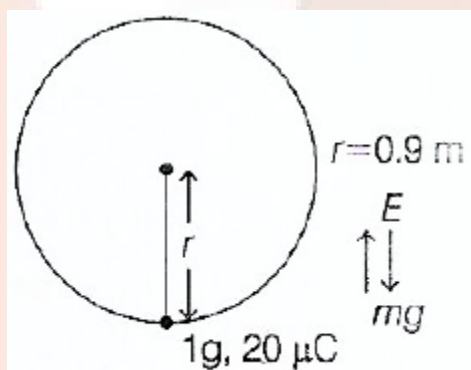
So the net force is,

$$F_{net} = \frac{4kQa}{(a^2 + x^2)^{\frac{3}{2}}}$$

$$F_{net} = \frac{4kQa}{x^3 \left(1 + \frac{a^2}{x^2}\right)^{\frac{3}{2}}}$$

$$F_{net} = \frac{4kQa}{x^3} \quad \left(\text{As, } \frac{a^2}{x^2} = 0, \quad x \gg a \right)$$

105. Consider the free body diagram of the ball as shown below.



The expression of the net effective force on the ball is,

$$F_{eff} = F_e - F_g$$

$$mg_{eff} = mg - qE$$

Substitute values,

$$(10^{-3})g_{eff} = (10^{-3})(10) - (20 \times 10^{-6})(100)$$

$$g_{eff} = 8 \times 10^{-3} \text{ N}$$

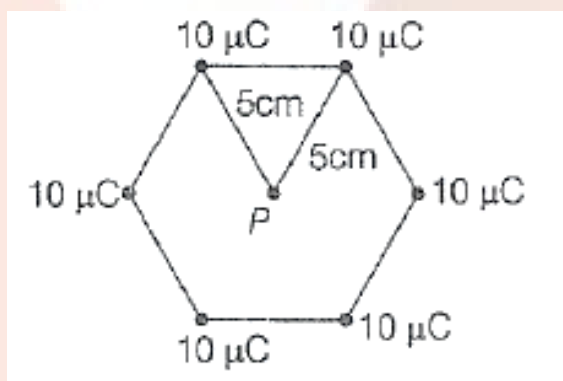
The minimum horizontal velocity that must be given to the ball at the lowest position can be calculated as,

$$v = \sqrt{5g_{eff}r}$$

$$v = \sqrt{5(8)\left(\frac{9}{10}\right)}$$

$$v = 6 \text{ m/s}$$

106. Consider the position of charges as shown below.



The net potential at P due to six charge is,

$$V_{net} = 6 \left(\frac{kq}{r} \right)$$

$$V_{net} = 6 \times \frac{(9 \times 10^9)(10 \times 10^{-6})}{5 \times 10^{-2}}$$

$$V_{net} = 6 \times 18 \times 10^5 \text{ N} \cdot \text{m}^2 / \text{C}^2$$

$$V_{net} = 1.08 \times 10^7 \text{ N} \cdot \text{m}^2 / \text{C}^2$$

107. The expression of the potential in an electric field is,

$$V = (x^2 - y^2)$$

The expression of the electric field is,

$$E = - \left[\frac{dV}{dx} \hat{i} + \frac{dV}{dy} \hat{j} \right]$$

Substitute values.

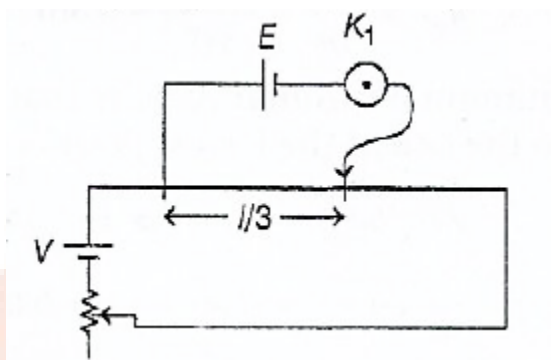
$$E = - \left[\frac{d(x^2 - y^2)}{dx} \hat{i} + \frac{d(x^2 - y^2)}{dy} \hat{j} \right]$$

$$E = - [2x\hat{i} - 2y\hat{j}]$$

$$E = -2x\hat{i} + 2y\hat{j}$$

Thus option (3) is correct.

108. Consider the diagram of the system as shown below.



The emf of the cell that gives balancing length $\frac{l}{3}$ is calculated

as,

$$E = K \left(\frac{l}{3} \right)$$

$$E = \frac{V}{l} \left(\frac{l}{3} \right)$$

$$E = \frac{V}{3} \quad \text{..... (I)}$$

When the length of potentiometer increases by $\frac{l}{2}$, the new length is calculated as,

$$l_1 = l + \frac{l}{2}$$

$$l_1 = \frac{3l}{2}$$

The new potential gradient is,

$$K' = \frac{V}{\left(\frac{3l}{2}\right)}$$

$$K' = \frac{2V}{3l}$$

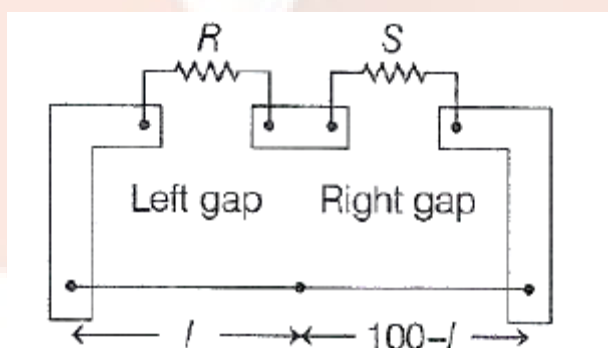
The emf is calculated as,

$$E = K'l'$$

$$\frac{V}{3} = \frac{2V}{3l} l'$$

$$l' = \frac{l}{2}$$

109. The meter bridge is shown below,



The equivalent resistance of $\frac{n}{2}$ series resistances is,

$$R_{eq} = R_1 + R_1 + \dots \frac{n}{2} \text{ times}$$

$$R_{eq} = \frac{nR_1}{2}$$

The equivalent resistance of $\frac{n}{2}$ parallel resistances is,

$$R'_{eq} = \frac{1}{R_1} + \frac{1}{R_1} + \dots \frac{n}{2} \text{ times}$$

$$R'_{eq} = \frac{n}{2R_1}$$

If l be the balancing length in the meter bridge wire, then,

$$\frac{R_{eq}}{R'_{eq}} = \frac{l}{100-l}$$

$$\frac{\left(\frac{nR_1}{2}\right)}{\left(\frac{2R_1}{n}\right)} = \frac{l}{100-l}$$

$$\frac{n^2}{4} = \frac{l}{100-l}$$

$$l = \frac{100n^2}{n^2 + 4}$$

110. The expression of the magnetic field normal to the plane of coil is,

$$B_{axis} = \frac{\mu_0 N i r^2}{2(r^2 + h^2)^{\frac{3}{2}}}$$

$$B_{axis} = \frac{\mu_0 N i r^2}{2r^3 \left(1 + \frac{h^2}{r^2}\right)^{\frac{3}{2}}}$$

$$B_{axis} = \frac{\mu_0 N i r^2}{2r^3} \left(1 + \frac{h^2}{r^2}\right)^{-\frac{3}{2}} \quad (\text{using binomial expansion})$$

$$B_{axis} = \frac{\mu_0 N i r^2}{2r^3} \left(1 - \frac{3}{2} \frac{h^2}{r^2}\right) \quad \dots (I)$$

The expression of the magnetic field at the centre of the coil is,

$$B_{centre} = \frac{\mu_0 N i}{2r} \quad \dots (II)$$

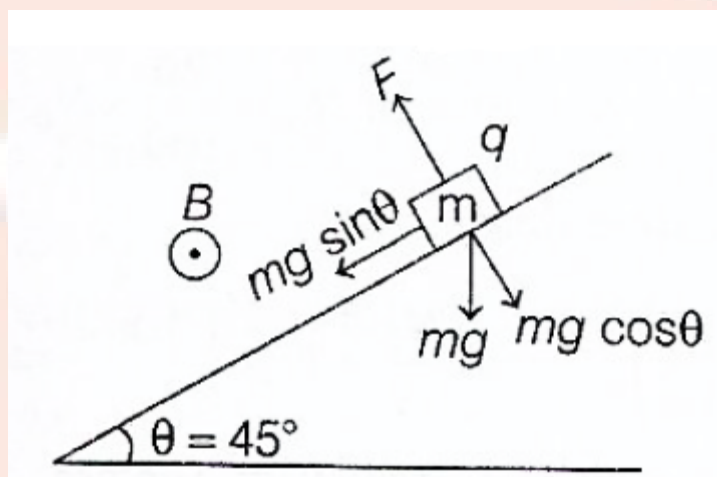
From equation (I) and (II)

$$B_{axis} = B_{centre} \left(1 - \frac{3}{2} \frac{h^2}{r^2}\right)$$

$$B_{axis} = B_{centre} - \left(\frac{3}{2} \frac{h^2}{r^2}\right) B_{centre}$$

Thus the magnetic field on axis is smaller than the field at the centre by a fraction $\frac{3}{2} \frac{h^2}{r^2}$.

111. The free body diagram of the system is shown below,



The magnetic force on the particle is,

$$F = Bqv$$

The equilibrium equation for the particle on inclined plane is,

$$F = mg \cos \theta$$

$$Bqv = mg \cos \theta$$

$$v = \frac{mg \cos \theta}{qB}$$

The time taken to reach at velocity v is,

$$v = u + at$$

$$\frac{mg \cos \theta}{qB} = 0 + (g \sin \theta)t$$

$$t = \frac{m \cot \theta}{qB}$$

Substituting the values, we get

$$t = \frac{0.02 \cot 45^\circ}{4 \times 10^{-3} \times 1}$$

$$t = 5 \text{ s}$$

112. Gauss's law states that,

$$\phi_E = \oint_s E dA = \frac{Q}{\epsilon_0}$$

Gauss's law of magnetism states that the total magnetic flux passing through a bar magnet around its enclosed surface is zero.

$$\phi_B = \int_s B dA = 0$$

According to Faraday's law of electromagnetic induction,

$$\varepsilon = -\frac{d\phi_B}{dt} \quad \text{..... (I)}$$

The electromotive force in a wire is the line integral, so

$$\varepsilon = \int E dl \quad \dots\dots (II)$$

From equation (I) and (II),

$$\int E dl = -\frac{d\phi_B}{dt}$$

In a time varying electric field, there exist an additional current to the conduction current known as displacement current. So, total current across the loop is,

$$i = i_c + i_d,$$

Here, i_c is the conductor current and i_d is the displacement current.

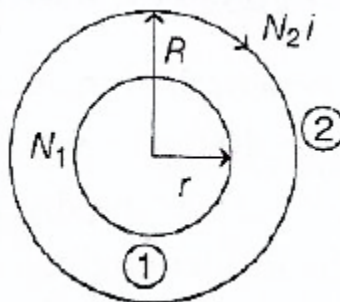
Thus, the modified form of the Ampere's law is,

$$\oint B dl = \mu_0 i$$

$$\oint B dl = \mu_0 [i_c + i_d]$$

Thus, option (3) is correct.

113. The two concentric coils are shown below.



The expression of the magnetic field at centre due to coil 2 is,

$$B_2 = \frac{\mu_0 N_2 I}{2R}$$

The expression of the magnetic flux through the coil is,

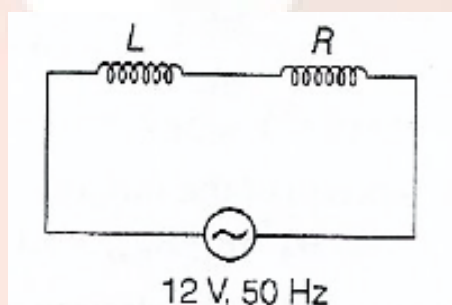
$$\phi_{B1} = N_1 B_2 A$$

$$MI = N_1 \left(\frac{\mu_0 N_2 I}{2R} \right) (\pi r^2)$$

$$M = \left(\frac{N_1 \mu_0 N_2 I \pi}{2} \right) \frac{r^2}{R}$$

$$M \propto \frac{r^2}{R}$$

114. Consider the figure of system as shown below.



The impedance of the AC circuit is calculated as,

$$Z = \frac{V_{rms}}{I}$$

$$Z = \frac{12}{0.5}$$

$$Z = 24 \Omega$$

The resistance can be calculated using the power factor as,

$$\cos \phi = \frac{R}{Z}$$

$$\cos \frac{\pi}{3} = \frac{R}{24}$$

$$R = 12 \Omega$$

115. From the expression of the intensity of EM wave,

$$I = \frac{P}{4\pi r^2}$$

$$\frac{1}{2} \epsilon_0 E_0^2 c = \frac{P}{4\pi r^2}$$

$$E_0^2 = \frac{P}{2\pi r^2 \epsilon_0 c}$$

Substituting the values, we get

$$E_0^2 = \frac{960}{2\pi(4)^2(8.85 \times 10^{-12})(3 \times 10^8)}$$

$$E_0 = 60 \text{ Vm}^{-1}$$

116. The energy of the incident light is calculated as,

$$E = \frac{hc}{\lambda}$$

$$E = \frac{6.6 \times 10^{-34} \times 3 \times 10^8}{4 \times 10^{-7}} \text{ J}$$

$$E = \frac{4.95 \times 10^{-19}}{1.6 \times 10^{-19}} \text{ eV}$$

$$E \approx 31 \text{ eV}$$

The acceleration (retardation) of the emitted electrons in the electric field is,

$$a = \frac{qE}{m}$$

The initial speed of the electron is,

$$v^2 = u^2 - 2as$$

$$(0)^2 = u^2 - 2\left(\frac{qE}{m}\right)(1)$$

$$u^2 = \frac{2qE}{m}$$

The maximum kinetic energy of the electron is calculated as,

$$K_{\max} = \frac{1}{2}mu^2$$

$$K_{\max} = \frac{1}{2}m\left(\frac{2qE}{m}\right)$$

$$K_{\max} = qE$$

$$K_{\max} = (1.6 \times 10^{-19})(2)$$

$$K_{\max} = 3.2 \times 10^{-19} \text{ J} = 2 \text{ eV}$$

So, the work function of the surface is,

$$W_0 = E - K_{\max}$$

$$W_0 = 3.1 - 2$$

$$W_0 = 1.1 \text{ eV}$$

117. The orbital period of electron in Bohr's first orbit is calculated as,

$$T_1 = \frac{2\pi r_1}{v_1}$$

$$T_1 = \frac{2\pi(0.53 \times 10^{-10})}{2.18 \times 10^6}$$

$$T_1 = 1.52 \times 10^{-16} \text{ s}$$

From using the formula for time period of n^{th} orbit,

$$T_n = n^3 T_1$$

$$4.1 \times 10^{-15} = n^3 (1.52 \times 10^{-16})$$

$$n = 3$$

118. The value of n can be calculated as,

$$N = N_0 \left(\frac{1}{2} \right)^{\frac{t}{t_{1/2}}}$$

$$\frac{N_0}{5} = N_0 \left(\frac{1}{2} \right)^{\frac{n}{138.6}}$$

$$\frac{1}{5} = \left(\frac{1}{2} \right)^{\frac{n}{138.6}}$$

$$\ln \left(\frac{1}{5} \right) = \frac{-n}{138.6} \ln 2$$

$$n \approx 322 \text{ days}$$

119. The total gain of the three amplifiers is calculated as,

$$A = A_1 \times A_2 \times A_3$$

$$A = 10 \times 20 \times 30$$

$$A = 6000$$

The output voltage is calculated as,

$$V_0 = AV_{in}$$

$$V_0 = (6000)(1 \times 10^{-3})$$

$$V_0 = 6 \text{ V}$$

120. The required coverage range of TV tower is,

$$d = \sqrt{2R_e h}$$

$$d = \sqrt{2(6.4 \times 10^6)(160)}$$

$$d \approx 45255 \text{ m}$$

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121. The wavelength is calculated as:

$$\begin{aligned}\lambda &= \frac{h}{\sqrt{2m(KE)}} \\ &= \frac{6.63 \times 10^{-34} \text{ Js}}{\sqrt{2 \times 9.0 \times 10^{-31} \text{ kg} \times 2 \times 10^{-25} \text{ J}}} \\ &= 11.043 \times 10^{-6} \text{ m}\end{aligned}$$

The conversion of wavelength into nm is shown below.

$$\begin{aligned}\lambda &= \frac{11.043 \times 10^{-6}}{10^{-9}} \text{ nm} \\ &= 1104.3 \text{ nm}\end{aligned}$$

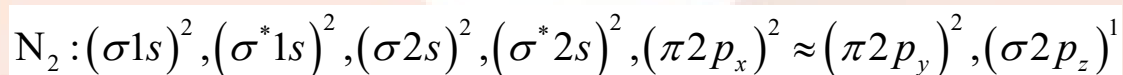
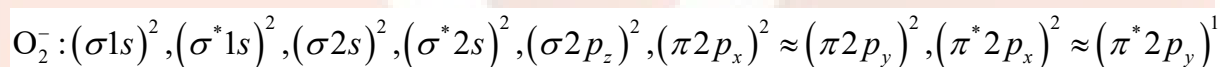
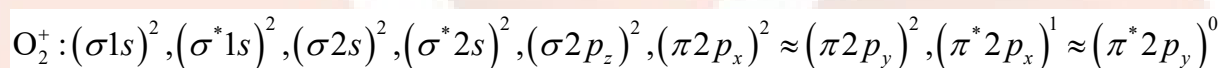
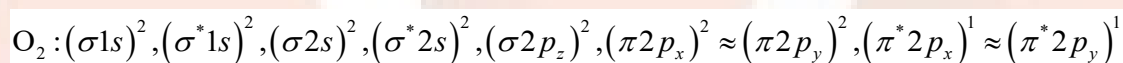
122. The uncertainty in velocity can be derived by using Heisenberg uncertainty principle as shown below.

$$\begin{aligned}\Delta x \cdot \Delta v &= \frac{h}{4\pi m} \\ &= \frac{6.634 \times 10^{-34} \text{ Js}}{4 \times \pi \times 9.1 \times 10^{-31} \text{ kg} \times 1.098 \times 10^{-9} \text{ m}} \\ &= \frac{1.6565 \times 10^4}{\pi}\end{aligned}$$

123. In 4-*f* series, due to the lanthanide contraction, the atomic number goes on increasing from La to Lu. As a result, atomic and ionic radii decreases. Therefore, the correct order is:

$$\text{Pr}^{3+} > \text{Gd}^{3+} > \text{Tm}^{3+}$$

124. The electronic configuration of the species which are having paramagnetic property (unpaired electrons) are shown below:



125. (A) The empirical formula of BrF_5 is AB_5E . Its hybridization is

sp^3d^2 and shape is square pyramidal.

(B) The empirical formula of SF_4 is AB_4E . Its hybridization is

sp^3d and shape is see saw.

(C) The empirical formula of XeF_4 is AB_4E_2 . Its hybridization

is sp^3d^2 and shape is square planar.

(D) The empirical formula of ClF_3 is AB_3E_2 . Its hybridization is sp^3d and shape is T-shape.

Therefore,

A-III, B-I, C-II, D-IV

126. We can calculate the root mean square velocity by the ratio of:

$$v_{\text{rms}} : v_{\text{avg}} : v_{\text{most probable}}$$

$$1 : 0.92 : 0.82$$

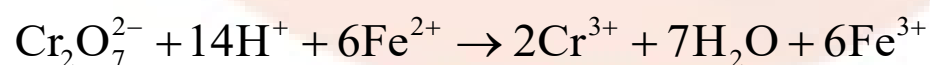
$$v_{\text{rms}} = \frac{v_{\text{most probable}}}{0.82}$$

$$= \frac{400}{0.82}$$

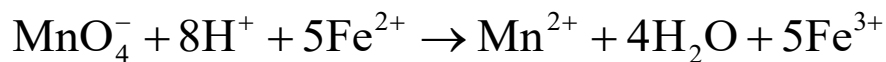
$$= 487.8$$

$$\approx 490 \text{ ms}^{-1}$$

127. The reaction taking place in acidic medium between Fe^{2+} and $\text{Cr}_2\text{O}_7^{2-}$ is shown below.



The reaction taking place in acidic medium between Fe^{2+} and MnO_4^- is shown below.



By applying the dilution formula:

$$M_1 \times V_1 \times n_1 = M_2 \times V_2 \times n_2$$

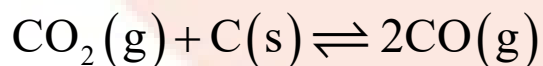
$$0.01 \times 20 \times 6 = M_2 \times 20 \times 5$$

$$\begin{aligned} M_2 &= \frac{0.01 \times 6}{5} \\ &= 0.012 \text{ M} \end{aligned}$$

128. The standard Gibbs energy change is calculated as:

$$\begin{aligned} \Delta G^\circ &= -RT \ln K_p \\ &= -0.008314 \text{ kJ/mol} \cdot \text{K} \times 400 \text{ K} \times \ln(1.0 \times 10^{-30}) \\ &= -3.3256 \text{ kJ/mol} \times -69.0775 \\ &= 229.724 \text{ kJ/mol} \\ &\approx 229.8 \text{ kJ/mol} \end{aligned}$$

129. The given reaction is:



$$\begin{aligned} \Delta n &= \text{Moles of gaseous products} - \text{Moles of gaseous reactants} \\ &= 2 - 1 \\ &= 1 \end{aligned}$$

First we need to calculate K_p as shown below.

$$\begin{aligned}
 K_p &= \frac{(p_{CO})^2}{(p_{CO_2})} \\
 &= \frac{(0.6)^2}{0.15} \\
 &= 2.4
 \end{aligned}$$

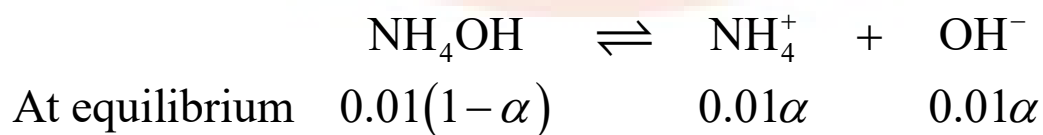
Now, the value of K_c is calculated as:

$$\begin{aligned}
 K_p &= K_c (RT)^{\Delta n} \\
 2.4 &= K_c (0.0821 \times 1000)^1 \\
 K_c &= \frac{2.4}{0.0821 \times 1000} \\
 &= 2.89 \times 10^{-2}
 \end{aligned}$$

130. The ionization constant of conjugate acid is calculated as:

$$\begin{aligned}
 K_a(NH_4^+) \cdot K_b(NH_3) &= 10^{-14} \\
 K_a(NH_4^+) &= \frac{10^{-14}}{K_b(NH_3)} = \frac{10^{-14}}{2.5 \times 10^{-5}} \\
 &= 4 \times 10^{-10}
 \end{aligned}$$

The dissociation reaction of ammonia is as follows:



The degree of ionization is:

$$\begin{aligned}\alpha &= \sqrt{\frac{K_b}{C}} \\ &= \sqrt{\frac{2.5 \times 10^{-5}}{0.01}} \\ &= 0.05\end{aligned}$$

Therefore concentration of hydroxide ion becomes:

$$\begin{aligned}[\text{OH}^-] &= C\alpha \\ &= 0.01 \times 0.05 \\ &= 0.0005\end{aligned}$$

The pH of the ammonia solution is calculated as:

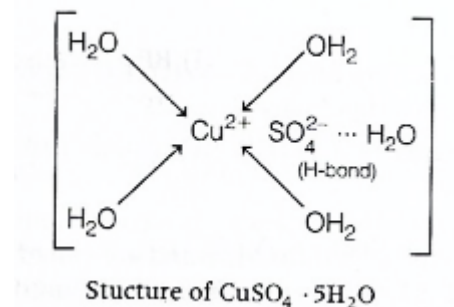
$$\begin{aligned}\text{pH} &= 14 - \text{pOH} \\ &= 14 - (-\log[\text{OH}^-]) \\ &= 14 + \log(0.0005) \\ &= 14 + (-3.3) \\ &= 10.7\end{aligned}$$

Therefore, pH is 10.7 and ionization constant is equal to

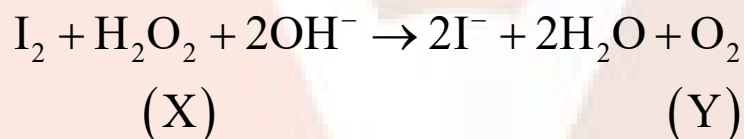
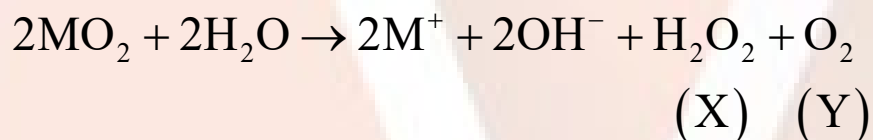
$$4 \times 10^{-10}$$

131. Copper sulphate pentahydrate is represented by the molecular formula of $[\text{Cu}(\text{H}_2\text{O})_4]\text{SO}_4 \cdot \text{H}_2\text{O}$. It has five molecules of water of crystallization. Four molecules of water are coordinated

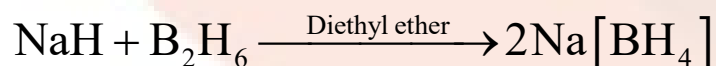
with Cu^{2+} and remaining one molecule of water coordinates through hydrogen bonding as shown below:



132. The reactions taking place are shown below.



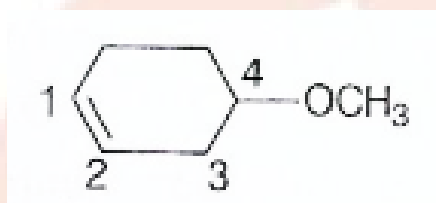
133. When sodium hydride undergoes a reaction with diborane, it forms sodium tetrahydroxoborate (X) as shown below.



134. Dry ice is used as refrigerant for ice cream and frozen foods. It preserves food items by using non cyclic refrigeration.

135. The presence of SO_2 and NO_2 in the atmosphere causes acid rain. These substances react with water, oxygen and other harmful chemicals to produce acid rain.

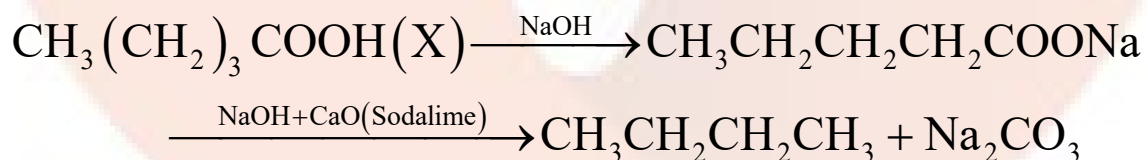
136. The IUPAC name of the given structure is 4-methoxycyclohexene as shown below.



137. The Wurtz reaction is shown below.



When sodium salt of pentanoic acid (X) reacts with sodalime and heated, it forms n-butane as shown below.

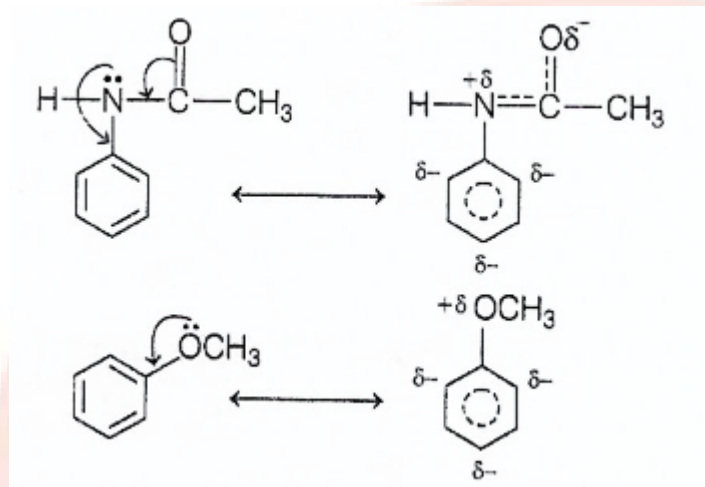


Therefore, X is $\text{CH}_3(\text{CH}_2)_3\text{COOH}$

138. The ortho and para directing groups are

$-\text{NHCOCH}_3$ and $-\text{OCH}_3$. These groups are known as electron

donating groups as these groups increases the density of electrons especially at ortho and para positions of the aromatic ring as follows:



139. The number of atoms is calculated as:

$$\begin{aligned}\text{For } 0.5 \text{ mol} &= 0.5 \times 6.022 \times 10^{23} \\ &= 3.011 \times 10^{23}\end{aligned}$$

Tetrahedral voids is equal to $2 \times$ Number of atoms of metal

Octahedral voids is equal to number of atoms of metal

Therefore,

$$\begin{aligned}\text{Number of tetrahedral voids} &= 2 \times 3.011 \times 10^{23} \\ &= 6 \times 10^{23} \\ &= 1 \text{ mol}\end{aligned}$$

$$\begin{aligned}\text{Number of octahedral voids} &= 3.011 \times 10^{23} \\ &= 0.5 \text{ mol}\end{aligned}$$

The total number of voids is $1 + 0.5 = 1.5 \text{ mol}$

140. The mole fraction of toluene (A) is calculated as:

$$\begin{aligned}x_A &= \frac{\frac{w_A}{M_A}}{\frac{w_A}{M_A} + \frac{w_B}{M_B}} \\ &= \frac{\frac{460}{92}}{\frac{460}{92} + \frac{390}{78}} \\ &= 0.5\end{aligned}$$

The mole fraction of benzene (B) is calculated as:

$$\begin{aligned}x_B &= 1 - x_A \\ &= 1 - 0.5 \\ &= 0.5\end{aligned}$$

The total pressure becomes equal to

$$\begin{aligned}p_{\text{total}} &= p_A^{\circ} x_A + p_B^{\circ} x_B \\ &= (32 \times 0.5) + (40 \times 0.5) \\ &= 36\end{aligned}$$

Mole fraction of toluene in vapour pressure (Y_T) is derived as:

$$p_T^\circ x_T = Y_T \cdot p_{\text{total}}$$

$$Y_T = \frac{p_T^\circ x_T}{p_{\text{total}}}$$

$$= \frac{32 \times \frac{1}{2}}{36}$$
$$= 0.444$$

141. The change in boiling point is calculated as shown below.

$$\Delta T_b = K_b [M_X + M_Y]$$
$$= K_b \left[\frac{w_X}{M_X} + \frac{w_Y}{M_Y} \right] \times \frac{1000}{100}$$
$$= 0.52 \left[\frac{9}{180} + \frac{1}{50} \right] 10$$
$$= 0.36$$

The boiling point of solution will become as:

$$\begin{aligned}T_b &= T_b^\circ + \Delta T_b \\&= 100 + 3.6 \\&= 100.36 \\&\approx 100.4\end{aligned}$$

142. The equilibrium constant is calculated as:

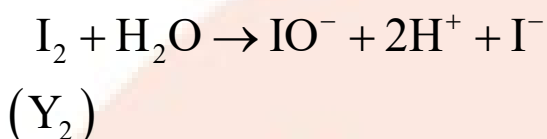
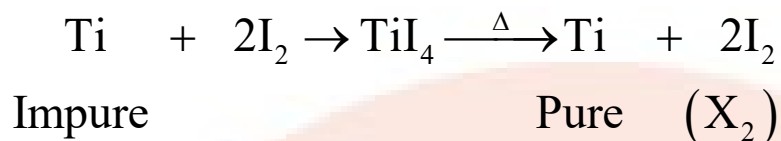
$$\begin{aligned}E_{\text{cell}}^\circ &= \frac{0.059}{n} \ln K \\0.59 &= \frac{2 \times 0.59}{0.059} \ln K \\\ln K &= 20 \\K &= 1 \times 10^{20}\end{aligned}$$

143. In collision theory, the term $e^{-E_a/RT}$ gives information about the fraction of molecules that are having energy equal to or greater than activation energy.

Therefore statement (1) is correct.

144. The enzyme trypsin changes proteins into amino acids. It activates the pancreatic enzymes. First, it convert proteins into dipeptides and then finally into amino acids.

145. The reactions are shown below.

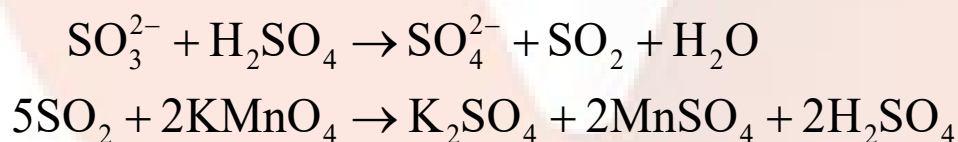


146. When phosphine reacts with CuSO_4 solution, it forms a black precipitate of cupric phosphide as shown below.



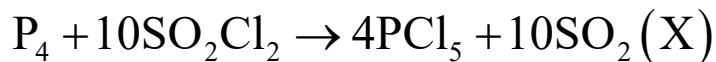
Therefore, statement (2) is not correct.

147. The reactions are shown below.

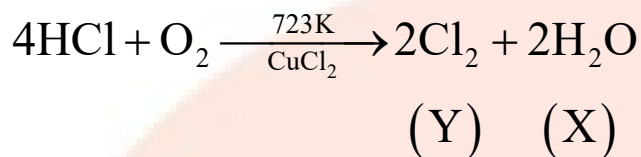


In the above reaction, the oxidation state of Mn in MnSO_4 is +2

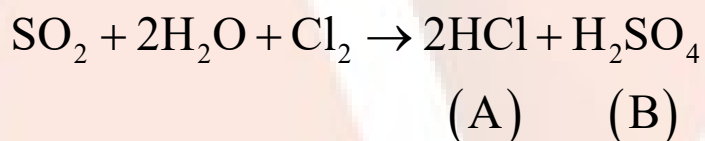
148. When thionyl chloride undergoes a reaction with white phosphorus, phosphorus pentachloride and sulphur dioxide is produced.



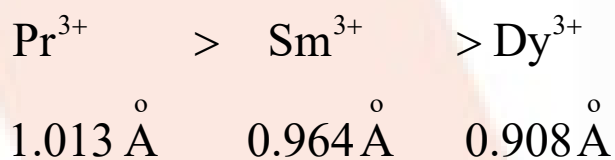
When HCl gas undergoes a reaction with oxygen in the availability of CuCl_2 , it produces water and chlorine gas.



Now, when Y reacts with X in water, it produces HCl (A) and H_2SO_4 (B)



149. (i) The correct order of ionic radii is shown below.



(ii) Eu^{2+} acts as strong reducing agent as +2 oxidation state oxidizes to +3 easily.

(iii) Pu shows +7 oxidation state. It can use maximum eight electrons because its electronic configuration is $5f^6 6d^0 7s^2$.

Therefore, (ii) and (iii) statements are correct.

150. The expression between tetrahedral and octahedral crystal field splitting is shown below.

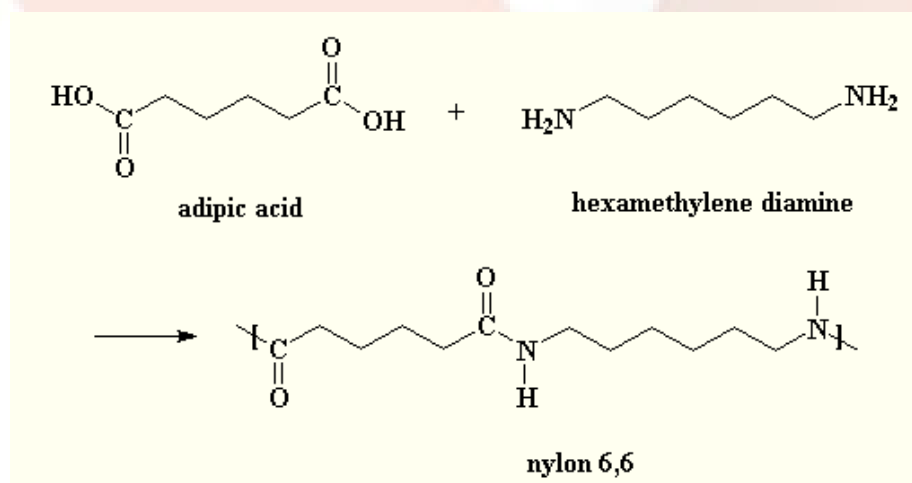
$$\Delta_t = \frac{4}{9} \Delta_o$$

$$\Delta_o = \frac{9}{4} \Delta_t$$

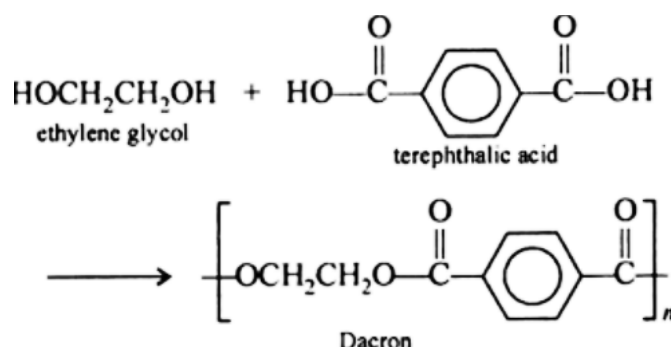
It is given that CFSE of tetrahedral complex of $[ML_4]^{n+}$ is x eV, therefore the CFSE of octahedral complex $[ML_6]^{n+}$ will become as:

$$\Delta_o = \frac{9x}{4}$$

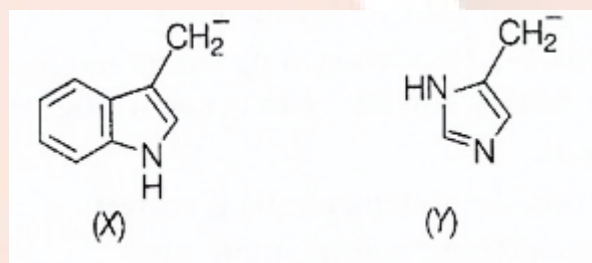
151. Nylon 6,6 is formed by the condensation reaction of hexamethylene diamine and adipic acid as shown below.



Dacron is formed by the condensation reaction of ethane-1,2-diol and terephthalic acid as shown below.



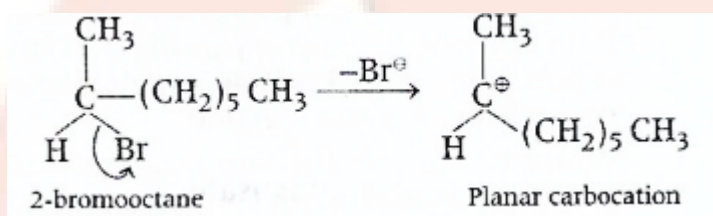
152. The group R in tryptophan (X) and histidine (Y) are shown below.



153. (i) The shape of the active sites is changed by the binding of inhibitor at allosteric site
- (ii) After attachment of chemical messenger to receptor, the shape of receptor gets changed.
- (iii) Without entering into the cell, a chemical messenger passes messages to the cell.
- (iv) Erythromycin is an example of bacteriostatic antibiotic.

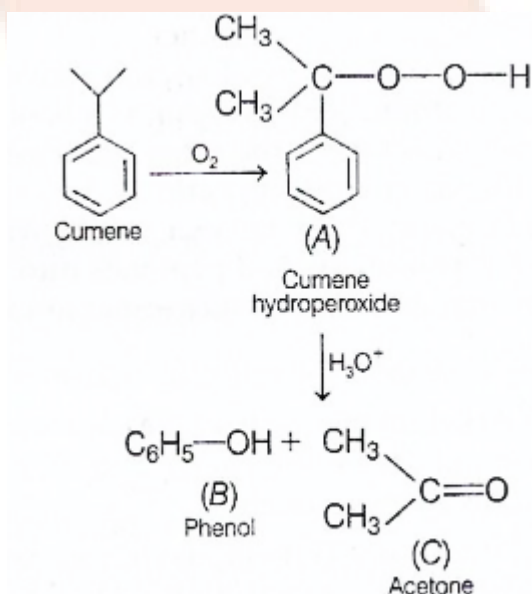
Therefore, statements (i) and (iv) are correct.

154. S_N1 hydrolysis of optically active 2-bromooctane forms $(\pm)$ -octan-2-ol as the reaction proceeds through a planar carbocation which can be attacked by the nucleophile from either side. The reaction is as follows:



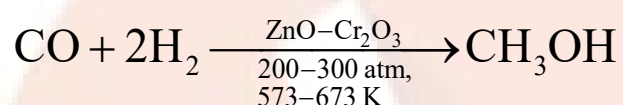
Therefore, Assertion (A) and Reason (R) is the correct explanation of (A)

155. The products (A), (B) and (C) formed are shown below



156. When carbon monoxide reacts with hydrogen in the presence of reagents such as $\text{ZnO-Cr}_2\text{O}_3$ at a pressure of 200-300 atm and at a temperature of 573-673 K, the formation of methanol takes place

The reaction is shown below.

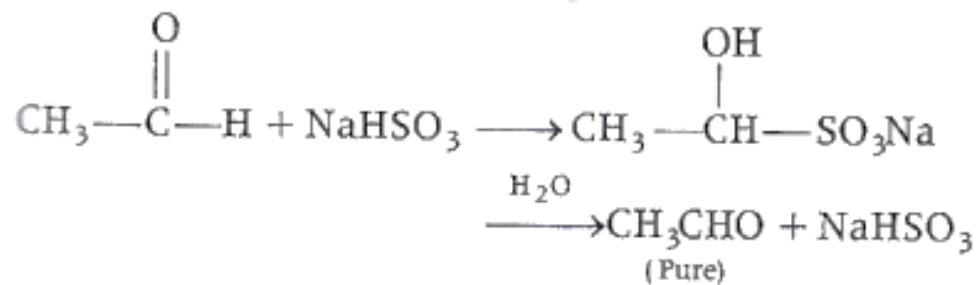


157. Alkenes (X) is electrophilic because alkenes contain π – electron cloud hence, they are highly reactive substances.

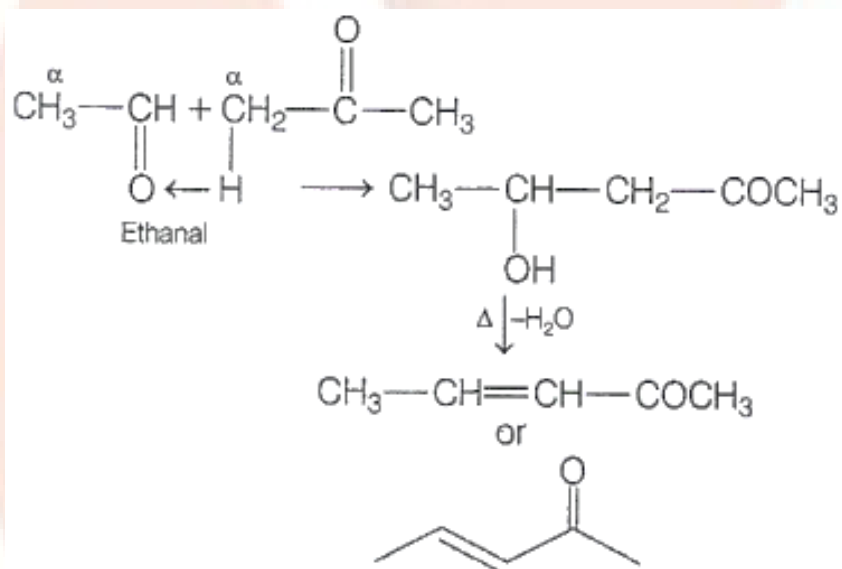
Therefore, X shows electrophilic reactions.

Carbonyl compounds (Y) is nucleophilic because carbonyl compounds are highly polar in nature. Therefore, Y shows nucleophilic reactions.

158. NaHSO_3 is used for the purification of aldehydes. The reaction is shown below.



159. The aldol condensation reaction taking place between ethanal and propanone is as follows:



$X = \text{CH}_3\text{CH}_2\text{OH}$

$\text{Ar N}_2^+ \text{Cl}^- \longrightarrow \boxed{\phantom{\text{Reaction}}} \longrightarrow \text{ArH}$

$\text{H}_3\text{PO}_2/\text{H}_2\text{O}$

Reaction (X):

$$\text{C}_6\text{H}_5\text{N}_2^+\text{Cl}^- + \text{C}_2\text{H}_5\text{OH} \longrightarrow \text{C}_6\text{H}_5\text{H} + \text{N}_2\uparrow + \text{H}_3\text{C}-\overset{\text{O}}{\parallel}{\text{C}}-\text{H} + \text{HCl}\uparrow$$

Reaction (Y):

$$\text{C}_6\text{H}_5\text{N}_2^+\text{Cl}^- + \text{H}_3\text{PO}_2 + \text{H}_2\text{O} \longrightarrow \text{C}_6\text{H}_5\text{H} + \text{N}_2\uparrow + \text{HCl}\uparrow + \text{H}_3\text{PO}_3$$

